

BIRATIONAL INVARIANTS FROM HODGE STRUCTURES AND QUANTUM MULTIPLICATION

Ludmil Katzarkov, Maxim Kontsevich, Tony Pantev, Tony Yue Yu

Abstract. We introduce new invariants of smooth complex projective varieties, called Hodge atoms. Their construction combines rational Gromov-Witten invariants with classical Hodge theory and relies on the notion of an F -bundle, which is a non-archimedean version of a non-commutative Hodge structure. The Hodge atoms arise from the spectral decomposition of the F -bundle under the Euler vector field action, and behave additively under blowups, in accordance with Iritani’s blowup theorem. We compute several examples and demonstrate applications to birational geometry. In particular, we prove that a very general cubic fourfold is not rational. We also obtain a new proof of the equality of Hodge numbers of birational Calabi-Yau manifolds in any dimension. Furthermore, we show that the framework naturally extends to representations of other motivic Galois groups. This enables the theory of atoms to produce new obstructions to rationality over non-algebraically closed fields of characteristic zero as well.

Contents

1	Introduction	2
2	Atoms of equivariant singularities	9
2.1	Isolated singularities of functions	9
2.2	Equivariant isolated singularities of functions	12
3	Generalities on F-bundles	17
3.1	F -bundles	17
3.2	Operations on F -bundles	18
3.3	Natural symmetries of F -bundles	19
3.4	Maximal and overmaximal F -bundles	19
3.5	The A-model F -bundle	22
4	Decomposition theorems	42
4.1	Spectral decomposition theorem	42
4.2	The blowup formula	43
4.3	The Leray-Hirsch formula	47
5	G-atoms	47
5.1	Mumford-Tate groups in the nc setting	48
5.2	G -atoms of projective manifolds	50
5.3	G -atomic F -bundles	58
5.4	Hodge atoms	61
5.5	Atoms from motivated motives	63

Date: August 7, 2025 (Revised March 5, 2026).

6	New obstructions to rationality	64
6.1	F -bundles and Givental's theorem	64
6.2	Four-dimensional cubics	69
6.3	Birational maps of Calabi-Yau manifolds	72
6.4	Enhanced atoms and further examples	74

1 Introduction

We propose new birational invariants of algebraic varieties that arise from a combination of classical Hodge theory and Gromov-Witten theory. Even though we were guided and inspired by ideas from Homological Mirror Symmetry (HMS), our arguments do not use the mirror duality directly. Instead of HMS we utilize motivic information encoded in the algebraic and symplectic geometry on the same projective manifold. This approach leads to new non-rationality results, and resolves a number of classical problems studied in the foundational works of Beauville, Donagi, Debarre, Hassett, Huybrechts, Kulikov, Kuznetsov, Pirutka, Thomas, Tschinkel, Voisin, and many others [8, 9, 19, 20, 21, 30, 31, 34, 35, 32, 42, 41, 53, 54, 55, 56, 57, 66, 5, 2, 77, 79, 78, 76]. In particular, we prove the non-rationality of very general cubic fourfolds over the complex numbers. We also study the non-rationality of varieties over non-algebraically closed fields, as well as applications and examples in higher dimensions.

In this work, we only scratch the surface of the realm of potential applications of quantum cohomology to birational geometry. Given a smooth complex projective variety X , our main invariants, called *Hodge atoms*, are pieces in an intrinsic decomposition of the so-called A -model F -bundle associated to X , which is a non-archimedean version of variation of A -model non-commutative Hodge structures (or nc -Hodge structures for short, see [48]). The A -model F -bundle consists of enumerative data (coming from Gromov-Witten theory) on top of the $\mathbb{Z}/2$ -weighted version¹ of the classical polarized $\mathbb{Q}$ -Hodge structure H of X . We show that at a general point in the locus of Hodge classes inside the base of the F -bundle (i.e. the Frobenius manifold), the generalized eigenspaces for the quantum multiplication by the Euler vector field are compatible with the Hodge structure on H . Concretely, we show that each generalized eigenspace decomposes canonically into $\overline{\mathbb{Q}}$ -linear representations of the Mumford-Tate group Hod of $\mathbb{Z}/2$ -graded polarizable pure Hodge structures, which we call *Hodge atoms*. We prove that the *chemical formula* or *atomic composition* of X , i.e. the collection of Hodge atoms appearing in the decomposition of H , behaves additively under blowups and can be used as an obstruction to birational equivalence.

The setup readily generalizes to other realizations of motives leading to variants of the notion of Hodge atoms. The formalism of motivic atoms we develop here is just the beginning of a more intricate story. Atoms are fairly coarse building blocks of nc -Hodge structures. For instance there are no natural morphisms of atoms, so they do not form a category and thus are not well-suited for studying specific birational maps. Atoms work well as first level obstructions to birational equivalence but are far from being complete birational invariants. We will explore other more enhanced theories in future works.

Here is a more detailed overview of our work. Suppose for concreteness X is a complex smooth projective variety. Let $H := H_B^\bullet(X, \mathbb{Q})$ be the rational Betti cohomology of X taken with its

¹Having $\mathbb{Z}/2$ -weights here means that we have folded the $\mathbb{Z}$ -grading of cohomology to a $\mathbb{Z}/2$ -grading, or equivalently, that we consider Hodge structures modulo Tate twists. The loss of the $\mathbb{Z}$ -grading is natural from the point of view of non-commutative geometry, see [48].

$\mathbb{Z}/2$ -grading, let $\{T_i\}$ be a homogeneous basis of H , and let $t = (t_i)$ be the dual coordinates on H . Write $\text{NE}(X, \mathbb{Z}) \subset \text{N}_1(X, \mathbb{Z}) = \text{im}(\text{CH}_1(X) \rightarrow H_2(X, \mathbb{Q}))$ for the monoid of numerically effective curve classes on X , and let $\mathbb{Q}[[q]]$ denote the completion of the monoid algebra $\mathbb{Q}[\text{NE}(X, \mathbb{Z})] = \mathbb{Q}[q^\beta \mid \beta \in \text{NE}(X, \mathbb{Z})]$ with respect to the maximal ideal $(q^\beta, \beta \neq 0)$. The quantum product for X

$$\star: H \otimes H \longrightarrow H[[q, t]].$$

is a deformation of the cup product on H over the big Novikov ring $\text{Nov}_X := \mathbb{Q}[[q, t]]$ of X . This deformation is given by Gromov-Witten invariants (see [52], which we will review in Section 3.5.1). It gives rise to a structure of formal Frobenius manifold on a quotient of $\text{Spf } \text{Nov}_X$ by a foliation given by the divisor axiom (see [26, 60]).

To extract birational invariants from symplectic data, we will work with a variant of the notion of Frobenius manifold, which we call a F-bundle (see Section 3.1). It is a non-archimedean version of several existing notions in the literature: TE-structure [37, 36], quantum D -module [43], and a variation of nc-Hodge structures [48]. We have to pass to non-archimedean geometry for two reasons. First, the complex analytic convergence of the series defining the quantum product is not known in general. Second, and more crucially, the spectral decomposition theorem does not hold over the complex numbers due to the Stokes phenomenon [48, 38].

Abstractly a $\mathbb{k}$ -analytic F-bundle is specified by data $(\mathcal{H}, \nabla)/B$, where B is a $\mathbb{k}$ -analytic supermanifold, $\mathcal{H}$ is a $\mathbb{k}$ -analytic vector bundle on $B \times \mathbb{D}$, where $\mathbb{D}$ denotes the germ at 0 in an analytic disk with coordinate u , and ∇ is a meromorphic connection on $\mathcal{H}$ with poles at most at $u = 0$, and such that $\nabla_{u^2 \partial_u}$ and $\nabla_{u\xi}$ have no poles for all $\mathbb{k}$ -analytic vector fields $\xi \in T_B$. Sections $\xi \in T_B$ act on $\mathcal{H}|_{u=0}$ by the residues of ∇_ξ at $u = 0$, and we say that the F-bundle is **overmaximal** if locally on B we can find a cyclic vector for the action, and further say it is **maximal** if the evaluation on the cyclic vector gives an isomorphism between T_B and $\mathcal{H}|_{u=0}$.

The A-model F-bundle $(\mathcal{H}, \nabla)/\mathcal{B}_X$ associated to X consists of a base $\mathcal{B}_X$ which is roughly a non-archimedean $\mathbb{k}$ -analytic space obtained from $\text{Spf } \mathbb{Q}[[q, t]]$, a trivial vector bundle $\mathcal{H}$ with fiber H over $\mathcal{B} \times \mathbb{D}$, and a meromorphic flat connection ∇ on $\mathcal{H}$ called the quantum connection:

$$\left\{ \begin{array}{l} \nabla_{u\partial_u} = u\partial_u - u^{-1}\text{Eu} \star (-) + \text{Gr}, \\ \nabla_{\partial_{t_i}} = \partial_{t_i} + u^{-1}T_i \star (-), \\ \nabla_{\xi q \partial_q} = \xi q \partial_q + u^{-1}\xi \star (-) \quad \text{for } \xi \in H^2(X, \mathbb{k}), \end{array} \right.$$

where $\text{Eu} \in \Gamma(\mathcal{B}, \mathcal{H})$ is the Euler element, Gr is the grading operator, and $\xi q \partial_q$ is the derivation of $\mathbb{k}[[q]]$ given by $\xi q \partial_q(q^\beta) = (\beta \cdot \xi)q^\beta$ (see Section 3.5 for the details).

The A-model F-bundle $(\mathcal{H}, \nabla)/\mathcal{B}_X$ is overmaximal and can be made maximal by removing redundant degree 2 variables on the base $\mathcal{B}_X$ (implied by the divisor axiom of Gromov-Witten invariants). Write $(\mathcal{H}, \nabla)/B_X$ for the resulting maximal F-bundle. For a rigid point $b \in B_X$, the generalized eigenspace decomposition of the $\text{Eu}_b \star (-)$ action on $\mathcal{H}_{b,0}$ extends to a subbundle decomposition of $\mathcal{H}|_{u=0}$ locally near b . The spectral decomposition theorem from [38] implies the base B_X also locally splits into a product $\prod B_i$, and the F-bundle $(\mathcal{H}, \nabla)/B$ splits into an external direct sum of maximal F-bundles $(\mathcal{H}_i, \nabla_i)/B_i$, extending the above generalized eigenspace decomposition.

Let us look at an example of the spectral decomposition in the case of blowups. Consider the blowup $\pi: \hat{X} \rightarrow X$ along a smooth closed subvariety $Z \subset X$ of codimension $r \geq 2$. We have a

classical decomposition of cohomology

$$H^\bullet(\widehat{X}, \mathbb{Q}) \simeq H^\bullet(X, \mathbb{Q}) \oplus \bigoplus_{i=1}^{r-1} H^\bullet(Z, \mathbb{Q})[-2i].$$

This decomposition is in fact the eigenspace decomposition of the Euler vector field action at the limiting point b_∞ of B corresponding to the pullback π^* of an ample class on X . The eigenvalues are 0 and $(r-1)e^{\frac{\pi\sqrt{-1}(2j-1)}{r-1}}$ for $j = 1, \dots, r-1$. The 0-eigenspace is isomorphic to $H^\bullet(X, \mathbb{Q})$, and the others all isomorphic to $H^\bullet(Z, \mathbb{Q})$. As we move away from the limiting point, each eigenvalue above may further split, as shown in Figure 1. In Section 4.2, we identify the factors resulting from the spectral decomposition with the A-model F-bundle of X and $(r-1)$ copies of the A-model F-bundles of Z , using Iritani's blowup formula [44].

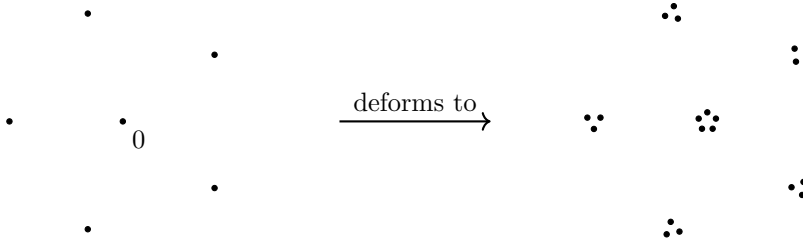


Figure 1: Eigenvalues of the Eu-action at b_∞

Next we explain the construction of Hodge atoms. Let again X be a smooth complex projective variety. Consider the subspace of its even cohomology spanned by the Hodge classes

$$H(X)^{\text{Hod}} := \bigoplus_i H^{i,i}(X) \cap H^{2i}(X, \mathbb{Q}).$$

It gives a closed $\mathbb{k}$ -analytic subspace $B_X^{\text{Hod}} \subset B_X$ of the base of the maximal A-model F-bundle of X . Let $\widetilde{B}_X \rightarrow B_X$ be the ramified spectral covering given by the eigenvalues of the $\text{Eu} \star (-)$ -action, let be $U_X \subset B^{\text{Hod}}$ the locus where the number of eigenvalues of the $\text{Eu} \star (-)$ -action is maximal, and $\widetilde{U}_X := \widetilde{B}_{X,\text{red}} \times_B U_X$ be the unramified part of the reduced spectral cover. The **set of local Hodge atoms** of X is the set $\pi_0(\widetilde{U}_X)$ of connected components of this unramified spectral cover. The spectral decomposition theorem (see [38] and Theorem 4.1), blowup theorem (see [44] and Theorem 4.5), and the projective bundle decomposition theorem (see [45, 38] and Theorem 4.11) imply that there are natural correspondences between the local Hodge atoms of X and the local Hodge atoms of a blowup of X or the local Hodge atoms of a projective bundles over X . Using these correspondences we define the **set of Hodge atoms of smooth complex projective varieties** as the quotient set

$$\text{HAtoms} := \left(\bigsqcup_{[X]} \pi_0(\widetilde{U}_X) / \text{Aut}(X) \right) / \sim,$$

where the union is taken over isomorphism classes of complex smooth projective varieties, and the equivalence relation $\sim$ is generated by the elementary equivalences corresponding to disjoint unions, blowups with smooth centers, and projective bundles.

The **atomic composition** or **chemical formula** of a particular variety X is then the multiset encoded in the finite support function $\text{CF}(X) = \sum_{\alpha \in \pi_0(\tilde{U}_X)/\text{Aut}(X)} \text{mult}_\alpha \delta_\alpha : \text{HAtoms} \rightarrow \mathbb{Z}_{\geq 0}$, where mult_α is the degree of the cover $\tilde{U}_{X,\alpha} \rightarrow U_X$ corresponding to the connected component α , and $\delta_\alpha : \text{HAtoms} \rightarrow \mathbb{Z}_{\geq 0}$ is the delta function supported at the atom $\alpha \in \text{HAtoms}$ which is represented by the local atom α .

Example. If the canonical class K_X of X is nef, then $\text{CF}(X)$ is a singleton, because the Eu -action has a single eigenvalue in this case (see Lemma 5.24).

The above discussion on blowups implies that $\text{CF}(\hat{X}) = \text{CF}(X) + (r-1)\text{CF}(Z)$. Since birational equivalences between smooth projective varieties are generated by blowups with smooth centers of codimension ≥ 2 we arrive at the following.

Non-rationality criterion: *Let X be a d -dimensional smooth complex project variety (with $d \geq 2$). If there is an Hodge atom in the Hodge atomic composition of X that does not appear in the Hodge atomic composition of any variety of dimension less than or equal to $d-2$, then X is not rational.*

Next we apply this criterion to show the non-rationality of certain 4-dimensional varieties. To that end, we have to understand the Hodge atoms coming from varieties of dimension less than or equal to 2, i.e. from points, curves and surfaces. Moreover, for each birational class of surfaces it is sufficient to consider the atomic composition of one representative in the class.

For every Hodge atom α represented by a local atom $\alpha \in \pi_0(\tilde{U}_X)/\text{Aut}(X)$, we get a well defined F-bundle $(\mathcal{H}^\alpha, \nabla)/U_X$, which is a generalized eigenspace for the action of $\text{Eu} \star (-)$ on the maximal A-model F-bundle $(\mathcal{H}, \nabla)/B_X$ of X . The atoms formalism assures that this generalized eigenspace F-bundle comes equipped with a natural action of the universal Mumford-Tate group Hod controlling the $\mathbb{Z}/2$ -weighted, pure, rational Hodge structures. The isomorphism class of the finite dimensional Hod -representation $\mathcal{H}^\alpha|_{b,u=0}$ is well-defined independent of the choice the representative local atom α or of the subsequent choice of a point b . In particular, we consider the following invariants of α :

- (1) **dimension ρ_α of the space of Hodge classes:** the dimension of the Hod -fixed subspace of the fiber $\mathcal{H}^\alpha|_{b,u=0}$.
- (2) **Hodge polynomial $P_\alpha \in \mathbb{Z}[t, t^{-1}]$:** the coefficient at t^k is equal to the dimension of the subspace of the fiber $\mathcal{H}^\alpha|_{b,u=0}$ with $(p-q)$ -degree equal to k .

Let us calculate these invariants for atoms coming from points, curves and surfaces.

- (1) For any Hodge atom α coming from points or curves, we have $\text{Coeff}_{t^2}(P_\alpha) = 0$.
- (2) For any Hodge atom α coming from a surface X with $h^{2,0} = 0$, we have $\text{Coeff}_{t^2}(P_\alpha) = 0$.
- (3) By the classification of surfaces, we know that a minimal surface with a geometric genus $\neq 0$ is always birational to a smooth S for which $K_S \geq 0$. Hence by the above example, a surface with geometric genus $\neq 0$ will contribute a single Hodge atom α with $\rho_\alpha \geq 3$ (containing the classes in H^0 , H^4 and the first Chern class of an ample line bundle in H^2).

Let X be a very general cubic 4-fold. Its Hodge diamond splits into the primitive part and an irreducible transcendental part (see [75]).

Let $(\mathcal{H}, \nabla)/B_X$ be the maximal **A**-model F-bundle associated to X . By Givental's calculation [27] of the quantum product at a particular point $b \in B$, the **Eu**-action has eigenvalues $\{0, 9, 9e^{2\pi i/3}, 9e^{4\pi i/3}\}$. The generalized eigenspace V corresponding to the eigenvalue 0 has dimension 24, containing the whole transcendental part. We have $\dim V^{\text{Hod}} = 2$, $\dim V^{p-q=-2} = \dim V^{p-q=2} = 1$ and $\dim V^{p-q=0} = 22$. The other three eigenspaces are 1-dimensional. As we move from b to generic points in U_X , V may further split, contributing more atoms, but there will always be one atom α with $\rho_\alpha \leq 2$ and $\text{Coeff}_{t^2}(P_\alpha) = 1$, which *can not* come from varieties of dimensions less than or equal to 2. We conclude that X can not be rational.

Theorem (see Theorem 6.8). *A very general 4-dimensional cubic hypersurface in $\mathbb{CP}^5$ is not rational.*

In Section 6.3, we give another application of Hodge atoms. We obtain a new proof of Batyrev's theorem [7, Corollary 6.29] asserting that birationally equivalent Calabi-Yau manifolds have equal Hodge numbers. The traditional proofs of this result [7, 80, 47] use either motivic integration, or p -adic Hodge theory and Čebotarev density. The proof via atoms is based on completely different technology.

Theorem (see Theorem 6.16). *Birationally equivalent Calabi-Yau manifolds have equal Hodge numbers.*

By design, the theory of Hodge atoms is motivic in nature. It is a special case of a theory of G -atoms governed by some proreductive algebraic group G which controls a particular realization of André motives. In the case of Hodge atoms the relevant group is the Galois group **Hod** of the Tannakian category of polarizable pure $\mathbb{Z}/2$ -weighted $\mathbb{Q}$ -Hodge structures. We develop the general theory of G -atoms for a general motivic Galois group G in Section 5. One variant of particular interest is the theory of **motivated atoms**, given by $G = \text{Mot}^{\mathcal{K}}$, the Galois group of the category of a $\mathbb{Z}/2$ -graded version of André's motives [3] equipped with the fiber functor of Betti cohomology.

Motivated atoms are useful for detecting non-rationality over a subfield $\mathcal{K} \subset \mathbb{C}$ which is not algebraically closed. To illustrate the idea, consider a smooth hypersurface X of multi-degree $(1, 1, 1, 1)$ in $\mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{P}^1$, defined over the algebraic closure $\bar{\mathcal{K}}$ of $\mathcal{K}$. It is isomorphic to the blowup of $\mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{P}^1$ at an elliptic curve E , and hence the motivated atomic decomposition of X contains 8 atoms associated to points and one more complicated atom α_E associated to E .

Now we consider a model $\mathfrak{X}$ of X defined over the non-closed field $\mathcal{K}$ such that the Galois group $\text{Gal}(\bar{\mathcal{K}}/\mathcal{K})$ acts by permuting transitively the 4 factors in the product $\mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{P}^1$. Let $(\mathcal{H}, \nabla)/B_{\mathfrak{X}}$ be the **A**-model F-bundle associated to $\mathfrak{X}$. At a particular point $b \in B_{\mathfrak{X}}$, the **Eu**-action

has 3 eigenvalues with multiplicities 1, 4 and 7 respectively. The last generalized eigenspace has Hodge polynomial $5 + t + t^{-1}$ and only 2 algebraic classes defined over $\mathcal{K}$. Then this representation of $\text{Mot}^{\mathcal{K}}$ cannot split further into representations coming from atoms associated to 0 and 1-dimensional varieties over $\mathcal{K}$. Hence we conclude the non-rationality of $\mathfrak{X}/\mathcal{K}$.

Many similar examples can be worked out in an analogous manner but we will leave these to the future work [14]. In addition to the Hodge atoms and motivated atoms, we will consider other enhanced versions of atoms, e.g. atoms which are furnished with the image of the topological K -theory in the nc-Hodge structure, or atoms enhanced with Mukai pairings or Serre automorphisms. The theory of enhanced atoms will be developed in the forthcoming work [49], where many more applications are worked out. Here we concentrate on the foundations of the theory of atoms and their most basic applications as obstructions to rationality.

This paper is organized as follows. In Section 2, we motivate the theory of F-bundles and atoms from a special case of the mirror symmetry B-model, in the elementary context of equivariant isolated singularities. The constructions in this section have no direct relation to the A-model invariants or rationality we discuss in the rest of the paper. In Section 3.1, we introduce and study F-bundles, which is a formal/non-archimedean version of non-commutative Hodge structure. In particular in Section 3.5.2 we introduce the non-archimedean A-model F-bundle of a complex smooth projective variety, the main object underpinning the construction of atoms. Section 4 discusses the spectral decomposition of F-bundles as well as the decomposition of F-bundles associated to blowups and projective bundles. In Section 5, we develop the general theory of G -atoms for any motivic Galois group G . Finally in Section 6, we apply the theory of atoms to obtain new obstructions to rationality in birational geometry. After recasting Givental's classical quantum cohomology computation in the language of F-bundles in Section 6.2, we use Hodge atoms to show that the very general cubic fourfold is not rational. In Section 6.3 we use Hodge atoms to give a new proof of the equality of Hodge numbers of birational Calabi-Yau manifolds in any dimension. Finally in Section 6.4 we discuss briefly enhancements of atoms and illustrate how enhanced atoms lead to further non-rationality results.

Acknowledgements

We would like to thank Denis Auroux, Hiroshi Iritani and Yuri Tschinkel for their interest and for their invaluable help at various stages of this work. We are grateful to Leonardo Cavenaghi, Sheel Ganatra, Mark Gross, Thorgal Hinault, Y.P. Lee, Ernesto Lupercio, James McKernan, Todor Milanov, Yong-Geun Oh, Yukinobu Toda, Chenyang Xu, Song Yu, Chi Zhang and Shaowu Zhang for inspiring discussions around the subject.

During the preparation of this paper Ludmil Katzarkov was supported by the National Science Fund of Bulgaria, the National Scientific Program “VIHREN” (Project no. KP-06-DV-7), the Institute of Mathematics and Informatics of the Bulgarian Academy of Sciences, the Simons Foundation (grant SFI-MPS-T-Institutes-00007697), the Ministry of Education and Science of the Republic of Bulgaria (contract DO1-239/10.12.2024), the Basic Research Program of the National Research University Higher School of Economics, the Simons Investigators Award (no. 003136), the Simons Collaboration on Homological Mirror Symmetry (award no. 003093), and by the NSF FRG grant DMS-2245099. Tony Pantev was supported by the Simons Collaboration on Homological mirror symmetry (award no. 347070) and the NSF FRG grants DMS-2244978 and DMS-2245099. Tony Yue Yu was partially supported by the NSF grants DMS-2302095 and DMS-2245099.

In addition, Ludmil Katzarkov and Maxim Kontsevich are grateful to Jeffrey Fuqua for his

continuous financial support of mathematical research throughout the years, and for his support and leadership in establishing the excellent creative environment at IMSA, University of Miami, where much of this work was done.

Notation and conventions

k	a field of characteristic zero, coefficient field of a Weil cohomology theory.
$\mathbb{k} \supset k$	an algebraically closed non-archimedean field whose absolute value is trivial on k .
$\mathcal{K}$	a field of characteristic zero, base field of smooth projective varieties, typically chosen to be a subfield of $\mathbb{C}$.
$\mathbb{D}$	the germ at 0 of a $\mathbb{k}$ -analytic affine line with coordinate u .
$(\mathcal{H}, \nabla)/B$	an F-bundle over a $\mathbb{k}$ -analytic space B .
$\kappa = \nabla_{u^2 \partial_u} _{\mathcal{H}_{u=0}}$	the residual endomorphism on $\mathcal{H}_{u=0}$.
$\circ$	the Frobenius product on T_B associated with a maximal F-bundle.
Eu	the Euler vector field on B associated with a maximal F-bundle.
$\mathcal{H}_o$	the restriction $\mathcal{H} _{u=0}$.
$\langle \gamma_1 \cdots \gamma_n \rangle_{g,n,\beta}$	Gromov-Witten invariants. We omit g and n from the notation when $g = 0$ and n is evident.
$\star$	the big quantum product on the cohomology $H^\bullet(X)$ of a smooth projective variety X .
Deg	the degree operator on $H^\bullet(X)$.
Gr	the grading operator on $H^\bullet(X)$.
$\text{NE}(X, \mathbb{Z})$	the monoid of numerical equivalence classes of effective curves.
$\text{NS}(X, \mathbb{Z})$	the Néron-Severi group of X .
$\text{N}^1(X, \mathbb{Z})$	the group of numerical equivalence classes of divisors in X , also equal to the maximal torsion-free quotient $\text{NS}(X, \mathbb{Z})_{\text{tf}}$ of the Néron-Severi group of X .
$\text{N}_1(X, \mathbb{Z})$	the group of numerical equivalence classes of curves in X .
$\mathcal{C}^\mathcal{K}$	the $\mathbb{Q}$ -linear Tannakian category of André motives of smooth projective $\mathcal{K}$ -varieties.
$\mathcal{C}^\mathcal{K}/\text{Tate}$	the $\mathbb{Q}$ -linear category of nc-André motives of smooth projective $\mathcal{K}$ -varieties.
$\text{G}^\mathcal{K}$	the pro-reductive motivic Galois group for André motives of smooth projective $\mathcal{K}$ -varieties.
(G, ϵ_G)	a proreductive algebraic group over k with a marked central element ϵ_G of order 2.
$(\text{H}, \epsilon_{\text{H}})$	the symmetry group of Hochschild graded Dolbeault cohomology.
$(\text{Hod}, \epsilon_{\text{Hod}})$	the Mumford-Tate group of $\mathbb{Z}/2$ -weighted pure Hodge structures.
$(\text{Mot}, \epsilon_{\text{Mot}})$	the Galois group of the $\mathbb{Z}/2$ -weighted version of André's motivated motives.
$\tilde{U}_{\mathfrak{X}}$	the unramified part of the reduced spectral cover associated with the operator κ attached to the A-model $\mathbb{k}$ -analytic F-bundle of $\mathfrak{X}$.
$\text{Atoms}_G^\mathcal{K}$	the set of G -atoms of smooth projective $\mathcal{K}$ -varieties.

$\text{CF}_G(\mathfrak{X})$	the chemical formula or G -atomic composition of $\mathfrak{X}$, i.e. the multiset of G -atoms of $\mathfrak{X}$ together with their multiplicities.
$\text{Atoms}_G^{\mathcal{K}, \text{F}}$	the set of geometric G -atomic F-bundles for smooth projective $\mathcal{K}$ -varieties.
$\text{CF}_G^{\text{F}}(\mathfrak{X})$	the multiset of geometric G -atomic F-bundles arising from $\mathfrak{X}$ together with their multiplicities.
HAtoms	the set of Hodge atoms of complex smooth projective varieties.

2 Atoms of equivariant singularities

To motivate the general theory we start with a particular B-model example dealing with simplified nc-Hodge theoretic invariants of equivariant isolated singularities of functions. We regard these simplified invariants as “baby atoms”. On its own this example is not related to any birational invariants but we include it since it provides valuable conceptual insights into the general theory we will develop later.

2.1 Isolated singularities of functions

Recall that if $f: (X, x_0) \rightarrow \mathbb{C}$ is an analytic germ of a function on a d -dimensional manifold X , with an isolated critical point x_0 , then the **Milnor number** μ_f of f is a basic numerical invariant of the germ, which is given by

$$\mu_f := \dim_{\mathbb{C}} (\mathcal{O}_{X, x_0} / J(f)),$$

where $J(f) = \text{image} \left[\mathcal{T}_{X, x_0} \xrightarrow{(-) \lrcorner df} \mathcal{O}_{X, x_0} \right]$ is the Jacobian ideal of f . We use $\mathcal{T}_X$ to denote the tangent sheaf, and T_X to denote the tangent bundle. Furthermore, since the singularity of f at x_0 is isolated, the Milnor number is finite and the function germ (X, x_0, f) has a **universal unfolding** which is a family of function germs

$$F: (X \times \mathfrak{U}_f, (x_0, u_0)) \longrightarrow \mathbb{C}$$

such that $F|_{(X, x_0) \times \{u_0\}} = f$, the family is parametrized by a germ $(\mathfrak{U}_f, u_0)$ of a μ_f -dimensional manifold, and the natural map $T_{\mathfrak{U}_f, u_0} \rightarrow \mathcal{O}_{X, x_0} / J(f)$ given by $v \mapsto [(\text{Lie}_v F)|_{X \times \{u_0\}}]$ is an isomorphism of vector spaces.

Explicitly, a model for the universal unfolding is constructed as follows. Let $\text{Milnor}_{f, x_0} := \mathcal{O}_{X, x_0} / J(f)$ be the **Milnor ring** of f at x_0 , and let $g_1, \dots, g_{\mu_f} \in \mathcal{O}_{X, x_0}$ be a collection of function germs whose images form a basis of Milnor_{f, x_0} as a complex vector space. Then we get a family of holomorphic functions

$$(2.1) \quad F: (X \times \mathbb{C}^{\mu_f}, (x_0, 0)) \longrightarrow \mathbb{C}, \quad (x, z) \longmapsto f(x) + \sum_{i=1}^{\mu_f} z_i g_i(x).$$

We can then take $(\mathfrak{U}_f, u_0)$ to be the germ of $(\mathbb{C}^{\mu_f}, 0)$, and F to be the germ of the function (2.1). The universal unfolding is a well-defined family of germs and the analytic automorphisms of the singularity (X, x_0, f) act naturally on the model universal unfolding $(X \times \mathfrak{U}_f, (x_0, u_0), F)$ defined above.

Intrinsically one can construct and work with the universal unfolding as follows. Let X be a d -dimensional complex manifold, $x_0 \in X$, and let $f: X \rightarrow \mathbb{C}$ be the germ at x_0 of a holomorphic function for which x_0 is an isolated critical point. Let $x_0 \in V \subset X$ be a Stein neighborhood on which f is defined, and such that x_0 is the only critical point of f in V . Let $x_0 \in \mathbb{B} \subset V$, be a smaller Stein neighborhood, which is an open Euclidean ball and such that the closure $\overline{\mathbb{B}}$ of $\mathbb{B}$ in X , is contained in V .

Consider the space

$$\mathfrak{M}_{\mathbb{B}} := \{ g \in \mathcal{O}(\overline{\mathbb{B}}) \mid dg_x \neq 0 \in T_{X,x}^{\vee}, \text{ for all } x \in \partial\overline{\mathbb{B}} \}.$$

Note that the algebra $\mathcal{O}(\overline{\mathbb{B}})$ of holomorphic functions on the Stein compact $\overline{\mathbb{B}}$ is naturally a nuclear Fréchet $\mathbb{C}$ -vector space with respect to the topology of compact convergence [29, Chapter V.6.1], and so the open subspace $\mathfrak{M}_{\mathbb{B}} \subset \mathcal{O}(\overline{\mathbb{B}})$ is naturally an infinite dimensional complex Fréchet manifold. This manifold represents the moduli functor which to every Stein set S assigns the set of all holomorphic functions on $S \times \mathbb{B}$ whose relative differential over S does not vanish anywhere on $S \times \partial\overline{\mathbb{B}}$.

At each point $g \in \mathfrak{M}$ the tangent space $T_{\mathfrak{M}_{\mathbb{B}},g}$ is naturally identified with $\mathcal{O}(\overline{\mathbb{B}})$, and it contains a natural closed subspace of finite codimension, namely the image

$$\mathfrak{L}_g := \text{image} \left[\mathcal{T}_X(\overline{\mathbb{B}}) \xrightarrow{\text{Lie}_{(-)} g} \mathcal{O}(\overline{\mathbb{B}}) \right]$$

of the “evaluation on g ” map $\xi \rightarrow \text{Lie}_{\xi} g$ from holomorphic vector fields to holomorphic functions on $\mathbb{B}$. The codimension of $\mathfrak{L}_g$ in $T_{\mathfrak{M}_{\mathbb{B}},g}$ is equal to the degree of the natural map

$$(2.2) \quad \begin{array}{ccc} m_g : & \partial\overline{\mathbb{B}} & \longrightarrow (\mathbb{C}^d - \{0\})/\mathbb{R}_{>0} \\ & \parallel & \parallel \\ & S^{2d-1} & S^{2d-1} \end{array} \quad , \quad m_g(x) := \frac{dg_x}{\|dg_x\|},$$

and so $\text{codim}(\mathfrak{L}_g \subset T_{\mathfrak{M}_{\mathbb{B}},g})$ is equal to the sum $\sum_{x \in \text{crit}(g)} \mu_{g,x}$ of the Milnor numbers of g at all critical points $\text{crit}(g) = \text{zero}(dg)$ of g in $\mathbb{B}$.

In particular $\mathfrak{L} \subset T_{\mathfrak{M}_{\mathbb{B}}}$ is a holomorphic foliation of finite codimension on $\mathfrak{M}_{\mathbb{B}}$. Given a point $g \in \mathfrak{M}_{\mathbb{B}}$ we will write $[g]$ for the leaf of this foliation through g . The universal unfolding $\mathfrak{U}_f$ of the singularity germ (X, x_0, f) is canonically the germ of the leaf space $[\mathfrak{M}_{\mathbb{B}}/\mathfrak{L}]$ of this foliation at the point $[f]$. The explicit model we considered above is a holomorphic slice through f to the leaves of $\mathfrak{L}$. Note that this approach allows us to define a canonical universal unfolding $\mathfrak{U}_g$ for any $g \in \mathfrak{M}_{\mathbb{B}}$ by simply taking $\mathfrak{U}_g$ to be the germ at $[g]$ of the leaf space $[\mathfrak{M}_{\mathbb{B}}/\mathfrak{L}]$.

For $[g] \in \mathfrak{U}_f$, we still have the identification $T_{\mathfrak{U}_f,[g]} \cong \text{Milnor}_g = \mathcal{O}(\mathbb{B})/J(g)$, where $J(g) = \text{image} \left[T(\mathbb{B}) \xrightarrow{(-) \lrcorner dg} \mathcal{O}(\mathbb{B}) \right]$ is the global Jacobian ideal of g on $\mathbb{B}$. This identification induces an analytic commutative associative product

$$\circ : T_{\mathfrak{U}_f} \otimes_{\mathcal{O}_{\mathfrak{U}_f}} T_{\mathfrak{U}_f} \longrightarrow T_{\mathfrak{U}_f},$$

which is just K. Saito’s Frobenius manifold structure [69, 70, 71, 26, 68] on $\mathfrak{U}_f$. Being open in a Fréchet vector space, the infinite dimensional manifold $\mathfrak{M}$ will have a natural holomorphic Euler vector field corresponding to the scaling action of $\mathbb{C}$ on $\mathcal{O}(\overline{\mathbb{B}})$. The projection of this vector field in $T_{\mathfrak{M}_{\mathbb{B}}}/\mathfrak{L}$ gives a holomorphic vector field Eu on the universal unfolding $\mathfrak{U}_f$.

The operator $\kappa := \text{Eu} \circ (-): T_{\mathfrak{U}_f} \rightarrow T_{\mathfrak{U}_f}$ is an analytic vector bundle endomorphism, and for any point $[g] \in \mathfrak{U}_f$ we have that $\kappa_{[g]}: T_{\mathfrak{U}_f, [g]} \rightarrow T_{\mathfrak{U}_f, [g]}$ preserves the natural decomposition $T_{\mathfrak{U}_f, [g]} = \text{Milnor}_g = \bigoplus_{x \in \text{crit}(g)} \text{Milnor}_{g,x}$, where $\text{Milnor}_{g,x} = \mathcal{O}_{X,x}/J(g)$. Furthermore, for each $x \in \text{crit}(g)$, the subspace $\text{Milnor}_{g,x} \subset \text{Milnor}_g$ is the generalized eigenspace of $\kappa_{[g]}$ of eigenvalue $g(x)$.

Saito's theory of Frobenius structures of singularities [69], [68, Chapter VI] implies that we have the following decomposition

Theorem 2.3. *Let (X, x_0, f) be the germ of an isolated singularity of a holomorphic function on a d -dimensional manifold. Let $g \in \mathfrak{M}_{\mathbb{B}}$ be a point close to $f \in \mathfrak{M}_{\mathbb{B}}$. Then the germ $\mathfrak{U}_f([g])$ of the universal unfolding of f at the point $[g] \in \mathfrak{U}_f$ decomposes canonically into a product*

$$\mathfrak{U}_f([g]) \cong \prod_{x \in \text{crit}(g)} \mathfrak{U}_{x,g},$$

where $(X, x, {}^xg)$ is the germ of g at x . Moreover this decomposition is compatible with Saito's Frobenius structures and with Euler vector fields.

Let $\text{Aut}(f)$ denote the group of biholomorphic automorphisms of the germ (X, x_0, f) . It is a Fréchet analytic Lie group which naturally acts on the germ $\mathfrak{M}_f$ of the universal unfolding. Its connected component $\text{Aut}^0(f)$ acts trivially and $\text{Aut}(f)$ acts through its quotient $\text{Aut}(f) \rightarrow \pi_0(\text{Aut}(f))$. By the approximation theorem, the group $\pi_0(\text{Aut}(f))$ will act with finite stabilizers and so the moduli $\mathcal{M}_f := [\mathfrak{M}_f / \pi_0(\text{Aut}(f))]$ of deformations of f will be a germ of a complex analytic orbifold.

The intrinsic construction of $\mathfrak{U}_f$ works equally well to produce a “global” universal unfolding parametrizing functions on d -dimensional manifolds with isolated singularities and a given Milnor number μ . More precisely, we can consider the subset $\mathfrak{M}_{\mathbb{B}, \mu} \subset \mathfrak{M}_{\mathbb{B}}$ comprised of all $g \in \mathfrak{M}_{\mathbb{B}}$ such that $\deg m_g = \mu$. Then $\mathfrak{M}_{\mathbb{B}, \mu}$ is a union of connected components of $\mathfrak{M}_{\mathbb{B}}$. By the discussion above the leaf space $[\mathfrak{M}_{\mathbb{B}, \mu} / \mathfrak{L}]$ is a smooth complex analytic orbifold $\mathcal{U}_{d, \mu}$ with finite stabilizers (possibly non-separated), which we will think of as the universal unfolding of d -dimensional germs functions with isolated singularities with Milnor number μ . Again the tangent bundle $T_{\mathcal{U}_{d, \mu}}$ is equipped with a Frobenius product, we have an Euler vector field Eu on $\mathcal{U}_{d, \mu}$, and the germ of $\mathcal{U}_{d, \mu}$ at any point $[g]$ decomposes into a product of pieces compatible with Frobenius products and Euler fields, so that each piece corresponds to the generalized eigenspace of the operator $\text{Eu}_{[g]} \circ (-)$.

The moduli $\mathcal{M}_{d, \mu}$ of d -dimensional germs of functions with a single isolated singularity of Milnor number μ is then a closed, possibly singular, analytic substack in $\mathcal{U}_{d, \mu}$. Note that $\mathcal{M}_{d, \mu}$ is in fact an algebraic Deligne-Mumford stack defined over $\mathbb{Q}$. The formal completion $\mathcal{U}_{d, \mu}^\wedge$ of $\mathcal{U}_{d, \mu}$ along $\mathcal{U}_{d, \mu}$ is a formal stack over $\overline{\mathbb{Q}}$ (in general non-algebraizable) which is a smooth thickening of $\mathcal{M}_{d, \mu}$.

Remark 2.4. By the theory of Frobenius manifolds [60] we know that the Frobenius powers $\text{Eu}^k := \text{Eu}^{\circ k}$ of the Euler vector field satisfy the relations of the Witt Lie algebra of polynomial vector fields on the affine line, i.e.

$$(2.5) \quad [\text{Eu}^k, \text{Eu}^l] = (l - k) \text{Eu}^{k+l-1}, \quad \text{for all } k, l \geq 0,$$

where we view $\text{Eu}^{\circ k}$ as a holomorphic vector field on $\mathcal{U}_{d, \mu}$, and the bracket in (2.5) is the Lie bracket on vector fields on $\mathcal{U}_{d, \mu}$.

In an open dense domain of $\mathcal{U}_{d,\mu}$ the function has μ isolated Morse singularities, and the local coordinates on $\mathcal{U}_{d,\mu}$ are given by the critical values $x_1, \dots, x_\mu$. The Euler field and its powers are given by

$$\text{Eu} = \sum_i x_i \partial_{x_i}, \quad \text{Eu}^k = \sum_i x_i^k \partial_{x_i}.$$

Suppose $[f] \in \mathcal{U}_{d,\mu}$ is a germ of a function with isolated singularities with total Milnor number μ . One can show that there is a well defined germ of a complex analytic submanifold $\mathcal{W} \subset \mathcal{U}_{d,\mu}$ at point $[f]$ such that $\text{span}_{\mathbb{C}} \left(\left\{ \text{Eu}_{[g]}^k \right\}_{k \geq 0} \right) = T_{\mathcal{W},[g]}$ for all $[g] \in \mathcal{W}$ close to $[f]$. Moreover, the critical points of $[g]$ are in canonical bijection with the critical points of $[f]$ and that bijection preserves the multiplicities.

The picture becomes even more interesting when we consider equivariant isolated singularities of functions.

2.2 Equivariant isolated singularities of functions

For ease of reference let us introduce the following

Definition 2.6. Let G be a finite group. A d -dimensional complex analytic G -equivariant function germ with isolated singularities is the data $(G \curvearrowright X, f)$, where

- (1) Y is a possibly disconnected complex d -dimensional manifold .
- (2) Y is equipped with a holomorphic G -action $G \curvearrowright Y$.
- (3) $\mathbf{w}: Y \rightarrow \mathbb{C}$ is a G -invariant holomorphic function, with isolated critical points, such that the G -action on $\text{crit}(\mathbf{w})$ is transitive.
- (4) (X, f) is the G -equivariant germ of $(Y, \mathbf{w})$ at the subset $\text{crit}(\mathbf{w})$.

Note that our condition requiring that $\mathbf{w}$ is invariant and that the G -action on $\text{crit}(\mathbf{w})$ is transitive implies that a G -equivariant germ always has a single critical value.

Example 2.7. Consider the action of $\mathbb{Z}/N$ on $\mathbb{C}$ given by $k \mapsto (z \mapsto \zeta^k \cdot z)$ for some primitive N -th root of unity ζ . The function $\mathbb{C} \rightarrow \mathbb{C}$, $z \mapsto z^N$ is $\mathbb{Z}/N$ -invariant, and its derivative Nz^{N-1} vanishes only at $z = 0$. Therefore if we take X to be the germ of $\mathbb{C}$ at $0 \in \mathbb{C}$ and f to be the germ of $z \mapsto z^N$ at 0, then the data $(\mathbb{Z}/N \curvearrowright X, f)$ is a $\mathbb{Z}/N$ -equivariant function germ with isolated singularities.

Remark 2.8. (1) If $G = \{\mathbf{e}\}$ is the trivial group, then a d -dimensional G -equivariant function germ with isolated singularities is just a germ (X, f) of a holomorphic function on a d -dimensional manifold at an isolated critical point of the function.

(2) Suppose $(G \curvearrowright X, f)$ is a G -equivariant function germ with isolated singularities and let $p \in X$ be a critical point of f . Let $H \subset G$ be the stabilizer of $p \in X$, let $X(p)$ be the connected component of X containing p , and let ${}^p f = f|_{X(p)}$. Equivalently $p \in \text{crit}(\mathbf{w}) \subset Y$, $X(p)$ is the germ of Y at p , and ${}^p f$ is the germ of $\mathbf{w}$ at $p \in Y$. In this situation H acts on $X(p)$ by analytic isomorphisms and $(H \curvearrowright X(p), {}^p f)$ is an H -equivariant function germ with a single isolated singularity.

(3) Conversely, suppose $(H \curvearrowright X, f)$ is a d -dimensional H -equivariant function germ with a single isolated singularity. Suppose $H \subset G$ of H is embedded as a subgroup in some finite group G . Then we can construct an induced d -dimensional G -equivariant germ $(G \curvearrowright \text{ind}_H^G(X), \text{ind}_H^G(f))$. Here

- (1) $\text{ind}_H^G(X) := G \times_H X = (G \times X)/H$, where $h \in H$ acts on $(a, x) \in G \times X$ by $h \cdot (a, x) = (ah^{-1}, h(x))$.
- (2) $\text{ind}_H^G(X) = G \times_H X$ is equipped with the left translation G -action, i.e. $a \in G$ acts on $[(b, x)]$ by $a \cdot [(b, x)] := [(ab, x)]$.
- (3) $\text{ind}_H^G(f): \text{ind}_H^G(X) \rightarrow \mathbb{C}$ is the holomorphic function given by $\text{ind}_H^G(f)([(a, x)]) = f(x)$. This function is well defined since f is H -invariant.

The group H acts freely on $G \times X$ and so $\text{ind}_H^G(X)$ is a complex manifold germ. It maps holomorphically onto the zero dimensional complex manifold G/H and in fact $\text{ind}_H^G(X)$ is (non-canonically) identified with a disjoint union of $\#(G/H)$ -copies of X .

Definition 2.9. Let $(G \curvearrowright X, f)$ be a G -equivariant function germ with isolated singularities. We define the G -equivariant Milnor number μ_f^G of the germ to be the dimension of the G -invariants in the Milnor ring $\text{Milnor}_f(X)$ of f .

By definition $\text{Milnor}_f(X)$ is the ring of global analytic function on the closed analytic subgerm in X defined by the sheaf of Jacobian ideals $\mathcal{J}(f) := \text{image} \left[\mathcal{T}_X \xrightarrow{(-) \lrcorner df} \mathcal{O}_X \right]$ of f . Since by assumption f has isolated critical points, we have that

$$\text{Milnor}_f(X) = \bigoplus_{x \in \text{crit}(f)} \text{Milnor}_{f,x},$$

and so

$$\mu_f^G = \dim_{\mathbb{C}} (\text{Milnor}_f(X))^G = \dim_{\mathbb{C}} \left(\bigoplus_{x \in \text{crit}(f)} \text{Milnor}_{f,x} \right)^G.$$

Equivalently, μ_f^G is the dimension of the G -invariants $T_{\mathfrak{U}_f, u_0}^G$ in the tangent space of the base of the universal unfolding at the base point $u_0 \in \mathfrak{U}_f$. Note that by definition we have $0 \leq \mu_f^G \leq \mu_f$.

Example 2.10. Let as before $((\mathbb{Z}/N) \curvearrowright X, f)$ denote the germ of $(\mathbb{C}, 0, z^N)$ viewed as $\mathbb{Z}/N$ -equivariant germ for the action given by a primitive N -th root of unity. In this case the Milnor ring is $\mathbb{C}\{z\}/(z^{N-1}) \cong \mathbb{C}^{N-1}$ and $(\mathbb{C}\{z\}/(z))^{N-1} \cong \mathbb{C}$, and so $\mu_f = N - 1$ and $\mu_f^{\mathbb{Z}/N} = 1$.

In the G -equivariant context, if the finite group G acts on the germ X preserving f , then, as explained in the previous section, the group G will act on the universal unfolding $(\mathfrak{U}_f, u_0)$. In particular, the linear action of G on the Milnor ring $\text{Milnor}_f(X)$ is compatible with the linear action of G on the tangent space $T_{\mathfrak{U}_f, u_0}$.

With this picture in mind we now have the following

Definition 2.11. We will say that a G -equivariant germ $(G \curvearrowright X, f)$ of a function with isolated singularities is **strongly atomic** if, for a general point $g \in \mathfrak{U}_f^G$, the G -invariant function $g: X \rightarrow \mathbb{C}$ has a unique critical point, and we will say that $(G \curvearrowright X, f)$ is **atomic** if, for a general point $g \in \mathfrak{U}_f^G$, the function g has a single G -orbit of critical points.

Note that strongly atomic equivariant germs of functions are automatically atomic. In the other direction, if a G -equivariant germ $(G \curvearrowright X, f)$ is atomic, then the single singular fiber of f will be a G -stable hypersurface with finitely many isolated singularities which are transitively permuted by

the action of G . In other words, f will have finitely many isolated critical points which form a single G -orbit with stabilizer $H \subset G$ (defined up to conjugation), and such that the germ of (X, f) at each critical point $s \in \text{crit}(f) \subset X$ is an H -equivariant germ of a function with isolated singularities. In other words an atomic G -equivariant germ has a generic G -equivariant unfolding which splits into a G -orbit of H -equivariant strongly atomic singularity germs. This constrained behavior of the G -equivariant unfolding of atomic singularities reflects the minimality of the deformation and is tightly related to the structure of the universal deformation.

Thus, unfolding spaces — particularly the base of the universal deformation — provide the natural geometric framework for understanding the infinitesimal deformations of singularities, while the G -action further refines this picture by isolating those deformation directions that are compatible with the symmetry. The notion of atomicity for a G -equivariant isolated hypersurface singularity relies on the universal deformation structure since it ensures that, in a generic G -symmetric unfolding, the critical locus remains as simple as possible.

The first general observation we have is the following.

Proposition 2.12. *(i) The moduli problem for atomic G -equivariant isolated hypersurface singularities of fixed dimension d , fixed Milnor number μ , and fixed equivariant Milnor number μ^G , is representable by an algebraic Deligne–Mumford stack of finite type $\mathcal{M}_{d,\mu,\mu^G}$, defined over $\mathbb{Q}$.*

(ii) The moduli stack $\mathcal{M}_{d,\mu,\mu^G}$ can be thickened canonically $\mathcal{M}_{d,\mu,\mu^G} \subset \mathcal{U}_{d,\mu,\mu^G}$ to an analytic smooth orbifold $\mathcal{U}_{d,\mu,\mu^G}$ of dimension μ , equipped with a G -action, and such that

$$\mathcal{M}_{d,\mu,\mu^G} = (\mathcal{U}_{d,\mu,\mu^G})^G.$$

(iii) The formal completion

$$\mathcal{U}_{d,\mu,\mu^G}^\wedge$$

of the thickened moduli stack $\mathcal{U}_{d,\mu,\mu^G}$ along $\mathcal{M}_{d,\mu,\mu^G}$ is a smooth generally non-algebraizable formal Deligne–Mumford stack of finite type over $\mathbb{Q}$.

Proof. (i) The moduli stack $\mathcal{M}_{d,\mu,\mu^G}$ is constructed as the quotient of the G -invariant universal unfolding base space by the automorphism group of the germ. By Definition 2.6, the automorphisms preserving the G -action form a proper groupoid with finite stabilizers, ensuring the stack is Deligne–Mumford. Finite type over $\overline{\mathbb{Q}}$ follows from the fact that the ring of G -invariant jets is Noetherian.

(ii) The thickened moduli $\mathcal{U}_{d,\mu,\mu^G}$ is the quotient of the germ of the universal unfolding base space along its G -invariant part. The G -action on the universal unfolding induces the structure of a G -equivariant analytic germ.

(iii) Smoothness follows from the unobstructedness of G -equivariant deformations. Non-algebraizability arises as the thickening captures formal but non-convergent deformations. The stack structure persists via formal GAGA for Artin stacks. $\square$

Next we look at the special structures one has on these moduli.

Proposition 2.13. *(i) The moduli space $\mathcal{U}_{d,\mu,\mu^G}$ carries a natural structure of a smooth F -stack in the sense of Hertling–Manin. In particular its tangent bundle T is equipped with a commutative, associative product $\circ : T \times T \rightarrow T$ and an Euler vector field $\text{Eu} \in \Gamma(\mathcal{U}_{d,\mu,\mu^G}, T)$.*

(ii) The F -stack structure on $\mathcal{U}_{d,\mu,\mu^G}$ is G -equivariant and hence the tangent bundle of the moduli stack

$$\mathcal{M}_{d,\mu,\mu^G} = (\mathcal{U}_{d,\mu,\mu^G})^G$$

inherits a well-defined multiplication and an Euler field $\mathbf{eu} := \mathbf{Eu}|_{\mathcal{M}_{d,\mu,\mu^G}}$, such that the reduced spectrum of the operator $\mathbf{eu} \circ (-) : T_{\mathcal{M}_{d,\mu,\mu^G}} \rightarrow T_{\mathcal{M}_{d,\mu,\mu^G}}$ consists of a single point.

Proof. (i) The tangent space at a point $(G \curvearrowright X, f)$ of $\mathcal{U}_{d,\mu,\mu^G}$ is the Milnor algebra $\mathcal{O}(X)/J(f)$ of f and the value of $\mathbf{Eu}$ at this point is given by $f \in \mathcal{O}(X)/J(f)$. Saito's Frobenius structure on the base of the universal unfolding provides the multiplication, and everything is compatible with the G -action. The Euler field $\mathbf{Eu}$ scales deformations homogeneously and is preserved under G .

(ii) On the G -fixed locus, the multiplication reduces to the tensor product of invariant subspaces. The restriction $\mathbf{eu} \circ (-)$ of the multiplication by Euler vector field to the subbundle of G -invariants acts with a single eigenvalue as deformations are graded by G -invariant monomials, collapsing the spectrum. $\square$

We will write $\mathcal{M}_{d,\mu,\mu^G}^G$ for the union of all components of $\mathcal{M}_{d,\mu,\mu^G}$ that parametrize strongly atomic G -equivariant germs of functions.

With these structures in place we can now define **the set of atoms of G -equivariant isolated hypersurface singularities** by setting

$$\mathbf{Atoms}_G^{\text{sing}} = \pi_0 \left(\bigsqcup_{H \subset G} \bigsqcup_{d,\mu,\mu^H} \mathcal{M}_{d,\mu,\mu^H}^H \right) / G.$$

where G acts on the set of its subgroups by conjugation.

The set $\mathbf{Atoms}_G^{\text{sing}}$ is countable and to every G -equivariant isolated hypersurface singularity $(G \curvearrowright X, f)$, one can associate its atomic composition which consists of all atoms represented by atomic singularities appearing in the G -invariant unfolding of f . Thus we get a chemical formula of $(G \curvearrowright X, f)$ which is a finite combination of atoms with non-negative integer coefficients.

Given two G -equivariant germs $(G \curvearrowright X, f)$ and $(G \curvearrowright Y, y_0, g)$ of isolated hypersurface singularities, we will denote their external sum by $(X \times Y, (x_0, y_0), f \boxplus g)$, where $(f \boxplus g)(x, y) := f(x) + g(y)$ for all $(x, y) \in X \times Y$. The external sum is again a germ of an isolated hypersurface singularity, which is G -equivariant for the diagonal action of G on $X \times Y$. The Thom-Sebastiani formula for the external sum of potentials implies that we have an isomorphism of complex vector spaces

$$\text{Milnor}_{f \boxplus g} = \text{Milnor}_f \otimes_{\mathbb{C}} \text{Milnor}_g,$$

which is compatible with G -actions.

Splitting the external sum of G -equivariant singularity germs into its atomic composition gives a well defined map

$$(2.14) \quad \mathbf{Atoms}_G^{\text{sing}} \times \mathbf{Atoms}_G^{\text{sing}} \longrightarrow \text{Maps}_{\text{fsupp}} \left(\mathbf{Atoms}_G^{\text{sing}}, \mathbb{Z}_{\geq 0} \right)$$

where $\text{Maps}_{\text{fsupp}} \left(\mathbf{Atoms}_G^{\text{sing}}, \mathbb{Z}_{\geq 0} \right)$ denotes all maps from $\mathbf{Atoms}_G^{\text{sing}}$ to $\mathbb{Z}_{\geq 0}$ with finite support.

Explicitly the map (2.14) is defined as follows. Suppose $H_1, H_2 \subset G$ are two subgroups and let $\alpha_1 \in \mathcal{M}_{d_1,\mu_1,\mu_1^{H_1}}$ and $\alpha_2 \in \mathcal{M}_{d_2,\mu_2,\mu_2^{H_2}}$ be two G -atoms of singularities, that are represented by

equivariant germs $(H_1 \circlearrowleft X_1, f_1)$ and $(H_2 \circlearrowleft X_2, f_2)$ respectively. Then the map (2.14) sends the pair (α_1, α_2) to the formal sum of atoms with multiplicities appearing in the atomic composition of the Thom-Sebastiani sum

$$\left(G \circlearrowleft \left(\text{ind}_{H_1}^G(X_1) \times \text{ind}_{H_2}^G(X_2) \right), \text{ind}_{H_1}^G(f_1) \boxplus \text{ind}_{H_1}^G(f_1) \right).$$

Extending the map (2.14) by multiplicativity then induces a product structure on $\text{Maps}_{\text{fsupp}} \left(\text{Atoms}_G^{\text{sing}}, \mathbb{Z}_{\geq 0} \right)$ whose structure constants are integers. This equips the additive semi-group $\text{Maps}_{\text{fsupp}} \left(\text{Atoms}_G^{\text{sing}}, \mathbb{Z}_{\geq 0} \right)$ with the structure of a unital commutative semiring and the abelian group $\text{Maps}_{\text{fsupp}} \left(\text{Atoms}_G^{\text{sing}}, \mathbb{Z} \right)$ with the structure of a unital commutative ring.

Example 2.15. Fix a positive integer $N > 1$. Let $X_1 = \mathbb{C}$ with coordinate x_1 , $X_2 = \mathbb{C}$ with coordinate x_2 , and let $f_1(x_1) = x_1^N$ and $f_2(x_2) = x_2^N$. The group G of N -th roots of unity acts on X_1 and X_2 by

$$\zeta \mapsto (x_1 \rightarrow \zeta x_1) \quad \text{and} \quad k \mapsto (x_2 \rightarrow \zeta^{-1} x_2).$$

The equivariant germs $(G \circlearrowleft X_1, f_1)$ and $(G \circlearrowleft X_2, f_2)$ are atomic and their external sum is $(G \circlearrowleft X_1 \times X_2, x_1^N + x_2^N)$. Consider the following G -equivariant (nonuniversal) unfolding of $(G \circlearrowleft X_1 \times X_2, x_1^N + x_2^N)$

$$F : X_1 \times X_2 \times \mathbb{C}^2 \longrightarrow \mathbb{C}, \quad F(x, z) = x_1^N + x_2^N - z_1 x_1 x_2 + z_2.$$

Fix a general point $z \in \mathbb{C}^2$, s.t. $z_1 \neq 0$. With this fixed z we can view $F(x, z)$ as a function of x whose Jacobian ideal is generated by

$$\partial_{x_1} F(x, z) = x_1^{N-1} - (z_1/N)x_2 \quad \text{and} \quad \partial_{x_2} F(x, z) = x_2^{N-1} - (z_1/N)x_1.$$

Thus $F(x, z)$ has $(N-1)^2$ Morse critical points which map to $N-1$ critical values which are equal to $z_2, z_2 - (N-2)c_1, \dots, z_2 - (N-2)c_{N-2}$ where $c_1, \dots, c_{N-2}$ label the distinct $(N-2)$ nd roots of $(z_1/N)^N$. Over the critical value z_2 there is a single critical point $(x_1, x_2) = (0, 0)$, and over each of the other critical values there is a set of N critical points on which G acts simply transitively. More precisely, over the critical value $z_2 - (N-2)c_i$ we have N Morse critical points with coordinates $(x_1, x_2) = (\beta, (z_1/N)\beta^{-1})$, where β runs over the N th roots of c_i . Thus in the atomic composition of $(G \circlearrowleft X_1 \times X_2, x_1^N + x_2^N)$ we have two atoms. One is isomorphic to the G -equivariant 2-dimensional Morse singularity $(\mathbb{C}^2, x_1 x_2)$ equipped with the G -action $\zeta : (x_1, x_2) \mapsto (\zeta x_1, \zeta^{-1} x_2)$, and the other is isomorphic to the 2-dimensional Morse singularity $(\mathbb{C}^2, x_1 x_2)$ viewed as an H -equivariant germ of function with isolated singularities, where $H = \{1\} \subset G$ is the trivial subgroup. In the chemical formula of $(G \circlearrowleft X_1 \times X_2, x_1^N + x_2^N)$, the first atom enters with multiplicity 1, while the second atom enters with multiplicity $N-2$.

Remark 2.16. (1) The structure theory of atoms of equivariant singularities can be upgraded to the setting of F-bundles discussed in the next section.

(2) In the case of the trivial group G , we recover the discussion from the previous section: we have a unique atom corresponding to the Morse quadratic singularity $x_1^2 + \dots + x_d^2 = 0$ since Morsification shows that each isolated singularity (X, x_0, f) unfolds to a disjoint union of μ_f quadratic Morse singularities.

(3) One can stabilize the above definition of atoms of G equivariant isolated singularities by adding Morse singularities with trivial G -action.

(4) The analogue of Remark 2.4 also holds for G -equivariant germs of functions with isolated singularities. More precisely, if $[f] \in \mathcal{M}_{d,\mu,\mu^G} = (\mathcal{U}_{d,\mu,\mu^G})^G$ and we know that the Frobenius powers of Eu span $T_{\mathcal{M}_{d,\mu,\mu^G},[f]}$, then the atomic composition of $[f]$ can not split further for germs $[g]$ in a neighborhood of $[f]$ in $\mathcal{M}_{d,\mu,\mu^G}$.

3 Generalities on F -bundles

In this section, we introduce the notion of an F -bundle, which captures variations of linearizations of non-commutative geometry, given by the formal part of non-commutative Hodge structures (see [48]).

3.1 F -bundles

Our main objects of interest will be defined in the non-archimedean analytic setting.

3.1.1 Definition of F -bundles

We fix a non-archimedean normed field $\mathbb{k}$ of characteristic 0. Let $\mathbb{D}$ denote the germ at 0 in a $\mathbb{k}$ -analytic unit disk with coordinate u .

Definition 3.1. A **non-archimedean $\mathbb{k}$ -analytic F -bundle**, or simply an **F -bundle** is a triple $(\mathcal{H}, \nabla)/B$, in which

- (1) B is a smooth $\mathbb{k}$ -analytic super variety or a germ of smooth $\mathbb{k}$ -analytic super variety along an even closed smooth $\mathbb{k}$ -analytic subvariety.
- (2) $\mathcal{H}$ is a $\mathbb{k}$ -analytic super vector bundle over $B \times \mathbb{D}$.
- (3) ∇ is a meromorphic flat connection on $\mathcal{H}$ with poles at most at $B \times \{0\}$, such that for any locally defined vector field ξ on B the operators $\nabla_{u^2 \partial_u}$ and $\nabla_{u\xi}$ are regular, i.e. have no poles.

Remark 3.2. (i) The letter F in the name F -bundle comes from *Frobenius*, since F -bundles are variants of the Frobenius manifolds introduced by Dubrovin [26] for the study of WDVV equations. An F -bundle is also a non-archimedean version of the notion of TE-structures introduced by Hertling [37]. The complex analytic TE-structures are not suitable for our A-model considerations without the convergence assumptions on the Gromov-Witten potential. Furthermore, the spectral decomposition theorem (Theorem 4.1) does not hold in the complex analytic setting due to the Stokes phenomenon. By working in the non-archimedean setting, we avoid these issues, but retain most of the intuition coming from the complex picture.

(ii) From the perspective of non-commutative geometry, an F -bundle is a non-archimedean analytic version of the de Rham part of a variation of non-commutative Hodge structures (see [48, §2.1.5, §2.3.1]). Examples of F -bundles arise naturally from deformations of A_∞ categories, from Gromov-Witten invariants, and from deformations of singularities.

(iii) In examples one sometimes encounters F -bundles which have additional tame poles along divisors in B . So, in practice, it is sometimes necessary to work with **logarithmic F -bundles**, i.e. with triples $(\mathcal{H}, \nabla)/B$ in which B and $\mathcal{H}$ are as before but ∇ is a meromorphic flat connection on $\mathcal{H}$ with poles at most at $(B \times \{0\}) \cup (\mathfrak{D} \times \mathbb{D})$, for some even strict normal crossings divisor $\mathfrak{D} \subset B$. The logarithmic F -bundle condition then is that for all locally defined vector fields ξ on

B which are tangent to $\mathfrak{D}$ the operators $\nabla_{u^2\partial_u}$ and $\nabla_{u\xi}$ should have no poles. Of course, if B is a non-archimedean super manifold (rather than a germ) and $(\mathcal{H}, \nabla)/B$ is a logarithmic F-bundle with log poles along $\mathfrak{D}$ we can eliminate the log singularity by simply puncturing B along $\mathfrak{D}$, i.e. by restricting $(\mathcal{H}, \nabla)$ to $(B - \mathfrak{D}) \times \mathbb{D}$. This is another advantage of the non-archimedean analytic setting.

3.1.2 Variants: formal and complex analytic F -bundles

While historically prevalent in the mirror symmetry and Gromov-Witten literature, the complex analytic or formal analogues of an F-bundle will not be useful for constructing birational invariants directly. For disambiguation and ease of reference we will name these variants differently.

Suppose B is a complex analytic super manifold or a germ of a complex analytic super manifold. Let $\mathbf{D}$ denote a small analytic disk in $\mathbb{C}$ centered at zero, or the germ of $\mathbb{C}$ at zero, and let u denote the coordinate on $\mathbf{D}$. A **complex analytic F-bundle** consists of $(\mathcal{H}, \nabla)/B$ where $\mathcal{H}$ is a complex analytic super vector bundle on $B \times \mathbf{D}$ and ∇ is a meromorphic flat connection on $\mathcal{H}$ with poles at most at $u = 0$, such that for all locally defined vector fields on B , the operators $\nabla_{u^2\partial_u}$ and $\nabla_{u\xi}$ have no poles. We will say that $(\mathcal{H}, \nabla)/B$ is a **logarithmic complex analytic F-bundle** if B and $\mathcal{H}$ are as before, and ∇ is a meromorphic flat connection on $\mathcal{H}$ with poles at most at $(B \times \{0\}) \cup (\mathfrak{D} \times \mathbf{D})$ for some even strict normal crossings divisor $\mathfrak{D} \subset B$, such that for all locally defined vector fields ξ on B which are tangent to D , we have that the operators $\nabla_{u^2\partial_u}$ and $\nabla_{u\xi}$ have no poles.

To be consistent with the common mirror symmetry terminology from [48] or the singularity theory terminology from [37] we will also call such a triple a **(logarithmic) de Rham part of a nc variation of Hodge structures** or a **(logarithmic) TE structure**.

Suppose B is a smooth formal super scheme over a field k , $\text{char}(k) = 0$. Then a **formal F-bundle defined over k** consists of $(\mathcal{H}, \nabla)/B$, where $\mathcal{H}$ is a super vector bundle on $B \times \text{Spf } k[[u]]$ and ∇ is a meromorphic flat connection on $\mathcal{H}$ with poles at most at $u = 0$, such that the operators $\nabla_{u^2\partial_u}$ and $\nabla_{u\xi}$ have no poles. Similarly, a **formal logarithmic F-bundle defined over k** consists of $(\mathcal{H}, \nabla)/B$ with B and $\mathcal{H}$ as before, and ∇ a meromorphic flat connection on $\mathcal{H}$ with poles at most along $(B \times \{0\}) \cup (\mathfrak{D} \times \text{Spf } k[[u]])$ for an even strict normal crossing divisor $\mathfrak{D} \subset B$, and such that for all locally defined vector fields ξ on B which are tangent to D , we have that the operators $\nabla_{u^2\partial_u}$ and $\nabla_{u\xi}$ have no poles.

3.2 Operations on F -bundles

For a given base B , F-bundles over B form a category $\mathbf{Fbun}(B)$ which is a $\mathbb{k}$ -linear, additive, Karoubi closed tensor category with duals. The unit object in the category $\mathbf{Fbun}(B)$ is the trivialized rank 1 bundle on B with the trivial connection. Note that the category $\mathbf{Fbun}(B)$ is not abelian, it is only quasi-abelian (hence an **exact category**). An example of a morphism in $\mathbf{Fbun}(\text{Sp } \mathbb{k})$ without an adequate kernel is the operator of multiplication by u :

$$(\mathcal{O}, \nabla_{\partial_u} = \partial_u) \xrightarrow{u \cdot (-)} \left(\mathcal{O}, \nabla_{\partial_u} = \partial_u - \frac{1}{u} \right), \quad u \cdot \partial_u(f) = \left(\partial_u - \frac{1}{u} \right)(uf) \quad \forall f \in \mathbb{k}[[u]]$$

For any map $\phi: B' \rightarrow B$, and any F-bundle $(\mathcal{H}, \nabla)/B$ over B as above, we define the pullback to be the F-bundle over B' given by $(B', \tilde{\phi}^*\mathcal{H}, \tilde{\phi}^*(\nabla))$ where $\tilde{\phi}$ is $\varphi \times \text{id}_{\mathbb{D}}$. The pullback is an exact tensor functor from $\mathbf{Fbun}(B)$ to $\mathbf{Fbun}(B')$.

Definition 3.3. For two F-bundles $(B_1, \mathcal{H}_1, \nabla_1)$ and $(B_2, \mathcal{H}_2, \nabla_2)$, we define their **external sum** to be the F-bundle over $B_1 \times B_2$ given by

$$(B_1, \mathcal{H}_1, \nabla_1) \boxplus (B_2, \mathcal{H}_2, \nabla_2) := (B_1 \times B_2, \text{pr}_{13}^* \mathcal{H}_1 \oplus \text{pr}_{23}^* \mathcal{H}_2, \text{pr}_{13}^* \nabla_1 \oplus \text{pr}_{23}^* \nabla_2),$$

where pr_{13} and pr_{23} denote the projections from $B_1 \times B_2 \times \mathbb{D}$ to $B_1 \times \mathbb{D}$ and $B_2 \times \mathbb{D}$ respectively.

Similarly, we define their **external product** to be the F-bundle over $B_1 \times B_2$ as

$$(B_1, \mathcal{H}_1, \nabla_1) \boxtimes (B_2, \mathcal{H}_2, \nabla_2) := (B_1 \times B_2, p_{13}^* \mathcal{H}_1 \otimes p_{23}^* \mathcal{H}_2, p_{13}^* \nabla_1 \otimes \text{id} + \text{id} \otimes p_{23}^* \nabla_2).$$

Remark 3.4. One can define external sums and products of any finite collections of F-bundles. For the external sums and products of finitely many copies of the same F-bundle, one can replace the finite indexing set labeling the copies by a reduced finite scheme (or finite analytic space) over $\mathbb{k}$, i.e. by the finite set of its $\bar{\mathbb{k}}$ -points endowed with the action of the Galois group $\text{Gal}(\bar{\mathbb{k}}/\mathbb{k})$.

3.3 Natural symmetries of F-bundles

Write $\text{Aff}(1) = \mathbb{G}_m \ltimes \mathbb{G}_a$ for the group of affine linear automorphisms of the analytic affine line $\mathbb{A}_{\mathbb{k}}^{1,\text{an}}$. The group $\text{Aff}(1) \times \mathbb{G}_a$ acts on the category $\text{Fbun}(B)$ of F-bundles over a base B as follows. We denote elements in the group as triples $(\lambda, c_{-2}, c_{-1}) \in (\mathbb{G}_m \ltimes \mathbb{G}_a) \times \mathbb{G}_a = \text{Aff}(1) \times \mathbb{G}_a$.

Dilation: The component $\lambda \in \mathbb{G}_m \subset \text{Aff}(1)$ acts on $\text{Fbun}(B)$ by pullback via $\text{id}_B \times d_\lambda: B \times \mathbb{D} \rightarrow B \times \mathbb{D}$, where $d_\lambda: u \mapsto \lambda u$ denotes the dilation of the coordinate u . We denote the action by D_λ .

Exponential shift: The component $c_{-2} \in \mathbb{G}_a \subset \text{Aff}(1)$ acts on $\text{Fbun}(B)$ by tensoring with the rank-one F-bundle

$$(3.5) \quad \left(B; \mathcal{O}_{B \times \mathbb{D}}, \nabla = d + \frac{c_{-2}}{u^2} du \right).$$

We denote the action by $S_{c_{-2}}^e$.

Power shift: The component c_{-1} in the second copy of $\mathbb{G}_a$ acts on $\text{Fbun}(B)$ by tensoring with the rank-one F-bundle

$$(3.6) \quad \left(B; \mathcal{O}_{B \times \mathbb{D}}, \nabla = d + \frac{c_{-1}}{u} du \right).$$

We denote the action by $S_{c_{-1}}$.

Remark 3.7. The exponential shift corresponds to multiplying the flat sections in an F-bundle with $e^{\frac{c_{-2}}{u}}$, and the power shift corresponds to multiplying with $u^{c_{-1}}$, whence their names.

3.4 Maximal and overmaximal F-bundles

Let $(\mathcal{H}, \nabla)/B$ be an F-bundle (resp. a logarithmic F-bundle with ∇ having log poles along an even strict normal crossings divisor $\mathfrak{D} \subset B$). For any vector field ξ on B (resp. a vector field on B tangent to $\mathfrak{D}$ along $\mathfrak{D}$), the restriction of the operators $\nabla_{u\xi}$ to $\mathcal{H}_{|u=0}$ gives a map

$$\mu: T_B \longrightarrow \text{End}(\mathcal{H}_{|u=0}), \quad \text{respectively } \mu: T_B(-\log \mathfrak{D}) \longrightarrow \text{End}(\mathcal{H}_{|u=0}).$$

For any geometric point b in B , we have fiberwise maps

$$\mu_b: T_{B,b} \rightarrow \text{End}(\mathcal{H}_{(b,0)}), \quad \text{respectively } \mu_b: T_{B,b}(-\log \mathfrak{D}) \longrightarrow \text{End}(\mathcal{H}_{(b,0)}).$$

The flatness of ∇ implies that the image of μ_b consists of commuting operators.

Definition 3.8. An F-bundle $(\mathcal{H}, \nabla)/B$ over B is **maximal** (resp. **over-maximal**) at a geometric point $b \in B$ if there exists a cyclic vector for the action μ_b , i.e. a vector $h \in \mathcal{H}_{(b,0)}$ such that the map $\text{ev}_h \circ \mu_b: T_{B,b} \rightarrow \mathcal{H}_{(b,0)}$ is an isomorphism (resp. epimorphism).

Similarly a logarithmic F-bundle $(\mathcal{H}, \nabla)/B$ over B is **maximal** (resp. **over-maximal**) at a geometric point $b \in B$ if there exists a cyclic vector for the action μ_b , i.e. a vector $h \in \mathcal{H}_{(b,0)}$ such that the map $\text{ev}_h \circ \mu_b: T_{B,b}(-\log \mathfrak{D}) \rightarrow \mathcal{H}_{(b,0)}$ is an isomorphism (resp. epimorphism).

A (logarithmic) F-bundle is maximal (resp. over-maximal) if it is maximal (resp. over-maximal) everywhere.

Note that being maximal (resp. over-maximal) is an open condition. The significance of being over-maximal or maximal is captured by the following

Lemma 3.9. *For a F-bundle $(\mathcal{H}, \nabla)/B$ which is over-maximal at $b \in B$, the image of μ_b is a unital commutative associative super algebra over $\mathbb{k}$, and the fiber $\mathcal{H}_{(b,0)}$ is a free rank one module over $\text{im } \mu_b \simeq T_{B,b}/\ker \mu_b$.*

Similarly, for a logarithmic F-bundle $(\mathcal{H}, \nabla)/B$ which is over-maximal at $b \in B$, the image of μ_b is a unital commutative associative super algebra over $\mathbb{k}$, and the fiber $\mathcal{H}_{(b,0)}$ is a free rank one module over $\text{im } \mu_b \simeq T_{B,b}(-\log \mathfrak{D})/\ker \mu_b$.

Proof. The claim in the lemma boils down to the following linear algebra statement: Let T and H be $\mathbb{k}$ -vector spaces, and $\mu: T \rightarrow \text{End}(H)$ a linear map satisfying the following conditions:

- (1) The image $\mu(T)$ consists of commuting operators.
- (2) There is a vector $h \in H$ such that the map $\text{ev}_h \circ \mu: T \rightarrow H$ is surjective.

Then $\mu(T) \subset \text{End}(H)$ is a unital commutative associative subalgebra and H is a free rank one module over $\mu(T)$.

To see that this statement holds we need to check that if $A \in \text{End}(H)$ is an operator that commutes with all elements in $\mu(T)$, then $A \in \mu(T)$. By Condition (2), there exists $v \in T$ such that $Ah = \mu(v)h$. Then for any $w \in T$, by Condition (1) we have

$$A\mu(w)h = \mu(w)Ah = \mu(w)\mu(v)h = \mu(v)\mu(w)h.$$

By Condition (2) again, we deduce that $A = \mu(v)$, in particular, $A \in \mu(T)$.

Note that for any $v, w \in T$, the product $\mu(v)\mu(w)$ commutes with all elements in $\mu(T)$, so it lies in $\mu(T)$ by the above reasoning. For the same reason, $\text{id} \in \mu(T)$. Hence $\mu(T) \subset \text{End}(H)$ is a unital commutative associative subalgebra.

By Condition (2), $h \in H$ is a generator of H as $\mu(T)$ -module. For any $A \in \mu(T)$, by Condition (1), if $Ah = 0$, then for any $v \in T$, $A\mu(v)h = \mu(v)Ah = 0$, so $A = 0$ by Condition (2). Therefore, H is a free rank one module over $\mu(T)$ generated by h . This completes the proof of the lemma. $\square$

For an overmaximal F-bundle $(\mathcal{H}, \nabla)/B$ the subbundle $\ker \mu$ of the tangent bundle T_B is an integrable distribution on B , i.e. it defines a foliation $\mathcal{R}$ on B . We will call $\mathcal{R} = \ker \mu$ the **redundancy foliation** of the overmaximal F-bundle $(\mathcal{H}, \nabla)/B$.

Lemma 3.10. *Suppose $(\mathcal{H}, \nabla)/B$ is an overmaximal F-bundle. Then for any rigid point $b \in B$, locally near b the F-bundle $(\mathcal{H}, \nabla)/B$ is pullback of a maximal F-bundle on a non-archimedean analytic super manifold which is a transversal slice of the redundancy foliation $\mathcal{R}$ through b . Equivalently, globally $(\mathcal{H}, \nabla)/B$ is a pullback of a maximal formal F-bundle on the formal stack $\mathcal{M} := [B/\mathcal{R}]$ of leaves of $\mathcal{R}$.*

Proof. A vector field ξ on B lies in the sub bundle $\ker \mu$ if and only if the operator ∇_ξ is regular (i.e. has no poles) along $u = 0$. If ξ_1, ξ_2 are two such vector fields, then so is $[\xi_1, \xi_2]$, because

$$\nabla_{[\xi_1, \xi_2]} = [\nabla_{\xi_1}, \nabla_{\xi_2}]$$

by the flatness of ∇ .

By construction, along the leaves of the foliation $\mathcal{R}$, the connection ∇ is regular at $u = 0$. So it trivializes the bundle $\mathcal{H}$ along the leaves. This implies the second statement. $\square$

Let $(\mathcal{H}, \nabla)/B$ be a maximal F-bundle (resp. maximal logarithmic F-bundle) over B . It follows from Eq. (3.9) that the tangent bundle T_B (resp. the logarithmic tangent bundle $T_B(-\log \mathfrak{D})$) inherits the structure of a unital commutative associative superalgebra from its embedding in $\text{End}(\mathcal{H}_{u=0})$.

Definition 3.11. We denote the induced product by $\circ$ and call it **the Frobenius product associated with the maximal F-bundle**.

Definition 3.12. Let $(\mathcal{H}, \nabla)/B$ be a maximal F-bundle (resp. maximal logarithmic F-bundle) on B . The **Euler vector field** of $(\mathcal{H}, \nabla)/B$ is the unique even vector field Eu on B which under the action μ maps to the residual endomorphism $\kappa := \nabla_{u^2 \partial_u} |_{\mathcal{H}|_{u=0}}$ of $\mathcal{H}|_{u=0}$ associated with $(\mathcal{H}, \nabla)/B$, that is $\mu(\text{Eu}) = \kappa$. In the logarithmic case Eu is automatically a section in $T_B(-\log \mathfrak{D}) \subset T_B$.

The definition makes sense because operator κ commutes with the all operators from $\mu(T_B)$, as follows from the flatness of ∇ .

Remark 3.13. (a) If $(\mathcal{H}, \nabla)/B$ is not maximal but just overmaximal, then we still get a sheaf $\mu(T_B) = T_B/\mathcal{R}$ of unital, associative, supercommutative algebras on B , where again the product structure on $\mu(T_B)$ is inherited from the product of endomorphisms, through the embedding $\mu(T_B) \subset \text{End}(\mathcal{H}_{u=0})$. Furthermore, $\mathcal{H}|_{u=0}$ is a locally free $\mu(T_B)$ -module of rank one.

(b) Suppose now $(\mathcal{H}, \nabla)/B$ is an overmaximal F-bundle which has a global cyclic vector² $h \in \Gamma(B, \mathcal{H}|_{u=0})$, then $\text{ev}_h: \mu(T_B) \cong \mathcal{H}|_{u=0}$ is an isomorphism of coherent sheaves, which is compatible with the $\mu(T_B)$ -actions on source and target.

Transporting the product on $\mu(T_B)$ via ev_h gives an associative supercommutative product on $\mathcal{H}|_{u=0}$ which we will again denote by $\circ$. The image of the operator $\kappa \in \Gamma(B, \mu(T_B))$ via this isomorphism is an element in $\text{ev}_h(\kappa) \in \Gamma(B, \mathcal{H}|_{u=0})$, which we will again denote by Eu and will call the Euler element (or section) associated with $(\mathcal{H}, \nabla)/B$. By construction we have $\kappa = \text{Eu} \circ (-)$, and when $(\mathcal{H}, \nabla)/B$, this is precisely the Euler vector field defined above.

The Euler vector field interacts in an interesting way with the Frobenius product on T_B . Indeed, using $\circ$ we can define an infinite sequence of vector fields on B by taking powers of Eu with respect to $\circ$:

$$\begin{aligned} \text{Eu}^0 &= \mathbf{1}_B := \text{unit for } \circ, \\ \text{Eu}^1 &= \text{Eu} \\ \text{Eu}^k &= \underbrace{\text{Eu} \circ \text{Eu} \circ \cdots \circ \text{Eu}}_{k \text{ times}} \end{aligned}$$

²For instance, the A-model F-bundle we consider in the next section is of this type.

The reasoning of [60, Proposition 3.6.2, Theorem 5.6] shows that the vector fields $\{\mathbf{Eu}^k\}_{k \geq 0}$ give an action of the Witt Lie algebra $\mathbb{k}[t]\partial_t$ on B :

$$\forall k, l \geq 0: [\mathbf{Eu}^k, \mathbf{Eu}^l] = (l - k) \mathbf{Eu}^{k+l-1} \quad (\text{cf. } [t^k \partial_t, t^l \partial_t] = (l - k) t^{k+l-1} \partial_t).$$

The analogue of Remark 2.4 holds in this setting as well.

Remark 3.14. let $(\mathcal{H}, \nabla)/B$ be a maximal F-bundle, and let $b \in B^{\text{ev}} \subset B$ be a rigid even point of B . Then there exists a unique germ $W \subset B^{\text{ev}}$ at b of a purely even closed analytic submanifold in B s.t. $\text{span}_{\mathbb{k}}(\{\mathbf{Eu}_w^k\}_{k \geq 0}) = T_{W,w}$ for all $w \in W$ close to b . The number of distinct eigenvalues of $\kappa|_W : T_W \rightarrow T_W$ and the list of their multiplicities are constant on a neighborhood of $b \in W$. Moreover, more fine results can be proved, like, e.g., if $(\mathcal{H}, \nabla)/B$ is of exponential type (see [48] for the definition of exponential type) at some point of W , then it is of exponential type at *all* points of W . This implies that the isomorphism classes of formal meromorphic connections with regular singularities over $\text{Spf } \mathbb{k}[[u]]$ corresponding to the eigenvalues of $\kappa|_W$ via Hukuhara-Levelt-Turrittin decomposition are constant along W .

3.5 The A-model F -bundle

In this section we will describe the main example of the paper: the F-bundle underlying the A-model variation of nc-Hodge structures. We start with the formal picture, which we subsequently modify to get the non-archimedean analytic version which we will use in the applications.

3.5.1 Formal logarithmic A-model F -bundles

Fix an algebraically closed field $\mathcal{K} = \overline{\mathcal{K}}$ of characteristic zero and let X be a smooth projective variety over $\mathcal{K}$. Gromov-Witten classes are cohomology classes of algebraic cycles, so to talk about them we have to fix a cohomology theory. For convenience, in this section we will work with de Rham cohomology since it does not depend on any external choices. Later on we will study the dependence on the choice of cohomology theory more carefully.

As in the complex analytic setting, for every $n \in \mathbb{Z}_{\geq 0}$, and every

$$\beta \in \text{CH}_1^{\text{hom}}(X) = \text{CH}_1(X)/\sim_{\text{hom}} = \text{im} [\text{CH}_1(X) \rightarrow H_{dR}^{2 \dim X - 2}(X/\mathcal{K})]$$

we will write

$$\delta(n, \beta) = n + (\dim X - 3) + \int_{\beta} c_1(T_X)$$

for the expected dimension of the moduli stack $\overline{\mathcal{M}}_{0,n}(X, \beta)$ parametrizing stable genus zero n -marked maps to X . Concretely $\overline{\mathcal{M}}_{0,n}(X, \beta)$ is the moduli of maps $\varphi: (C, p_1, \dots, p_n) \rightarrow X$, where:

- (1) C is a connected nodal, genus 0 curve, defined over $\mathcal{K}$.
- (2) $p_1, \dots, p_n$ are distinct smooth $\mathcal{K}$ -points of C .

(3) φ is a morphism satisfying $\varphi_*[C] = \beta$ and the stability condition that if φ contracts a component of C to a point, then the number of marked points and nodes on this component is ≥ 3 . The stack $\overline{\mathcal{M}}_{0,n}(X, \beta)$ is a proper Deligne-Mumford stack/ $\mathcal{K}$. It is equipped (see [52, 10, 11]) with a **virtual fundamental cycle class**

$$[\overline{\mathcal{M}}_{0,n}(X, \beta)]_{\text{vir}} \in \text{CH}_{\delta(n, \beta)}(\overline{\mathcal{M}}_{0,n}(X, \beta))$$

of dimension $\delta(n, \beta)$, which further maps to a **virtual fundamental class** modulo homological equivalence

$$[\overline{\mathcal{M}}_{0,n}(X, \beta)]_{\text{vir}}^{\text{hom}} \in \text{CH}_{\delta(n, \beta)}^{\text{hom}}(\overline{\mathcal{M}}_{0,n}(X, \beta))$$

We consider the **Gromov-Witten cycle class** $I_{n, \beta}(X)$ defined as the image

$$\begin{aligned} I_{n, \beta}(X) &= \text{ev}_* [\overline{\mathcal{M}}_{0,n}(X, \beta)]_{\text{vir}}^{\text{hom}} \in \text{CH}_{\delta(n, \beta)}^{\text{hom}}(X^{\times n}) \otimes \mathbb{Q} \\ &\subset H_{dR}^{2n \dim X - 2\delta(n, \beta)}(X^{\times n}/\mathcal{K}) \end{aligned}$$

under the natural evaluation map

$$\text{ev}: \overline{\mathcal{M}}_{0,n}(X, \beta) \rightarrow X^{\times n}, \quad (C, p_1, \dots, p_n, \varphi) \mapsto (\varphi(p_1), \dots, \varphi(p_n)).$$

Similarly, if J is any finite set of cardinality n , we can consider the moduli stack of genus zero stable maps with marked points labeled by the elements in J , and define in the same way Gromov-Witten classes $I_{J, \beta}(X) \in \text{CH}_{\delta(n, \beta)}^{\text{hom}}(X^J) \otimes \mathbb{Q}$.

Properties of the Gromov-Witten classes [52]:

- (i) $I_{n, \beta}(X) \neq 0$ implies that β is the class of an effective one cycle. For future reference we will write c .
- (ii) When $\beta = 0$, then $I_{n, 0}(X) \neq 0$ implies that $n = 3$, and $I_{3, 0}(X) = (X \hookrightarrow X^{\times 3})_*[X]$, where $X \hookrightarrow X^{\times 3}$ is the diagonal embedding.
- (iii) $I_{n, \beta}(X)$ is invariant under the permutation action of the symmetric group S_n on $X^{\times n}$. Thus, for a finite set J of cardinality n , the class $I_{J, \beta}(X)$ can be identified unambiguously with $I_{n, \beta}(X)$. Indeed, any ordering of the elements in J identifies $I_{J, \beta}(X)$ with $I_{n, \beta}(X)$ and the S_n -invariance guarantees that this identification is independent of the choice of order.
- (iv) $I_{n, \beta}(X)$ satisfy the **unit and divisor axioms**. To state these tersely we will introduce some notation. Suppose $n, m \geq 0$ are integers and let $\xi \in \text{CH}_{\bullet}^{\text{hom}}(X^{\times(n+m)})$ and $\eta \in \text{CH}^{\text{hom}, \bullet}(X^{\times m})$. We define a new class $\int_{\xi} \eta \in \text{CH}_{\bullet}^{\text{hom}}(X^{\times n})$ by setting

$$\int_{\xi} \eta = \left(X^{\times(n+m)} \rightarrow X^{\times n} \right)_* \left(\xi \frown \left(X^{\times(n+m)} \rightarrow X^{\times m} \right)^* \eta \right)$$

where $X^{\times(n+m)} \rightarrow X^{\times n}$ is the projection on the first n factors and $X^{\times(n+m)} \rightarrow X^{\times m}$ is the projection the last m factors. With this notation we can now state that $I_{\beta, n}(X)$ satisfy the following two axioms:

(Unit axiom) For all $n \geq 0$, and $\beta \neq 0$ we have $\int_{I_{n+1, \beta}(X)} \mathbf{1}_X = 0 \in \text{CH}_{\bullet}^{\text{hom}}(X^{\times n})$, where $\mathbf{1}_X \in \text{CH}^{\text{hom}, \bullet}(X)$ denotes the unit.

(Divisor axiom) For any line bundle $\mathcal{L}$ on X we have

$$\int_{I_{n+1, \beta}(X)} c_1(\mathcal{L}) = \left(\int_{\beta} c_1(\mathcal{L}) \right) \cdot I_{n, \beta}(X) \in \text{CH}_{\bullet}^{\text{hom}}(X^{\times n}).$$

(v) For all $n \geq 4$ and all β the Gromov-Witten classes obey the **Witten-Dijkgraaf-Verlinde-Verlinde (WDVV) equation**:

$$\sum_{\substack{\beta=\beta_1+\beta_2 \\ \{5,6,\dots,n\}=J_1 \sqcup J_2}} \int_{I_{\{1,2\} \sqcup J_1 \sqcup \{\text{left}\}, \beta_1}(X) \times I_{\{3,4\} \sqcup J_2 \sqcup \{\text{right}\}, \beta_2}(X)} \left[X \xrightarrow{\text{diag}} X \times X \right] = \\ - \sum_{\substack{\beta=\beta_1+\beta_2 \\ \{5,6,\dots,n\}=J_1 \sqcup J_2}} \int_{I_{\{1,3\} \sqcup J_1 \sqcup \{\text{left}\}, \beta_1}(X) \times I_{\{2,4\} \sqcup J_2 \sqcup \{\text{right}\}, \beta_2}(X)} \left[X \xrightarrow{\text{diag}} X \times X \right]$$

Here, we are capping with the class $[X \xrightarrow{\text{diag}} X \times X] \in \mathbf{CH}^{\dim X}(X \times X)$ of the diagonal in the Cartesian square $X \times X$ which parametrizes the pairs of points with labels **(left, right)**.

Remark 3.15. If $\mathfrak{X}$ is a smooth projective variety defined over an arbitrary field $\mathcal{K}$ of characteristic zero, then by the above procedure, we can construct Gromov-Witten classes

$$I_{n,\beta}(\mathfrak{X} \otimes \overline{\mathcal{K}}) \in \mathbf{CH}_{\delta(n,\beta)}^{\text{hom}}(\mathfrak{X}^{\times n} \otimes \overline{\mathcal{K}}) \otimes \mathbb{Q}.$$

The construction is manifestly invariant under the action of the Galois group $\text{Gal}(\overline{\mathcal{K}}/\mathcal{K})$, i.e. for every $g \in \text{Gal}(\overline{\mathcal{K}}/\mathcal{K})$ we have $g_* I_{n,\beta}(\mathfrak{X} \otimes \overline{\mathcal{K}}) = I_{n,g_*\beta}(\mathfrak{X} \otimes \overline{\mathcal{K}})$.

One largely unexplored aspect of Gromov-Witten theory is its algebraic flexibility. The algebraic nature of Gromov-Witten invariants makes them a powerful tool for encoding and analyzing cycle-theoretic and motivic information. Above we summarized the constructions and properties of Gromov-Witten invariants in the classical setting where we were dealing with a variety over a field of characteristic zero, and the homological equivalence was defined with respect to the algebraic de Rham cohomology. For the applications to birational geometry, we would like to take full advantage of this algebraic flexibility, and so we will need the following variant of Gromov-Witten invariants which makes sense for arbitrary fields of definition and for homological equivalence extracted from a cycle map to a reasonably general Weil cohomology theory.

Variant 3.16. Fix two fields $\mathcal{K}$ and k , both of characteristic zero. Let $H^\bullet$ be a classical Weil cohomology theory defined on non-empty smooth projective $\mathcal{K}$ -varieties, with values in k -vector spaces (see e.g. [64, Definition 1.2.13] and [6, Section 45.7, tag 0FGS] for the definition).

Example 3.17. For us standard choices will be:

(a) **(de Rham setting):** $\mathcal{K}$ any field, $\text{char } \mathcal{K} = 0$, $k = \mathcal{K}$, and $H^\bullet$ is de Rham cohomology over $k = \mathcal{K}$. That is for any non-empty smooth projective variety $\mathfrak{X}$ over $\mathcal{K}$ we have $H^\bullet(\mathfrak{X}) := H_{dR}^\bullet(\mathfrak{X}/k)$.

(b) **(Dolbeault setting):** $\mathcal{K}$ any field, $\text{char } \mathcal{K} = 0$, $k = \mathcal{K}$, and $H^\bullet(-)$ is the Dolbeault cohomology over $k = \mathcal{K}$. That is, for any $\mathcal{K}$ -variety $\mathfrak{X}$ we have $H^\bullet(\mathfrak{X}) := H_{Dol}^\bullet(\mathfrak{X}/k) = \bigoplus_{p,q} H^q(\mathfrak{X}, \Omega_{\mathfrak{X}/k}^p)$.

(c) **(Betti setting):** $\mathcal{K} \subset \mathbb{C}$, $k = \mathbb{Q}$, and $H^\bullet$ is the Betti cohomology of the associated complex manifold. Indeed, for every $\mathcal{K}$ -variety $\mathfrak{X}$ we set $H^\bullet(\mathfrak{X}) = H_B^\bullet(\mathfrak{X}(\mathbb{C})^{\text{an}}, \mathbb{Q})$, where $\mathfrak{X}(\mathbb{C})^{\text{an}}$ is the complex analytic space of $\mathbb{C}$ -valued points of the base change $\mathfrak{X} \times_{\overline{\mathcal{K}}} \mathbb{C}$, taken with the complex topology.

(d) **(ℓ -adic setting):** $\mathcal{K}$ any field of $\text{char } \mathcal{K} = 0$, $k = \mathbb{Q}_\ell$, and $H^\bullet(-)$ is the ℓ -adic cohomology. That is, we fix an algebraic closure $\overline{\mathcal{K}} \subset \overline{\mathbb{Q}_\ell}$, and for any $\mathcal{K}$ -variety $\mathfrak{X}$ we set $H^\bullet(\mathfrak{X}) = H_{\text{ét}}^\bullet(\mathfrak{X} \times_{\mathcal{K}} \overline{\mathcal{K}}, \mathbb{Q}_\ell) = \varprojlim H_{\text{ét}}^\bullet(\mathfrak{X} \times_{\mathcal{K}} \overline{\mathcal{K}}, \mathbb{Z}/\ell^k \mathbb{Z}) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$.

To be specific, from now on we will focus on the Betti setting. Let $\mathfrak{X}/\mathcal{K}$ be a smooth projective variety. Let, as above, $\mathcal{K} \subset \bar{\mathcal{K}} \subset \mathbb{C}$ be the natural algebraic closure of $\mathcal{K}$ in $\mathbb{C}$ and let $X = \mathfrak{X} \times_{\mathcal{K}} \bar{\mathcal{K}} := \mathfrak{X} \times_{\text{Spec } \mathcal{K}} \text{Spec } \bar{\mathcal{K}}$ denote the base change of $\mathfrak{X}$ to $\bar{\mathcal{K}}$. The Chow group $GH_1(X)$ of 1-dimensional algebraic cycles on X maps by the cycle class map to the second singular homology $H_2(X(\mathbb{C})^{\text{an}}, \mathbb{Q})$ of the associated complex manifold, and as above we set

$$\text{CH}_1^{\text{hom}}(X) := \text{CH}_1(X)/\sim_{\text{hom}} = \text{im}[\text{CH}_1(X) \rightarrow H_2(X(\mathbb{C})^{\text{an}}, \mathbb{Q})].$$

The group $\text{CH}_1^{\text{hom}}(X)$ is a subgroup of finite index in the group of all degree two rational homology classes on X that pair integrally with the Néron-Severi lattice $\text{NS}(X, \mathbb{Z}) \subset H_B^2(X(\mathbb{C})^{\text{an}}, \mathbb{Q})$. In particular $\text{CH}_1^{\text{hom}}(X) \cong \mathbb{Z}^{\rho}$ is a free abelian group of rank ρ equal to the Néron-Severi rank of X .

Now, for any $\beta \in \text{CH}_1^{\text{hom}}(X)$ we can follow the above procedure and define Gromov-Witten cohomology classes

$$I_{n,\beta}(X) \in \text{CH}_{\delta(n,\beta)}^{\text{hom}}(X^{\times n}) \otimes \mathbb{Q} \subset H_B^{2n-2\delta(n,\beta)}(X^{\times n}, \mathbb{Q}).$$

The reasoning of [52] applies here and implies that these classes satisfy the properties (i)-(v).

Note also that since Betti cohomology is a Weil cohomology theory, the Chow group integration defined in the description of property (iv) transfers verbatim to the cup product of the associated cycle classes. Concretely, suppose Y is a smooth projective $\bar{\mathcal{K}}$ -variety, and let $\text{cl}: \text{CH}_{\bullet}(Y) = \text{CH}^{\dim Y - \bullet}(Y) \rightarrow H_B^{2 \dim Y - 2\bullet}(Y, \mathbb{Q})$ denote the cycle class map to Betti cohomology. Now if $\xi \in \text{CH}_{\bullet}(X^{\times(n+m)})$ and $\eta \in \text{CH}^{\bullet}(X^{\times n})$ we will have

$$\text{cl}\left(\int_{\xi} \eta\right) = \int_{\text{cl}(\xi)} \text{cl}(\eta) := (\text{pr}_n^{n+m})_*(\text{cl}(\xi) \smile (\text{pr}_m^{n+m})^*\text{cl}(\eta)),$$

where $\text{pr}_n^{n+m}: X^{\times(n+m)} \rightarrow X^{\times n}$ and $\text{pr}_m^{n+m}: X^{\times(n+m)} \rightarrow X^{\times m}$ are the projections on the first n and the last m factors respectively, $(\text{pr}_m^{n+m})^*$ is the pullback map on $H_B^{\bullet}(-, \mathbb{Q})$, and $(\text{pr}_n^{n+m})_*$ is the Gysin map on $H_B^{\bullet}(-, \mathbb{Q})$ which is the adjoint to $(\text{pr}_n^{n+m})^*$ with respect to the Poincaré pairing on $H_B^{\bullet}(-, \mathbb{Q})$.

With all of this in place we can now describe the requisite formal logarithmic F-bundles. To streamline the exposition we will focus on the case $\mathcal{K} = \bar{\mathcal{K}} \subset \mathbb{C}$, k of characteristic zero, and $H^{\bullet} = H_B^{\bullet}(-, k)$. The case of varieties over a subfield $\mathcal{K} \subset \mathbb{C}$ which is not algebraically closed is also interesting and we will discuss it in the next section. For now, assume $\mathcal{K} = \bar{\mathcal{K}} \subset \mathbb{C}$ and consider a smooth projective $\mathcal{K}$ -variety X . Fix a non-empty collection of ample classes $\omega_1, \dots, \omega_m$ in $\text{CH}^{1, \text{hom}}(X) \subset H^2(X)$ on X which are linearly independent over $\mathbb{Q}$, and a collection $\{\Delta_j\} \subset H^{\bullet}(X)$ of pure degree cohomology classes, which completes the collection $\omega_1, \dots, \omega_m$ to a graded basis of the k -vector space $H^{\bullet}(X)$. Let $\{q_i\}_{i=1}^m \cup \{t_j\}$ be the dual linear coordinates on $H^{\bullet}(X)$. Focusing on the $\mathbb{Z}/2$ -grading of $H^{\bullet}(X)$, i.e. remembering only the parity of the cohomological degree, we can consider the associated super affine space $\mathfrak{a}_X := \text{Spec}(k[\text{Sym}(H^{\bullet}(X)^{\vee})])$. Next we consider the formal neighborhood of $0 \in \mathfrak{a}_X$. This is the smooth formal super scheme given by

$$B_X^{\mathfrak{f}} := \text{Spf } k[[(q_i), (t_j)]],$$

where all q_i are even, and the t_j 's are even or odd depending on their cohomological degree.

This geometric setting gives rise to an $\mathcal{O}_{B_X^{\mathfrak{f}}}$ -linear **quantum cup product**

$$\star: (\mathcal{H}_{u=0}^{\mathfrak{f}})^{\otimes 2} \rightarrow \mathcal{H}_{u=0}^{\mathfrak{f}}$$

on the trivial vector bundle $\mathcal{H}_{u=0}^{\mathbf{f}} = \mathcal{O}_{B_X^{\mathbf{f}}} \otimes_k H^\bullet(X)$, which in the current context is defined by the formula

$$(3.18) \quad \gamma_1 \star \gamma_2 := \gamma_1 \smile \gamma_2 + \sum_{\beta \neq 0} \prod_{i=1}^m q_i^{\int_{\beta} \omega_i} \cdot \sum_{n \geq 0} \frac{1}{n!} \int_{I_{n+3, \beta}(X)} \gamma_1 \boxtimes \gamma_2 \boxtimes \left(\sum_j t_j \Delta_j \right)^{\boxtimes n},$$

where $\gamma_1 \smile \gamma_2$ is the classical cup product on $H^\bullet(X)$, and the integrand is viewed as a class in $X \times X \times X^{\times n} = X^{\times(n+2)}$. The properties of the Gromov-Witten classes listed above imply that $\star$ is an associative unital product which can be viewed as a deformation along $B_X^{\mathbf{f}}$ of the classical cup product $\smile$ on the closed fiber $\mathcal{H}_{u=0}^{\mathbf{f}}|_{q_j=0, t_i=0} = H^\bullet(X)$.

We will also consider the trivial bundle $\mathcal{H}^{\mathbf{f}} := H^\bullet(X) \otimes \mathcal{O}_{B_X^{\mathbf{f}} \times \mathrm{Spf} k[[u]]}$. By definition we have $\mathcal{H}^{\mathbf{f}}|_{u=0} = \mathcal{H}_{u=0}^{\mathbf{f}}$ and also $\mathcal{H}^{\mathbf{f}} = \mathrm{pr}_{B_X^{\mathbf{f}}}^* \mathcal{H}_{u=0}^{\mathbf{f}}$. In particular the quantum product pulls back via $\mathrm{pr}_{B_X^{\mathbf{f}}}$ to an $\mathcal{O}_{B_X^{\mathbf{f}} \times \mathrm{Spf} k[[u]]}$ -linear product on $\mathcal{H}^{\mathbf{f}}$ which by abuse of notation we will also denote by $\star: (\mathcal{H}^{\mathbf{f}})^{\otimes 2} \rightarrow \mathcal{H}^{\mathbf{f}}$.

Note next that our choice of coordinates on $\mathfrak{a}_X$ gives an even, strict normal crossing divisor $\mathfrak{d} = \sum_{i=1}^m \mathfrak{d}_i$ where $\mathfrak{d}_i \subset \mathfrak{a}_X$ is given by the equation $q_i = 0$. This defines a natural log structure on $\mathfrak{a}_X$ and hence on $B_X^{\mathbf{f}}$ where the corresponding logarithmic tangent bundles $T_{\mathfrak{a}_X}(-\log \mathfrak{d})$ and $T_{B_X^{\mathbf{f}}}(-\log \mathfrak{d})$ are spanned by the vector fields $q_i \partial_{q_i}$, and ∂_{t_j} .

The quantum product now gives rise to a meromorphic connection on $\mathcal{H}^{\mathbf{f}} := H^\bullet(X) \otimes \mathcal{O}_{B_X^{\mathbf{f}} \times \mathrm{Spf} k[[u]]}$ defined by the formulas:

$$(3.19) \quad \begin{cases} \nabla_{\partial_u}^{\mathbf{f}} &= \partial_u - u^{-2} (\mathrm{Eu} \star (-)) + u^{-1} \frac{\mathrm{Deg} - \dim X \cdot \mathrm{id}}{2}, \\ \nabla_{q_i \partial_{q_i}}^{\mathbf{f}} &= q_i \partial_{q_i} + u^{-1} (\omega_i \star (-)), \\ \nabla_{\partial_{t_j}}^{\mathbf{f}} &= \partial_{t_j} + u^{-1} (\Delta_j \star (-)), \end{cases}$$

where $\mathrm{Deg}: \mathcal{H}^{\mathbf{f}} \rightarrow \mathcal{H}^{\mathbf{f}}$ is the degree operator $\mathrm{Deg} = \bigoplus_{a=0}^{2 \dim X} a \cdot \mathrm{id}_{H^a(X)}$ and Eu denotes the Euler vector field on $B_X^{\mathbf{f}}$, defined by

$$(3.20) \quad \mathrm{Eu}_\gamma = c_1(T_X) + \frac{\mathrm{Deg} - 2 \mathrm{id}}{2}(\gamma) \in (\mathcal{H}_{u=0}^{\mathbf{f}})_\gamma = \mathcal{H}_{\gamma \times 0}^{\mathbf{f}} \cong T_{B_X^{\mathbf{f}}, \gamma}(-\log \mathfrak{d}) \subset T_{B_X^{\mathbf{f}}, \gamma}.$$

Remark 3.21. Here we have adopted a different convention than the one used in [48] to describe the quantum connection. Specifically, the parameter u used here and the parameter (also called u) used in [48, Section 3.1] differ by a minus sign. If we set $q = q_1$ and if in the formula (3.19) we make the change of variable $u \mapsto -u$, then the restriction of $\nabla^{\mathbf{f}}$ to the purely even formal submanifold $\mathrm{Spf}[[q]] \subset B_X^{\mathbf{f}}$ reproduces the formula used to define the de Rham part of the one parameter A-model variation of nc-Hodge structures in [48, Section 3.1]. In addition, note that there is a typo in the description of the connection in [48, Section 3.1]: the class κ_X (used there to denote the Euler vector field) is the first Chern class of the tangent bundle of X , and not the first Chern class of the cotangent bundle, as written in [48, Section 3.1]. The formula (3.19) corrects this misprint and agrees with the common usage. In particular it agrees with the formulas used to compute the Γ -class correction in [48, Section 3.1], with the classical Dubrovin definition [26] of the quantum connection, and with the standard description of the quantum connection used recently in [44, 67, 16, 38].

By definition now $(\mathcal{H}^f, \nabla^f)/B_X^f$ is a formal logarithmic F-bundle for the given logarithmic structure on B_X^f . Furthermore, the properties of the Gromov-Witten classes listed above, imply that $(\mathcal{H}^f, \nabla^f)/B_X^f$ is a maximal formal logarithmic F-bundle, i.e. ∇^f induces a unital associative Frobenius product $\circ$ on $T_{B_X^f}(-\log \mathfrak{d})$ together with an Euler vector field $\text{Eu} \in \Gamma(B_X^f, T_{B_X^f}(-\log \mathfrak{d}))$ so that under the identification $\mathcal{H}^f \cong T_{B_X^f}(-\log \mathfrak{d})$ given by the frame $(q_i \partial_{q_i}, \partial_{t_j})$ the quantum product $\star$ is identified with the Frobenius product $\circ$, and the Euler vector field Eu is given by the formula (3.20).

For ease of reference we give the following

Definition 3.22. Fix $\mathcal{K} = \overline{\mathcal{K}} \subset \mathbb{C}$, k - a field of $\text{char } k = 0$, $H^\bullet = H_B^\bullet(-, k)$. Let X be a smooth projective variety, defined over $\mathcal{K}$, and let $\{\omega_i\}_{i=1}^m$, and $\{\Delta_j\}$ as above. Then the **formal A-model F-bundle** associated with $(X, \omega_1, \dots, \omega_m)$ is the formal logarithmic F-bundle $(\mathcal{H}^f, \nabla^f)/B_X^f$, where B_X^f is the formal neighborhood of 0 in the superscheme $\mathfrak{a}_X = \text{Spec } k[\text{Sym}(H^\bullet(X)^\vee)]$, $\mathcal{H}^f = \mathcal{O}_{B_X^f} \otimes H^\bullet(X)$, and ∇ is given by (3.19).

Remark 3.23. Traditionally, the quantum product is defined by the series (3.18) in which the sum is taken over *all* monomials q^β in the monoid algebra $k[\text{NE}(X, \mathbb{Z})]$. In the above construction we restricted our attention only to monomials corresponding to a *finite simplicial cone* of ample classes, so that we can work with formal power series rather than general elements in a Novikov ring, and so that we can control the logarithmic structures explicitly. We will come back to the issue of general versus restricted quantum products when we discuss the A-model analytic F-bundles in Section 3.5.2 and the analytic F-bundle version of the blowup formula in Section 4.2.

Remark 3.24. At a first glance this definition will not provide a very good invariant of a projective $\overline{\mathcal{K}}$ -variety X , as it seem to depend on several additional choices. However, this dependency on choices is very mild and can be disregarded. Indeed

(a) The first issue is that even when we fix the ample classes ω_i the definition of ∇^f seem to depend on the choice of the Δ_j 's that complete the ample ω_i 's to a graded basis of $H^\bullet(X)$. This is a mild problem since a different choice $\tilde{\Delta}_j$ of Δ_j 's will be related to the original choice by a graded linear automorphism of $H^\bullet$ and this linear automorphism will induce an automorphism of $\mathfrak{a}_X$ and a posteriori of the pair $(B_X^f, \mathcal{H}^f)$. The induced automorphism of $\mathfrak{a}_X$ acts trivially on the divisor $\mathfrak{d} \subset \mathfrak{a}_X$ and thus preserves the logarithmic structure of B_X^f . Also, by construction, the automorphism of $(B_X^f, \mathcal{H}^f)$ pulls back the logarithmic connection ∇ defined in terms of the Δ_j to the logarithmic connection $\tilde{\nabla}^f$ defined via the $\tilde{\Delta}_j$. Thus the two A-model F-bundles for $(X, \omega_1, \dots, \omega_m)$ that arise from the choices of $\{\Delta_j\}$ and $\{\tilde{\Delta}_j\}$ are naturally isomorphic.

(b) The second issue with Definition 3.22 is the explicit dependence on the choice of ample classes $\omega_1, \dots, \omega_m$. One way to get around this is to work with a maximal collection of such ω_i , i.e. choose a basis of the rational Néron-Severi group $\text{CH}^{1, \text{hom}}(X) \otimes \mathbb{Q}$ that consists of ample classes. Indeed, any two maximal collections of ample classes will be related by a change of basis automorphism of $H^\bullet(X)$. Therefore, the contribution of a monomial in q -coordinates for one picture coincides with the contribution of the corresponding monomial in another picture. Therefore, there is no loss of information for going between one basis to another.

(c) The independence of the choice of ample classes can be refined further by noticing that the formal A-model F-bundle corresponding to $(X, \omega_1, \dots, \omega_m)$ and the formal A-model F-bundle corresponding to (X, ω_1) contain the same information, i.e. can be reconstructed from one another, see [38, Lemma 2.24].

The formal logarithmic A-model F-bundle $(\mathcal{H}^f, \nabla^f)/B_X^f$ we just described, captures the motivic and Gromov-Witten information about X that we need. However the logarithmic structures and the formal variables make these F-bundles awkward to manipulate. We will circumvent this by repackaging the data encoded in $(\mathcal{H}^f, \nabla^f)/B_X^f$ into convergent non-archimedean analytic data, i.e. into a non-archimedean, analytic F-bundle.

3.5.2 Non-archimedean A-model F-bundles

We will use the Betti setup from the previous subsection. Fix two fields $\mathcal{K} \subset \mathbb{C}$ and k , with $\text{char } k = 0$, and the Weil cohomology theory $H^\bullet = H_B^\bullet(-, k)$. We will also fix an algebraically closed non-archimedean field $\mathbb{k} \supset k$, such that the valuation $\mathbf{v}: \mathbb{k} \rightarrow \mathbb{R} \cup \{\infty\}$ is trivial on k . A convenient choice is $\mathbb{k} = \bar{k}((\mathbf{y}^\mathbb{Q}))$.

3.5.2.1 The case of varieties over $\bar{\mathcal{K}}$. Let $X/\bar{\mathcal{K}}$ be a smooth projective variety, and let $H^\bullet(X) = H_B^\bullet(X(\mathbb{C})^{\text{an}}, k)$. Let $\mathbf{N}_1(X, \mathbb{Z}) = \text{CH}_1^{\text{hom}}(X)$ denote the group of classes of curves in X modulo Betti homological equivalence, and let $\text{NE}(X, \mathbb{Z}) \subset \mathbf{N}_1(X, \mathbb{Z})$ be the **the monoid of effective curve classes in X** .

Consider the monoid algebra $k[\text{NE}(X, \mathbb{Z})]$, i.e. the commutative unital k -algebra, whose underlying k -vector space has a basis labeled by the elements in $\text{NE}(X, \mathbb{Z})$, with a product given by the additive semigroup operation. Concretely, we will write q^β for the basis vector labeled by $\beta \in k[\text{NE}(X, \mathbb{Z})]$ and will call q^β the **standard monomial** corresponding to β . Thus we have

$$k[\text{NE}(X, \mathbb{Z})] = \bigoplus_{\beta \in \text{NE}(X, \mathbb{Z})} k \cdot q^\beta,$$

as a k -vector space, with a unit $\mathbf{1} = q^0$, and multiplication given by $q^{\beta_1} q^{\beta_2} := q^{\beta_1 + \beta_2}$. Write $k[[q]] = k[[\text{NE}(X, \mathbb{Z})]]$ for the completion of $k[\text{NE}(X, \mathbb{Z})]$ along its maximal monomial ideal, i.e. the ideal generated by q^β , with $\beta \neq 0 \in \text{NE}(X, \mathbb{Z})$.

Let $\{T_i\}_{i=0, \dots, r}$ be a homogeneous basis of the k -vector space $H^\bullet(X)$, such that $T_0 = \mathbf{1} \in H^0(X)$ is the unit in the cohomology algebra $H^\bullet(X)$, and such that the degree two basis vectors spanning $H^2(X)$ are chosen by first choosing separately bases of the subspace $H^2(X)_{\text{alg}} := \text{CH}^{1, \text{hom}}(X) \otimes k \subset H^2(X)$ and of the subspace $H^2(X)_{\text{trans}}$ of all transcendental classes in $H^2(X)$, i.e. all classes orthogonal with respect to the intersection pairing to cycle classes of algebraic curves. Let (t_i) be the corresponding system of k -linear coordinates on $H^\bullet(X)$.

Given $n \geq 0$, $\beta \in \text{NE}(X, \mathbb{Z})$, and classes $\gamma_1, \dots, \gamma_n \in H^\bullet(X)$, we will write

$$\langle \gamma_1 \cdots \gamma_n \rangle_\beta := \int_{I_{n, \beta}(X)} \gamma_1 \boxtimes \cdots \boxtimes \gamma_n \in k = H^0(X^{\times n}),$$

for the corresponding correlator. Equivalently, by the Künneth and Poincaré duality properties of $H^\bullet$ we have

$$\langle \gamma_1 \cdots \gamma_n \rangle_\beta := (I_{n, \beta}(X), \text{pr}_1^* \gamma_1 \otimes \cdots \otimes \text{pr}_n^* \gamma_n),$$

where we still write $(-, -)$ for the Poincaré pairing on $X^{\times n}$.

Definition 3.25. The $H^\bullet$ -valued **genus 0 Gromov-Witten potential** of X is the formal power series with $k[[q]]$ coefficients in the set of super variables $t_0, \dots, t_r$:

$$(3.26) \quad \Phi(q; t) := \sum_{n \geq 0, \beta \in \text{NE}(X, \mathbb{Z})} \frac{q^\beta}{n!} \sum_{i_1, \dots, i_n} \langle T_{i_1} \cdots T_{i_n} \rangle_\beta t_{i_1} \cdots t_{i_n} \in k[[q]][[t_0, \dots, t_r]].$$

Notation 3.27. Here $k[[q]][[t_0, \dots, t_r]] \cong k[\text{NE}(X, \mathbb{Z})] \widehat{\otimes} \widehat{\text{Sym}} H^\bullet(X)^\vee$ is a coordinated version of the **big Novikov ring for X** and we will sometimes denote it by Nov_X .

Note that by the unit axiom, the variable t_0 appears only in the terms with $\beta = 0$ and $n \leq 3$. In particular Φ is polynomial in t_0 .

Recall that a formal power series in a set of super variables is a usual formal series in even variables with coefficients in the exterior algebra of the odd variables. For the moment we view Φ as an R -valued function on the formal super scheme $/k$ which is the formal neighborhood of some (closed, even) base point of the algebraic super scheme $\text{Spec } k[\{t_i\}]$ whose k -points give the super vector space $H^\bullet(X)$. The third derivative $\partial^3 \Phi / \partial t_a \partial t_b \partial t_c$ of Φ in the coordinates $t_0, \dots, t_r$ defines a linear map $H^\bullet(X)^{\otimes 3} \rightarrow \text{Nov}_X$ and, by the standard Gromov-Witten lore [52], after contraction with the Poincaré pairing gives a quantum product

$$\star: H^\bullet(X)^{\otimes 2} \rightarrow H^\bullet(X) \otimes_k \text{Nov}_X,$$

which is the general (unrestricted by a choice of $\{\omega_i\}_{i=1}^m$) version of the product (3.18).

Consider next the torsion free part $\text{NS}(X, \mathbb{Z})_{\text{tf}} := \text{im} [\text{CH}^1(X) \rightarrow H^2(X)]$ of the Néron-Severi group of X . It is a free abelian group of finite rank ρ . Let

$$\mathcal{T}_{\mathbb{k}}(X) := \text{NS}(X, \mathbb{Z})_{\text{tf}} \otimes_{\mathbb{Z}} \mathbb{G}_{m, \mathbb{k}}$$

denote the split affine torus over $\mathbb{k}$ with cocharacter lattice $\text{NS}(X, \mathbb{Z})_{\text{tf}}$ and let $\mathcal{T}_{\mathbb{k}}(X)^{\text{an}}$ denote the associated non-archimedean $\mathbb{k}$ -analytic torus. We can also consider the $\mathbb{k}$ -vector space $\text{NS}(X, \mathbb{k}) := \text{NS}(X, \mathbb{Z})_{\text{tf}} \otimes_{\mathbb{Z}} \mathbb{k}$. By definition, the tangent space to $\mathcal{T}_{\mathbb{k}}(X)^{\text{an}}$ at each rigid point can be identified canonically with $\text{NS}(X, \mathbb{k})$.

With this notation we now define the following analytic $\mathbb{k}$ -spaces:

$B_{X, q} \subset \mathcal{T}_{\mathbb{k}}(X)^{\text{an}}$ is the preimage of the ample cone in $\text{NS}(X, \mathbb{R})$ under the valuation map $\mathcal{T}_{\mathbb{k}}(X)^{\text{an}} \rightarrow \text{NS}(X, \mathbb{R})$. Note that by construction $B_{X, q}$ is an admissible open in $\mathcal{T}_{\mathbb{k}}(X)^{\text{an}}$.

$\mathcal{B}_{X, t}^{\text{ev}}$ is the product of an analytic affine line with an open unit polydisk, where the affine line has coordinate t_0 , and the unit polydisk has coordinates t_i corresponding to the basis vectors T_i with $\deg T_i \in \{2, 4, 6, \dots\}$.

$B_{X, t}^{\text{ev}}$ is the product of the analytic affine line with coordinate t_0 and a unit polydisk with coordinates t_i corresponding to basis vectors T_i with $\deg T_i \in \{4, 6, \dots\}$, and to basis vectors $T_i \in H^2(X)_{\text{transc}}$ whenever $\deg T_i = 2$.

B_X^{odd} is the $\mathbb{k}$ -super analytic variety corresponding to the purely odd $\mathbb{k}$ -vector space $H^{\text{odd}}(X) \otimes_{\mathbb{k}} \mathbb{k}$, i.e. the super analytic variety whose underlying even analytic variety is a point and whose algebra of functions is the exterior algebra in the coordinates t_i for $\deg T_i \in \{1, 3, 5, \dots\}$.

Set

$$(3.28) \quad \mathcal{B}_X := B_{X,q} \times \mathcal{B}_{X,t}^{\text{ev}} \times B_X^{\text{odd}} \quad \text{and} \quad B_X := B_{X,q} \times B_{X,t}^{\text{ev}} \times B_X^{\text{odd}}.$$

By definition $\mathcal{B}_X$ is open in the $\mathbb{k}$ -super analytic variety $\mathcal{T}_{\mathbb{k}}(X)^{\text{an}} \times (H^\bullet(X) \otimes_{\mathbb{k}} \mathbb{k})^{\text{an}}$ and B_X is open in the $\mathbb{k}$ -super analytic variety $\mathcal{T}_{\mathbb{k}}(X)^{\text{an}} \times ((H^2(X)_{\text{transc}} \oplus H^{*\neq 2}(X)) \otimes_{\mathbb{k}} \mathbb{k})^{\text{an}}$. Note that the vector space inclusion map $H^2(X)_{\text{transc}} \otimes_{\mathbb{k}} \mathbb{k} \hookrightarrow H^2(X) \otimes_{\mathbb{k}} \mathbb{k}$ induces a closed analytic embedding $B_X \rightarrow \mathcal{B}_X$ of non-archimedean super analytic varieties.

With this notation, we now have the following important observation

Lemma 3.29. *The genus zero Gromov-Witten potential $\Phi(q, t)$ defines a $\mathbb{k}$ -valued analytic function on $\mathcal{B}_X$.*

Proof. Recall, that for any smooth projective $\bar{\mathcal{K}}$ -variety X we can find a finite collection of ample line bundles on X , so that their first Chern classes form a basis of the $\mathbb{Q}$ -Néron-Severi space $\text{NS}(X, \mathbb{Q}) := \text{CH}^{1, \text{hom}}(X) \otimes \mathbb{Q} \subset H^2(X)$. Let $L_1, \dots, L_m \in \text{Pic}(X)$ be such ample line bundles, and write $\omega_i := \text{cl}(c_1(L_i))$.

Consider the open simplicial cone

$$\sigma := \left\{ \sum_{i=1}^m a_i \omega_i \mid a_i \in \mathbb{R}_{>0} \right\} \subset \mathbb{R}\omega_1 \oplus \dots \oplus \mathbb{R}\omega_m.$$

Let $B_{\sigma,q} \subset \text{NS}(X, \mathbb{k})$ be the preimage of σ by the valuation map, and let $\mathcal{B}_{\sigma} := B_{\sigma,q} \times \mathcal{B}_{X,t}^{\text{ev}} \times B_X^{\text{odd}}$. Note that by definition $\mathcal{B}_{\sigma} \subset \mathcal{B}_X$ is an open analytic subvariety.

Now note, that the formula for Φ defines a $\mathbb{k}$ -valued analytic function on $\mathcal{B}_{\sigma}$. Indeed, let $q_1, \dots, q_m$ be the coordinates on $\mathcal{T}_{\mathbb{k}}(X)$ that are dual to the basis $\omega_1, \dots, \omega_m$. Using q_j and t_i as coordinates on the super analytic space $\mathcal{T}_{\mathbb{k}}(X)^{\text{an}} \times (H^\bullet(X) \otimes_{\mathbb{k}} \mathbb{k})^{\text{an}}$ we see that the restriction of the formal germ (3.26) to the open $\mathcal{B}_{\sigma}$ is given by the power series

$$\begin{aligned} \Phi|_{\mathcal{B}_{\sigma}} &= \sum_{n \geq 0, \beta} \frac{1}{n!} q_1^{(\beta, \omega_1)} \dots q_m^{(\beta, \omega_m)} \sum_{i_1, \dots, i_n} \langle T_{i_1} \dots T_{i_n} \rangle_{\beta} t_{i_1} \dots t_{i_n} \\ &\in k[[\{q_j\}, \{t_i\}]] \subset \mathbb{k}[[\{q_j\}, \{t_i\}]]. \end{aligned}$$

This series is polynomial in t_0 , and hence an analytic $\mathbb{k}$ -valued function on the non-archimedean analytic super domain $\mathcal{B}_{\sigma}$. Since the union of all such σ covers the ample cone of X , this implies that Φ can be viewed as an analytic function on $\mathcal{B}_X$ as claimed. $\square$

Let as before $\mathbb{D}$ denote the germ at 0 of an analytic disk with coordinate u . Write $\mathcal{O}_{\mathcal{B}_X \times \mathbb{D}}$ for the sheaf of analytic $\mathbb{k}$ -valued functions on the non-archimedean analytic supermanifold $\mathcal{B}_X \times \mathbb{D}$ and let $\mathcal{H} = H^\bullet(X) \otimes_{\mathbb{k}} \mathcal{O}_{\mathcal{B}_X \times \mathbb{D}}$ be the trivial analytic vector bundle on $\mathcal{B}_X \times \mathbb{D}$ with fiber $H^\bullet(X) \otimes_{\mathbb{k}} \mathbb{k}$. Note that our chosen basis of $H^\bullet(X)$ gives a global frame of $\mathcal{H}$, i.e. we have a canonical identification $\mathcal{H} = \oplus_{i=1}^r \mathcal{O}_{\mathcal{B}_X \times \mathbb{D}} \cdot T_i$. The inverse (g^{ij}) of the Gram matrix of the Poincaré pairing gives a section $\text{PD}^{-1} \in H^0(\mathcal{B}_X \times \mathbb{D}, \mathcal{H} \otimes \mathcal{H})$. The section is constant in the frame $\{T_i\}$ and therefore analytic.

The third derivatives of Φ with respect to the coordinates $\{t_i\}$ define a map of analytic super vector bundles

$$\mathcal{H}^{\otimes 3} \rightarrow \mathcal{O}, \quad T_a \otimes T_b \otimes T_c \mapsto \frac{\partial^3 \Phi}{\partial t_a \partial t_b \partial t_c}$$

and the contraction of this map with $\mathbf{PD}^{-1} \in \mathcal{H}^{\otimes 2}$ gives an analytic quantum product

$$\star: \mathcal{H} \otimes \mathcal{H} \longrightarrow \mathcal{H},$$

which is independent of u and tautologically satisfies the properties (i)-(v).

Furthermore, the quantum product $\star$ gives rise to a **non-archimedean analytic quantum connection** $\nabla: \mathcal{H} \rightarrow \mathcal{H} \otimes \Omega_{\mathcal{B}_X \times \mathbb{D}}^1[u, u^{-1}]$ which over an open chart $(\mathcal{B}_\sigma, (\{q_j\}, \{t_i\}))$ is defined by

$$(3.30) \quad \begin{cases} \nabla_{\partial_u} = \partial_u - u^{-2} (\text{Eu} \star (-)) + u^{-1} \frac{\text{Deg} - \dim X \cdot \text{id}}{2}, \\ \nabla_{\partial_{q_j}} = \partial_{q_j} + u^{-1} q_j^{-1} (\omega_j \star (-)), \\ \nabla_{\partial_{t_i}} = \partial_{t_i} + u^{-1} (T_i \star (-)), \end{cases}$$

Here again $\text{Deg}: \mathcal{H} \rightarrow \mathcal{H}$ is the degree operator $\text{Deg} = \bigoplus_{a=0}^{2 \dim X} a \cdot \text{id}_{H^a(X)}$ and Eu denotes the analytic Euler vector field, defined for every point $\gamma \in \mathcal{B}_\sigma \subset (H^\bullet(X) \otimes_k \mathbb{k})^{\text{an}}$ by

$$(3.31) \quad \text{Eu}_\gamma = c_1(T_X) + \frac{\text{Deg} - 2 \cdot \text{id}}{2}(\gamma) \in T_{\mathcal{B}_\sigma, \gamma} \subset \mathcal{H}_\gamma.$$

By construction $(\mathcal{H}, \nabla)/\mathcal{B}_X$ is an over maximal F-bundle over $\mathcal{B}_X$ and $B_X \subset \mathcal{B}_X$ is a closed analytic subvariety which meets transversally every leaf through a rigid point of the redundancy foliation of $(\mathcal{H}, \nabla)/\mathcal{B}_X$. Thus the restriction of $(\mathcal{H}, \nabla)|_{B_X \times \mathbb{D}}$ is the maximal F-bundle $(\mathcal{H}, \nabla)/B_X$ over B_X .

Definition 3.32. The F-bundle $(\mathcal{H}, \nabla)/\mathcal{B}_X$ will be called the **non-archimedean overmaximal A-model F-bundle associated to X , $\bar{\mathcal{K}} \subset \mathbb{C}$, $\mathbb{k} \supset k$, and $H^\bullet$** . The F-bundle $(\mathcal{H}, \nabla)/B_X$ will be called the **non-archimedean maximal A-model F-bundle associated to X , $\bar{\mathcal{K}} \subset \mathbb{C}$, $\mathbb{k} \supset k$, and $H^\bullet$** .

Remark 3.33. The analyticity of Φ implies that F-bundles $(\mathcal{H}, \nabla)/\mathcal{B}_X$ and $(\mathcal{H}, \nabla)/B_X$ are non-archimedean analytic in the sense of Definition 3.1.

3.5.2.2 The case of varieties over $\mathcal{K}$. Fix two fields k and $\mathcal{K}$, both of characteristic zero.

3.5.2.2.1 Preliminaries on motives. Let $\mathcal{C}^\mathcal{K}$ be the category of pure André motives (aka motivated motives) [3] of smooth projective $\mathcal{K}$ -varieties. This is a $\mathbb{Q}$ -linear, rigid, semisimple tensor category which is defined unconditionally. For a $\mathcal{K}$ -variety $\mathfrak{X}$ we will write $h(\mathfrak{X})$ for the associated André motive.

Fix an embedding $\mathcal{K} \subset \mathbb{C}$. Then we have a natural algebraic closure $\bar{\mathcal{K}} \subset \mathbb{C}$ of $\mathcal{K}$ inside $\mathbb{C}$, and a base change functor

$$(-) \times_{\mathcal{K}} \bar{\mathcal{K}}: \mathcal{C}^\mathcal{K} \longrightarrow \mathcal{C}^{\bar{\mathcal{K}}}, \quad h(\mathfrak{X}) \mapsto h(\mathfrak{X} \times_{\mathcal{K}} \bar{\mathcal{K}}),$$

which by definition is a monoidal functor of $\mathbb{Q}$ -linear tensor categories.

Taking Betti cohomology of the associated complex analytic space gives a Weil cohomology theory on $\bar{\mathcal{K}}$ -varieties [6]

$$(3.34) \quad H_B^\bullet(-): \mathcal{C}^{\bar{\mathcal{K}}} \longrightarrow \text{Vect}_{\mathbb{Q}}, \quad h(X) \longmapsto H_B^\bullet((X \times_{\bar{\mathcal{K}}} \mathbb{C})^{\text{an}}, \mathbb{Q}),$$

and by precomposing with base change gives also a compatible Weil cohomology theory on $\mathcal{K}$ -varieties (3.35)

$$\begin{array}{ccccccc} H_B^\bullet(-) : \mathcal{C}^\mathcal{K} & \longrightarrow & \mathbf{Vect}_\mathbb{Q} & , & h(\mathfrak{X}) & \longmapsto & h(X) \longmapsto H_B^\bullet((X \times_{\bar{\mathcal{K}}} \mathbb{C})^{\text{an}}, \mathbb{Q}) \\ & & & & \parallel & & \parallel \\ & & & & h(\mathfrak{X} \times_{\mathcal{K}} \bar{\mathcal{K}}) & & H_B^\bullet((\mathfrak{X} \times_{\mathcal{K}} \mathbb{C})^{\text{an}}, \mathbb{Q}). \end{array}$$

Abusing notation we denote both of these tensor functors by $H_B^\bullet(-)$ since it will be clear from the context which type of varieties serve as input of Betti cohomology.

These data are conveniently encoded in certain proalgebraic groups.

(i) The automorphism groups $\mathcal{G}^{\bar{\mathcal{K}}}$ and $\mathcal{G}^\mathcal{K}$ of the tensor functors (3.34) and (3.35) are the **motivic Galois groups of André motives** of smooth projective $\bar{\mathcal{K}}$ -varieties and $\mathcal{K}$ -varieties respectively. $\mathcal{G}^{\bar{\mathcal{K}}}$ and $\mathcal{G}^\mathcal{K}$ are proreductive algebraic group schemes over $\mathbb{Q}$. Tannaka duality identifies $\mathcal{C}^{\bar{\mathcal{K}}}$ and $\mathcal{C}^\mathcal{K}$ with the categories $\mathbf{Rep}_\mathbb{Q}(\mathcal{G}^{\bar{\mathcal{K}}})$ and $\mathbf{Rep}_\mathbb{Q}(\mathcal{G}^\mathcal{K})$ of representations of $\mathcal{G}^{\bar{\mathcal{K}}}$ and $\mathcal{G}^\mathcal{K}$ on finite dimensional $\mathbb{Q}$ -vector spaces.

(ii) The base change functor $(-) \times_{\mathcal{K}} \bar{\mathcal{K}} : \mathcal{C}^\mathcal{K} \rightarrow \mathcal{C}^{\bar{\mathcal{K}}}$ induces a normal embedding $\mathcal{G}^{\bar{\mathcal{K}}} \rightarrow \mathcal{G}^\mathcal{K}$.

(iii) The quotient group scheme $\mathcal{G}^\mathcal{K}/\mathcal{G}^{\bar{\mathcal{K}}}$ is the automorphism group of the restriction of the functor $H_B^\bullet(-)$ to the full tensor subcategory $\mathbf{Art}^\mathcal{K} \subset \mathcal{C}^\mathcal{K}$ of Artin motives, i.e. motives of zero dimensional smooth projective varieties over $\mathcal{K}$. In other words, $\mathcal{G}^\mathcal{K}/\mathcal{G}^{\bar{\mathcal{K}}}$ is the automorphism group of the induced fiber functor

$$H_B^\bullet(-)|_{\mathbf{Art}^\mathcal{K}} = H_B^0(-) : \mathbf{Art}^\mathcal{K} \longrightarrow \mathbf{Vect}_\mathbb{Q}.$$

This group is naturally identified with the Galois group $\text{Gal}(\bar{\mathcal{K}}/\mathcal{K})$, where $\text{Gal}(\bar{\mathcal{K}}/\mathcal{K})$ is viewed as a profinite group scheme over $\mathbb{Q}$ for the trivial scheme structure.

In summary, we have a short exact sequence

$$1 \longrightarrow \mathcal{G}^{\bar{\mathcal{K}}} \longrightarrow \mathcal{G}^\mathcal{K} \longrightarrow \text{Gal}(\bar{\mathcal{K}}/\mathcal{K}) \longrightarrow 1$$

of proreductive group schemes³ over $\mathbb{Q}$.

(iv) The group $\mathcal{G}^\mathcal{K}$ has a natural Lefschetz character $\mathcal{G}^\mathcal{K} \rightarrow \mathbb{G}_m$ given by its action on $H_B^2(\mathbb{P}_\mathcal{K}^1, \mathbb{Q})$, and we denote the kernel of this character by $\mathbf{Mot}^\mathcal{K}$. The restriction of the Lefschetz character to $\mathcal{G}^{\bar{\mathcal{K}}}$ is given by its action on $H_B^2(\mathbb{P}_{\bar{\mathcal{K}}}^1, \mathbb{Q})$ and we denote its kernel by $\mathbf{Mot}^{\bar{\mathcal{K}}}$.

Again $\mathbf{Mot}^\mathcal{K}$ and $\mathbf{Mot}^{\bar{\mathcal{K}}}$ are proreductive algebraic groups over $\mathbb{Q}$ which we view as the **motivic Galois groups of the category of nc André motives**. They can be interpreted as the Galois groups of the semisimple Tannakian $\mathbb{Q}$ -linear categories $\mathcal{C}^\mathcal{K}/\text{Tate}$ and $\mathcal{C}^{\bar{\mathcal{K}}}/\text{Tate}$ obtained as the orbit categories of $\mathcal{C}^\mathcal{K}$ and $\mathcal{C}^{\bar{\mathcal{K}}}$ for the action of the Tate twist.

³It is expected that $\mathcal{G}^{\bar{\mathcal{K}}}$ is the connected component of the identity of $\mathcal{G}^\mathcal{K}$ or equivalently that $\text{Gal}(\bar{\mathcal{K}}/\mathcal{K}) = \pi_0(\mathcal{G}^\mathcal{K})$ is the component group of $\mathcal{G}^\mathcal{K}$ but we will not need this fact.

Thus $\mathcal{C}^{\mathcal{K}}/\text{Tate} = \text{Rep}_{\mathbb{Q}}(\text{Mot}^{\mathcal{K}})$ and $\mathcal{C}^{\overline{\mathcal{K}}}/\text{Tate} = \text{Rep}_{\mathbb{Q}}(\text{Mot}^{\overline{\mathcal{K}}})$ and the quotient functors

$$\begin{array}{ccc} \mathcal{C}^{\mathcal{K}} & \longrightarrow & \mathcal{C}^{\mathcal{K}}/\text{Tate} \\ \parallel & & \parallel \\ \text{Rep}_{\mathbb{Q}}(\mathbf{G}^{\mathcal{K}}) & & \text{Rep}_{\mathbb{Q}}(\text{Mot}^{\mathcal{K}}) \end{array} \quad \text{and} \quad \begin{array}{ccc} \mathcal{C}^{\overline{\mathcal{K}}} & \longrightarrow & \mathcal{C}^{\overline{\mathcal{K}}}/\text{Tate} \\ \parallel & & \parallel \\ \text{Rep}_{\mathbb{Q}}(\mathbf{G}^{\overline{\mathcal{K}}}) & & \text{Rep}_{\mathbb{Q}}(\text{Mot}^{\overline{\mathcal{K}}}) \end{array}$$

correspond to the pullback of representations via the inclusions $\text{Mot}^{\mathcal{K}} \subset \mathbf{G}^{\mathcal{K}}$ and $\text{Mot}^{\overline{\mathcal{K}}} \subset \mathbf{G}^{\overline{\mathcal{K}}}$.

(v) The induced functor $\text{Art}^{\mathcal{K}} \subset \mathcal{C}^{\mathcal{K}} \rightarrow \mathcal{C}^{\mathcal{K}}/\text{Tate}$ from (Artin motives) to (André motives modulo Tate twists), is still fully faithful and so the composed homomorphism of group schemes over $\mathbb{Q}$

$$\text{Mot}^{\mathcal{K}} \subset \mathbf{G}^{\mathcal{K}} \rightarrow \text{Gal}(\overline{\mathcal{K}}/\mathcal{K})$$

is surjective. Thus $\text{Gal}(\overline{\mathcal{K}}/\mathcal{K})$ can also be identified with the quotient group scheme $\text{Mot}^{\mathcal{K}}/\text{Mot}^{\overline{\mathcal{K}}}$, i.e. we also have a short exact sequence

$$1 \longrightarrow \text{Mot}^{\overline{\mathcal{K}}} \longrightarrow \text{Mot}^{\mathcal{K}} \longrightarrow \text{Gal}(\overline{\mathcal{K}}/\mathcal{K}) \longrightarrow 1.$$

Suppose that $\mathfrak{X}$ is a smooth projective $\mathcal{K}$ -variety and let $X = \mathfrak{X} \times_{\mathcal{K}} \overline{\mathcal{K}}$ be its base change to $\overline{\mathcal{K}}$. By definition the Betti cohomologies of $\mathfrak{X}$ and X have the same underlying rational vector space. To distinguish notationally between the two Betti cohomology theories, our convention will be to write $H_B^{\bullet}(X)$ for the Betti cohomology of X , viewed as a $\text{Mot}^{\overline{\mathcal{K}}}$ -module, and will write $H_B^{\bullet}(\mathfrak{X})$ for the Betti cohomology of $\mathfrak{X}$, viewed as a $\text{Mot}^{\mathcal{K}}$ -module.

The Galois group $\text{Gal}(\overline{\mathcal{K}}/\mathcal{K})$ acts on X by automorphisms, and so by pullback acts on the maximal torsion free quotient

$$\text{NS}(X, \mathbb{Z})_{\text{tf}} := \text{NS}(X, \mathbb{Z})/(\text{torsion}) = \text{im} [\text{CH}_1(X) \rightarrow H_B^2((X \times_{\overline{\mathcal{K}}} \mathbb{C})^{\text{an}}, \mathbb{Q})]$$

of the Néron-Severi group of X . But $\text{NS}(X, \mathbb{Z})_{\text{tf}}$ is a free abelian group of finite rank ρ and hence we get a homomorphism

$$\text{Gal}(\overline{\mathcal{K}}/\mathcal{K}) \rightarrow \text{GL}(\text{NS}(X, \mathbb{Z})_{\text{tf}}) \cong \text{GL}_{\rho}(\mathbb{Z}).$$

The image of this homomorphism is a finite subgroup of $\text{GL}(\text{NS}(X, \mathbb{Z})_{\text{tf}}) \cong \text{GL}_{\rho}(\mathbb{Z})$ which we will denote by $\Gamma_{\mathfrak{X}}$.

Consider next the rational vector space

$$H_B^2(X) = H_B^2((X \times_{\overline{\mathcal{K}}} \mathbb{C})^{\text{an}}, \mathbb{Q}).$$

It decomposes into a direct sum of natural subspaces

$$(3.36) \quad H_B^2(X) = H_B^2(X)_{\text{alg}} \oplus H_B^2(X)_{\text{transc}},$$

where

- $H_B^2(X)_{\text{alg}} = \text{NS}(X, \mathbb{Q})$ is the Néron-Severi subspace spanned by $\text{NS}(X, \mathbb{Z})_{\text{tf}}$,
- $H_B^2(X)_{\text{transc}} \subset H_B^2(X)$ is the subspace of classes orthogonal to algebraic curves in X .

If we view $H_B^2(X)$ as a representation of $\text{Mot}^{\bar{\mathcal{K}}}$, then $H_B^2(X)_{\text{alg}} = H_B^2(X)^{\text{Mot}^{\bar{\mathcal{K}}}}$ is just the space of invariants, and $H_B^2(X)_{\text{transc}}$ is just the complementary $\text{Mot}^{\bar{\mathcal{K}}}$ subrepresentation.

More importantly, the $\text{Mot}^{\mathcal{K}}$ -action on $H_B^2(\mathfrak{X}) = H_B^2(X)$ also respects the decomposition (3.36).

Proposition 3.37. (a) *The natural action of $\text{Mot}^{\mathcal{K}}$ on $H_B^2(\mathfrak{X})$ respects the decomposition*

$$(3.38) \quad H_B^2(\mathfrak{X}) = H_B^2(X)_{\text{alg}} \oplus H_B^2(X)_{\text{transc}}$$

(b) *The induced action of $\text{Mot}^{\mathcal{K}}$ on $H_B^2(X)_{\text{alg}}$ factors through the homomorphism*

$$\text{Mot}^{\mathcal{K}} \rightarrow \text{Gal}(\bar{\mathcal{K}}/\mathcal{K}) \rightarrow \Gamma_{\mathfrak{X}} \subset \text{GL}(\text{NS}(X, \mathbb{Z})_{\text{tf}} \subset \text{GL}(H_B^2(X)_{\text{alg}}).$$

Proof. The Galois group $\text{Gal}(\bar{\mathcal{K}}/\mathcal{K})$ acts on the $\bar{\mathcal{K}}$ -points of X and thus acts on the cohomology space $H_B^2(X)$. If L is a line bundle on X , then the image ℓ of $c_1(L) \in \text{CH}^1(X)$ in $H_B^2(X)$ is a motivated cycle, and so is fixed by $\text{Mot}^{\bar{\mathcal{K}}}$. Therefore an element $g \in \text{Mot}^{\mathcal{K}}$ acts on $\ell \in H_B^2(X)$ through its image $\underline{g} \in \text{Gal}(\bar{\mathcal{K}}/\mathcal{K})$, i.e. $g(\ell) = \underline{g}(\ell)$, where on the right hand side we use the natural action of $\text{Gal}(\bar{\mathcal{K}}/\mathcal{K})$ on $H_B^2(X)$. Since the action of any element of $\text{Gal}(\bar{\mathcal{K}}/\mathcal{K})$ on X sends a divisor to a divisor, it follows that the action of $\text{Gal}(\bar{\mathcal{K}}/\mathcal{K})$ on $H_B^2(X)$ will preserve the subspace $\text{NS}(X, \mathbb{Q}) = H_B^2(X)_{\text{alg}}$. Hence the action of $\text{Mot}^{\mathcal{K}}$ on $H_B^2(X)$ will preserve $H_B^2(X)_{\text{alg}}$. This proves that $H_B^2(X)_{\text{alg}}$ is stable under $\text{Mot}^{\mathcal{K}}$ and, since $\text{Mot}^{\bar{\mathcal{K}}}$ acts trivially on $H_B^2(X)_{\text{alg}}$, this proves part **(b)** of the Proposition.

To finish the proof of part **(a)** we need to check that $H_B^2(X)_{\text{transc}}$ is also stable under $\text{Mot}^{\mathcal{K}}$. Let $\mathcal{O}_{\mathfrak{X}}(1)$ be a very ample line bundle on the $\mathcal{K}$ -variety $\mathfrak{X}$ and let $\mathbf{P} \in H_B^2(\mathfrak{X}) = H_B^2(X)$ be the image of $c_1(\mathcal{O}_{\mathfrak{X}}(1))$ in Betti cohomology. The choice of $\mathcal{O}_{\mathfrak{X}}(1)$ gives rise to a natural $\mathbb{Q}$ -valued symmetric bilinear form

$$\langle -, - \rangle: H_B^2(X) \otimes H_B^2(X) \longrightarrow \mathbb{Q}, \quad \langle \alpha, \beta \rangle = \int_X \alpha \smile \beta \smile \mathbf{P}^{d-2},$$

where d is the dimension of X . Since $\mathbf{P}$ is the class of line bundle defined over $\mathcal{K}$, this form is invariant under the action of $\text{Mot}^{\mathcal{K}}$. But for any $\ell \in H_B^2(X)_{\text{alg}}$ $\ell \cdot \mathbf{P}^{d-2}$ is the class of an algebraic curve in X , and so $H_B^2(X)_{\text{transc}} \subset H_B^2(X)$ is contained in the $\langle -, - \rangle$ -orthogonal complement of $H_B^2(X)_{\text{alg}}$ of $H_B^2(X)_{\text{alg}} \subset H_B^2(X)$.

On the other hand, if we write $\text{Prim}_{\text{alg}}^2$ for the $\langle -, - \rangle$ -orthogonal complement of $\mathbf{P}$ inside $H_B^2(X)_{\text{alg}}$, then we know that $\langle \mathbf{P}, \mathbf{P} \rangle > 0$ by the ampleness of $\mathcal{O}_{\mathfrak{X}}(1)$, and we know that the restriction of $\langle -, - \rangle$ to $\text{Prim}_{\text{alg}}^2$ is negative definite by the Hodge index theorem for $X \times_{\bar{\mathcal{K}}} \mathbb{C}$. In particular, the $\mathbb{Q}$ -valued form $\langle -, - \rangle$ is non-degenerate on the subspace $H_B^2(X)_{\text{alg}} \subset H_B^2(X)$, and so $H_B^2(X) = H_B^2(X)_{\text{alg}} \oplus H_B^2(X)_{\text{alg}}^{\perp}$. Thus $H_B^2(X)_{\text{transc}} = H_B^2(X)_{\text{alg}}^{\perp}$, and since the form $\langle -, - \rangle$ is $\text{Mot}^{\mathcal{K}}$ -invariant, we conclude that $H_B^2(X)_{\text{transc}}$ is $\text{Mot}^{\mathcal{K}}$ -stable. This concludes the proof of the Proposition. $\square$

Notation 3.39. For disambiguation we will write $H_B^2(\mathfrak{X})_{\bar{\mathcal{K}}-\text{alg}}$ and $H_B^2(\mathfrak{X})_{\bar{\mathcal{K}}-\text{transc}}$ for the rational vector spaces $H_B^2(X)_{\text{alg}}$ and $H_B^2(X)_{\text{transc}}$ together with the $\text{Mot}^{\mathcal{K}}$ -actions given by part **(a)** of the previous proposition. Note that these $\text{Mot}^{\mathcal{K}}$ -representations are different from the representations $H_B^2(\mathfrak{X})_{\text{alg}}$ and $H_B^2(\mathfrak{X})_{\text{transc}}$ consisting of $\mathcal{K}$ -algebraic and $\mathcal{K}$ -transcendental classes respectively.

3.5.2.2.2 The quantum cohomology algebra for varieties over $\mathcal{K}$ Let again $\mathfrak{X}$ be a smooth projective variety over $\mathcal{K}$, and let $X = \mathfrak{X} \times_{\mathcal{K}} \overline{\mathcal{K}}$ be its base change to $\mathrm{Spec} \overline{\mathcal{K}}$. Consider the rational vector space $H_B^\bullet(X)$ together with its structure of a $\mathrm{Mot}^{\overline{\mathcal{K}}}$ -module.

Let

$$\mathrm{NE}(X, \mathbb{Z}) \subset \mathrm{CH}_1^{\mathrm{hom}}(X) = \mathrm{im} [\mathrm{CH}_1(X) \rightarrow H_2((X \times_{\overline{\mathcal{K}}} \mathbb{C})^{\mathrm{an}}, \mathbb{Q})]$$

denote the monoid of numerically effective curve classes in X , and let, as before, Nov_X denote the big Novikov ring for X , i.e. the ring

$$\mathrm{Nov}_X := \mathbb{Q}[[\mathrm{NE}(X, \mathbb{Z})]] \widehat{\otimes}_{\mathbb{Q}} \widehat{\mathrm{Sym}} H_B^\bullet(X)^\vee,$$

where $\mathbb{Q}[[\mathrm{NE}(X, \mathbb{Z})]]$ is the completion of the monoid algebra $\mathbb{Q}[\mathrm{NE}(X, \mathbb{Z})]$ in its maximal ideal $\langle q^\beta | \beta \in \mathrm{NE}(X, \mathbb{Z}), \beta \neq 0 \rangle$, and similarly $\widehat{\mathrm{Sym}} H_B^\bullet(X)^\vee$ is the completion of the superalgebra of polynomials on $H_B^\bullet(X)$ along the maximal ideal of the origin.

The $\mathrm{Mot}^{\overline{\mathcal{K}}}$ -action on $H_B^\bullet(X)$ induces a canonical action on Nov_X , preserving the $\mathbb{Q}$ -algebra structure of Nov_X . Here $\mathrm{Mot}^{\overline{\mathcal{K}}}$ acts trivially on the factor $\mathbb{Q}[[\mathrm{NE}(X, \mathbb{Z})]]$ (since it acts trivially on the classes of effective curves defined over $\overline{\mathcal{K}}$) and tautologically (i.e. by pullback) on the power series algebra $\widehat{\mathrm{Sym}} H_B^\bullet(X)^\vee$.

As explained before, the $H_B^\bullet(-)$ -valued Gromov-Witten invariants of X equip the rational vector space $H_B^\bullet(X)$ with an associative, super commutative quantum product

$$\star: H_B^\bullet(X) \otimes H_B^\bullet(X) \longrightarrow H_B^\bullet(X) \otimes \mathrm{Nov}_X.$$

Furthermore, since the Gromov-Witten classes are cycle classes of $\overline{\mathcal{K}}$ -rational algebraic cycles in Cartesian powers of X , it follows that the quantum product $\star$ is $\mathrm{Mot}^{\overline{\mathcal{K}}}$ -equivariant for the natural actions of $\mathrm{Mot}^{\overline{\mathcal{K}}}$ on $H_B^\bullet(X)$ and Nov_X .

Since X arises as a base change of a variety $\mathfrak{X}$ over $\mathcal{K}$, the rational vector space underlying $H_B^\bullet(X)$ is the same as $H_B^\bullet(\mathfrak{X})$ and is therefore also equipped with a natural $\mathrm{Mot}^{\mathcal{K}}$ -action. Again this induces a tautological action (pulling back formal germs of functions at the origin) of $\mathrm{Mot}^{\mathcal{K}}$ on $\widehat{\mathrm{Sym}} H_B^\bullet(\mathfrak{X})^\vee$. As a $\mathrm{Mot}^{\mathcal{K}}$ -module this completed algebra is simply the completion $\widehat{\mathrm{Sym}} H_B^\bullet(\mathfrak{X})^\vee$. Also $\mathrm{Mot}^{\mathcal{K}}$ acts naturally on $\mathrm{NE}(X, \mathbb{Z})$ and the action factors through the homomorphism $\mathrm{Mot}^{\mathcal{K}} \twoheadrightarrow \Gamma_{\mathfrak{X}}$, where the action of $\Gamma_{\mathfrak{X}}$ on $\mathrm{NE}(X, \mathbb{Z})$ is dual to the action of $\Gamma_{\mathfrak{X}}$ on $\mathrm{NS}(X, \mathbb{Z})_{\mathrm{tf}}$. This, in turn, induces an action on the completed monoid algebra $\mathbb{Q}[[\mathrm{NE}(X, \mathbb{Z})]]$ and hence an action on the Novikov ring. We will write $\mathrm{Nov}_{\mathfrak{X}}$ for this $\mathrm{Mot}^{\mathcal{K}}$ -equivariant algebra. Note that again $\mathrm{Nov}_{\mathfrak{X}}$ and Nov_X are the same as $\mathbb{Q}$ -algebras (or seen as $\mathbb{Q}$ -algebras equipped with a $\mathrm{Mot}^{\overline{\mathcal{K}}}$ -action), but that the action of $\mathrm{Mot}^{\mathcal{K}}$ on $\mathrm{Nov}_{\mathfrak{X}}$ depends on the $\mathcal{K}$ -model $\mathfrak{X}$.

Proposition 3.40. *The quantum product viewed as map*

$$(3.41) \quad \star: H_B^\bullet(\mathfrak{X}) \otimes H_B^\bullet(\mathfrak{X}) \longrightarrow H_B^\bullet(\mathfrak{X}) \otimes \mathrm{Nov}_{\mathfrak{X}}$$

is $\mathrm{Mot}^{\mathcal{K}}$ -equivariant.

Proof. The Galois group $\mathrm{Gal}(\overline{\mathcal{K}}/\mathcal{K})$ acts on $\mathrm{NE}(X, \mathbb{Z})$ through its finite quotient $\mathrm{Gal}(\overline{\mathcal{K}}/\mathcal{K}) \twoheadrightarrow \Gamma_{\mathfrak{X}}$. Therefore the colimit of finite unions of $\Gamma_{\mathfrak{X}}$ -orbits in $\mathrm{NE}(X, \mathbb{Z})$ will be an ind-finite ind-affine smooth semigroup $\mathcal{K}$ -scheme $\underline{\mathrm{NE}}(X, \mathbb{Z})$ whose set of $\overline{\mathcal{K}}$ -points is the semigroup of effective curve classes in X .

For a $\mathcal{K}$ -variety $\mathfrak{X}$ let us write $h(\mathfrak{X}) \in \mathbf{ob} \mathcal{C}^{\mathcal{K}}$ for the corresponding André motive and similarly $h(X) \in \mathcal{C}^{\mathcal{K}}$ for the André motive of the base changed variety. By taking filtered colimits, we can then define $h(\underline{\mathbf{NE}}(X, \mathbb{Z}))$ as a commutative algebra in the ind completed tensor category $\mathbf{Ind} \mathcal{C}^{\mathcal{K}}$ of ind objects in $\mathcal{C}^{\mathcal{K}}$.

For any $\Gamma_{\mathfrak{X}}$ -orbit $\mathfrak{o} \subset \mathbf{NE}(X, \mathbb{Z})$ of effective curve classes on X , and any n , the collection of Gromov-Witten classes $(I_{n, \beta}(X))_{\beta \in \mathfrak{o}}$ can be viewed (see Remark 3.15) as a motivated cycle on the $\mathcal{K}$ -scheme $\underline{\mathbf{NE}}(X, \mathbb{Z}) \times \mathfrak{X}^{\times n}$, i.e. as a map from the trivial motive

$$I_{n, \mathfrak{o}} : \mathbb{1} \longrightarrow h(\underline{\mathbf{NE}}(X, \mathbb{Z}) \times \mathfrak{X}^{\times n}) = h(\underline{\mathbf{NE}}(X, \mathbb{Z})) \widehat{\otimes} h(\mathfrak{X})^{\otimes n}$$

in the category $\mathbf{Ind} \mathcal{C}^{\mathcal{K}}$. In particular, for every $n = m + 3$, after contracting with the Poincaré pairing on $h(\mathfrak{X})$, we get a map in $\mathbf{Ind} \mathcal{C}^{\mathcal{K}}$:

$$\star_{m, \mathfrak{o}} : \mathbb{1} \longrightarrow h(\underline{\mathbf{NE}}(X, \mathbb{Z})) \widehat{\otimes} h(\mathfrak{X})^{\vee \otimes m} \otimes h(\mathfrak{X})^{\vee \otimes 2} \otimes h(X)$$

which by construction is invariant under the action of $\mathbf{S}_m \times \mathbf{S}_2$. This is the $\mathfrak{o}$ -contribution to the m -th homogeneous piece of the quantum product, realized as a map of André motives. Summing up over all m and $\mathfrak{o}$ we get the quantum product map in $\mathbf{Ind} \mathcal{C}^{\mathcal{K}}$:

$$\star := \sum_{m \geq 0} \sum_{\mathfrak{o} \in \mathbf{NE}(X, \mathbb{Z}) / \Gamma_{\mathfrak{X}}} \frac{1}{m!} \star_{m, \mathfrak{o}} : h(\mathfrak{X})^{\otimes 2} \longrightarrow h(\mathfrak{X}) \otimes h(\underline{\mathbf{NE}}(X, \mathbb{Z})) \widehat{\otimes} \widehat{\mathbf{Sym}} h(\mathfrak{X})^{\vee}.$$

Now applying the fiber functor $H_B^{\bullet}(-) : \mathcal{C}^{\mathcal{K}} \rightarrow \mathbf{Vect}_{\mathbb{Q}}$ and passing to the quotient by Tate twists yields the statement of the Proposition. $\square$

Next suppose G is a proreductive group over k and $\epsilon_G \in G$ is a fixed central element of order two. The category of representations $\mathbf{Rep}_k(G)$ of G on finite dimensional G -vector spaces is a semisimple k -linear tensor category which is naturally $\mathbb{Z}/2$ -graded by the eigenvalues of the action of ϵ_G .

Suppose we are given a (G, ϵ_G) -**symmetric Weil cohomology theory on $\mathcal{K}$ -varieties**, i.e. a tensor functor

$$(3.42) \quad \mathcal{C}^{\mathcal{K}} / \text{Tate} \xrightarrow{\mathbf{H}^{\bullet}} \mathbf{Rep}_k(G),$$

of $\mathbb{Z}/2$ -graded categories, which commutes with duality.

Through the identification $\mathcal{C}^{\mathcal{K}} / \text{Tate} \cong \mathbf{Rep}_{\mathbb{Q}}(\mathbf{Mot}^{\mathcal{K}})$ given by the fiber functor $[\mathfrak{X}] \mapsto H_B^{\bullet}(\mathfrak{X})$ we can view $\mathbf{H}^{\bullet}$ as a functor $\mathfrak{h} : \mathbf{Rep}_{\mathbb{Q}}(\mathbf{Mot}^{\mathcal{K}}) \rightarrow \mathbf{Rep}_k(G)$ which respects $\mathbb{Z}/2$ -gradings commutes with duality. With this definition we have a strictly commutative diagram of functors

$$\begin{array}{ccc} \mathcal{C}^{\mathcal{K}} / \text{Tate} & \xrightarrow{\mathbf{H}^{\bullet}} & \mathbf{Rep}_k(G). \\ H_B^{\bullet}(-) \parallel & \nearrow \mathfrak{h} & \\ \mathbf{Rep}_{\mathbb{Q}}(\mathbf{Mot}^{\mathcal{K}}) & & \end{array}$$

Applying $\mathbf{H}^{\bullet}$ to the André motive version of the quantum cohomology algebra we get a super commutative, associative quantum product

$$(3.43) \quad \star : \mathbf{H}^{\bullet}(\mathfrak{X}) \otimes \mathbf{H}^{\bullet}(\mathfrak{X}) \longrightarrow \mathbf{H}^{\bullet}(\mathfrak{X}) \otimes_k \mathbf{Nov}_{\mathbf{H}^{\bullet}(\mathfrak{X})}$$

where to simplify notation we write $\mathbf{H}^{\bullet}(\mathfrak{X})$ instead of $\mathbf{H}^{\bullet}([\mathfrak{X}])$. By definition we have

- $\mathbf{H}^\bullet(\mathfrak{X}) = \mathfrak{h}(H_B^\bullet(\mathfrak{X}))$ is a $\mathbb{Z}/2$ -graded finite dimensional k -vector space equipped with an action of G ;
- $\text{Nov}_{\mathbf{H}^\bullet(\mathfrak{X})}$ is a complete super commutative k -algebra equipped with an action of G ;
- $\star$ is the image of the quantum product (3.41) under the functor $\mathfrak{h}$.

Remark 3.44. The G -equivariant variant $\text{Nov}_{\mathbf{H}^\bullet(\mathfrak{X})}$ of the big Novikov algebra is defined in the obvious way. If we write $\mathbb{M} = \langle q^\beta \mid \beta \in \text{NE}(X, \mathbb{Z}), \beta \neq 0 \rangle$ for the maximal ideal of the monoid algebra $\mathbb{Q}[\text{NE}(X, \mathbb{Z})]$ and $\mathfrak{m} \subset \text{Sym } H_B^\bullet(\mathfrak{X})^\vee$ for the maximal ideal of the origin in the $\mathbb{Q}$ vector space $H_B^\bullet(\mathfrak{X})$, then we have

$$\text{Nov}_{\mathfrak{X}} = \varinjlim_{a,b} (\mathbb{Q}[\text{NE}(X, \mathbb{Z})]/\mathbb{M}^a) \bigotimes_{\mathbb{Q}} ((\text{Sym } H_B^\bullet(\mathfrak{X})^\vee)/\mathfrak{m}^b),$$

as a limit of finite dimensional $\text{Mot}^{\mathcal{K}}$ -equivariant $\mathbb{Q}$ -algebras. Applying the functor $\mathfrak{h}$ yields

$$\mathfrak{h}((\text{Sym } H_B^\bullet(\mathfrak{X})^\vee)/\mathfrak{m}^b) = (\text{Sym}_k \mathbf{H}^\bullet(\mathfrak{X})^\vee)/\mathfrak{m}_k^b,$$

where $\mathfrak{m}_k$ is the maximal ideal in the polynomial algebra $\text{Sym } \mathbf{H}^\bullet(\mathfrak{X})^\vee$ and the right hand side is equipped with the action of G induced from the G action on $\mathbf{H}^\bullet(\mathfrak{X})$.

Thus we set

$$\begin{aligned} \text{Nov}_{\mathbf{H}^\bullet(\mathfrak{X})} &:= \varinjlim_{a,b} \mathfrak{h}(\mathbb{Q}[\text{NE}(X, \mathbb{Z})]/\mathbb{M}^a) \bigotimes_k ((\text{Sym}_k \mathbf{H}^\bullet(\mathfrak{X})^\vee)/\mathfrak{m}_k^b) \\ &= \left(\varinjlim_a \mathfrak{h}(\mathbb{Q}[\text{NE}(X, \mathbb{Z})]/\mathbb{M}^a) \right) \widehat{\bigotimes}_k (\widehat{\text{Sym}_k \mathbf{H}^\bullet(\mathfrak{X})^\vee}) \end{aligned}$$

To understand the G -equivariant k -algebra $\varinjlim_a \mathfrak{h}(\mathbb{Q}[\text{NE}(X, \mathbb{Z})]/\mathbb{M}^a)$ better, we will need to look at the way $\mathfrak{h}$ transports the rational $\text{Mot}^{\mathcal{K}}$ -representations $\mathbb{Q}[\text{NE}(X, \mathbb{Z})]/\mathbb{M}^a$ to k -linear G -representations. The transport mechanism depends on a relationship between the two Galois groups $\text{Gal}(\overline{\mathcal{K}}/\mathcal{K})$ and $\text{Gal}(\overline{k}/k)$ which we describe next.

3.5.2.2.3 Galois twists Composing the functor $\mathfrak{h}: \text{Rep}_{\mathbb{Q}}(\text{Mot}^{\mathcal{K}}) \rightarrow \text{Rep}_k(G)$ with the inclusion

$$\begin{array}{ccc} \text{Rep}_{\mathbb{Q}}(\text{Gal}(\overline{\mathcal{K}}/\mathcal{K})) & \hookrightarrow & \text{Rep}_{\mathbb{Q}}(\text{Mot}^{\mathcal{K}}) \\ \parallel & & \parallel \\ \text{Art}^{\mathcal{K}} & \hookrightarrow & \mathcal{C}^{\mathcal{K}}/\text{Tate} \end{array}$$

we get a tensor functor

$$(3.45) \quad \text{Rep}_{\mathbb{Q}}(\text{Gal}(\overline{\mathcal{K}}/\mathcal{K})) \longrightarrow \text{Rep}_k(G).$$

Hence, for any smooth projective $\mathcal{K}$ -variety $\mathfrak{X}$ we can compose the pullback functor

$$\text{Rep}_{\mathbb{Q}}(\Gamma_{\mathfrak{X}}) \longrightarrow \text{Rep}_{\mathbb{Q}}(\text{Gal}(\overline{\mathcal{K}}/\mathcal{K}))$$

with the functor (3.45) to get a tensor functor

$$(3.46) \quad \mathfrak{h}_{\mathfrak{X}}: \mathrm{Rep}_{\mathbb{Q}}(\Gamma_{\mathfrak{X}}) \longrightarrow \mathrm{Rep}_k(G).$$

Such a tensor functor can be viewed as homomorphism from G to a $\mathrm{Gal}(\bar{k}/k)$ -twisted form of $\Gamma_{\mathfrak{X}}$. Indeed, let us fix an algebraic closure $k \subset \bar{k}$. Suppose we are given a homomorphism of profinite groups

$$\varphi: \mathrm{Gal}(\bar{k}/k) \rightarrow \Gamma.$$

Such a φ gives rise to

(a) a tensor functor

$$F_{\varphi}: \mathrm{Rep}_{\mathbb{Q}}(\Gamma) \longrightarrow \mathrm{Vect}_k,$$

which to every $\mathbb{Q}$ -representation $\lambda: \Gamma \rightarrow \mathrm{GL}(V)$ assigns the k -vector space

$$(V \otimes_{\mathbb{Q}} \bar{k})^{\mathrm{Gal}(\bar{k}/k)},$$

where $\mathrm{Gal}(\bar{k}/k)$ acts on V via the homomorphism $\lambda \circ \varphi: \mathrm{Gal}(\bar{k}/k) \rightarrow \mathrm{GL}(V)$, and acts on $\bar{k}$ tautologically.

(b) a φ -twisted (inner) form of Γ , i.e. a finite group scheme Γ^{φ} over $\mathrm{Spec} k$, defined by

$$\Gamma^{\varphi} := \mathrm{Spec} \bar{k} \times_{\mathrm{Ad} \circ \varphi} \Gamma.$$

(c) A canonical functorial action

$$\mathrm{can}_{\varphi}(\lambda): \Gamma^{\varphi} \rightarrow \mathrm{GL}(F_{\varphi}(\lambda))$$

of Γ^{φ} on $F_{\varphi}(\lambda)$ for all $\mathbb{Q}$ -representations $\lambda: \Gamma \rightarrow \mathrm{GL}(V)$ of Γ . The action $\mathrm{can}_{\varphi}(\lambda)$ is the $\mathrm{Gal}(\bar{k}/k)$ -descent of the extension $\lambda \otimes \bar{k}: \Gamma \rightarrow \mathrm{GL}(V \otimes \bar{k})$ of λ by $\bar{k}$ -linearity.

With this notation we have the following

Lemma 3.47. *The assignment $\varphi \mapsto F_{\varphi}$ gives an equivalence of groupoids:*

$$\left(\begin{array}{l} \text{group homomorphisms} \\ \mathrm{Gal}(\bar{k}/k) \rightarrow \Gamma, \text{ with maps} \\ \text{given by conjugation by} \\ \text{elements in } \Gamma \end{array} \right) \xrightarrow{\cong} \left(\begin{array}{l} \text{tensor functors } \mathrm{Rep}_{\mathbb{Q}}(\Gamma) \rightarrow \mathrm{Vect}_k \\ \text{with maps given by isomorphisms} \\ \text{of functors} \end{array} \right).$$

Proof. We can construct the inverse functor as follows. Let $\nu: \mathrm{Rep}_{\mathbb{Q}}(\Gamma) \rightarrow \mathrm{Vect}_k$ be a fiber functor and let $\omega: \mathrm{Rep}_{\mathbb{Q}}(\Gamma) \rightarrow \mathrm{Vect}_{\mathbb{Q}}$ denote the forgetful functor. By the classification [63, Theorem II.8.1] of fiber functors in Tannaka duality, we know that the functor $A \mapsto \mathrm{Isom}^{\otimes}(\omega \otimes A, \nu)$ from commutative k -algebras to sets is represented by an affine scheme $\mathbf{I}(\omega \otimes k, \nu)$ which is a right fppf torsor over Γ viewed as a group scheme over k . Such torsors are classified by $H^1(\mathrm{Gal}(\bar{k}/k), \Gamma)$, i.e. the groupoid of such torsors is the groupoid with objects given by Γ valued 1-cocycles of $\mathrm{Gal}(\bar{k}/k)$ and morphisms given by coboundaries. But the 1-cocycles are simply homomorphisms $\varphi: \mathrm{Gal}(\bar{k}/k) \rightarrow \Gamma$ and the maps between cocycles are given by conjugations by elements in Γ . This proves the lemma. $\square$

Remark 3.48. Let $\nu: \text{Rep}_{\mathbb{Q}}(\Gamma) \rightarrow \text{Vect}_k$. The group scheme $\mathbf{I}(\omega \otimes k, \nu) \times_{\text{Ad}} \Gamma$ of automorphisms of the Γ -torsor $\mathbf{I}(\omega \otimes k, \nu)$ is an inner twisted form of Γ as a group scheme over $\text{Spec } k$ and $\mathbf{I}(\omega \otimes k, \nu)$ is a left fppf torsor over this group scheme. If ν corresponds to a homomorphism $\varphi: \text{Gal}(\bar{k}/k) \rightarrow \Gamma$, then this group scheme is canonically identified with Γ^φ .

The previous lemma has an obvious refinement. Indeed, suppose that in addition to the homomorphism φ we are given also a homomorphism $\psi: G \rightarrow \Gamma^\varphi$ of group schemes over k . Proposition with ψ gives a canonical action

$$\text{can}_\varphi(\lambda) \circ \psi: G \rightarrow \text{GL}(F_\varphi(\lambda))$$

of G on the k -vector space $F_\varphi(\lambda)$. This gives rise to a tensor functor

$$F_{\varphi, \psi}: \text{Rep}_{\mathbb{Q}}(\Gamma) \longrightarrow \text{Rep}_k(G), \quad \lambda \mapsto \text{can}_\varphi(\lambda) \circ \psi.$$

Also, if $\gamma \in \Gamma$, and we consider the γ -conjugate homomorphism

$$\varphi^\gamma := \text{Ad}_\gamma \circ \varphi: \text{Gal}(\bar{k}/k) \rightarrow \Gamma,$$

then the inner automorphism

$$\text{Ad}_\gamma: \Gamma \rightarrow \Gamma$$

intertwines the $\text{Gal}(\bar{k}/k)$ action $\text{Ad} \circ \varphi$ on the source copy of Γ with the $\text{Gal}(\bar{k}/k)$ action $\text{Ad} \circ \varphi^\gamma$ on the source copy of Γ .

In particular $\text{Ad}_\gamma: \Gamma \rightarrow \Gamma$, descends to a homomorphism

$$\mathbf{a}_\gamma: \Gamma^\varphi \rightarrow \Gamma^{\varphi^\gamma}$$

of finite group schemes over $\text{Spec } k$, and so we get an action of the group Γ on the set of pairs of homomorphisms (φ, ψ) given by

$$(3.49) \quad (\varphi, \psi) \rightarrow (\varphi^\gamma, \mathbf{a}_\gamma \circ \psi).$$

With this notations we now have the following

Lemma 3.50. *the assignment $(\varphi, \psi) \mapsto F_{\varphi, \psi}$ gives an equivalence of groupoids:*

$$\left(\begin{array}{l} \text{pairs of group homomorphisms} \\ (\varphi, \psi) \text{ with maps given by (3.49)} \end{array} \right) \xrightarrow{\cong} \left(\begin{array}{l} \text{tensor functors } \text{Rep}_{\mathbb{Q}}(\Gamma) \longrightarrow \text{Rep}_k(G), \\ \text{with maps given by isomorphisms of} \\ \text{functors.} \end{array} \right).$$

Proof. As in the proof of Lemma 3.47 we can construct the inverse functor explicitly. If $\nu: \text{Rep}_{\mathbb{Q}}(\Gamma) \rightarrow \text{Rep}_k(G)$ is a fiber functor, we can compose it with the forgetful functor $\text{for}: \text{Rep}_k(G) \rightarrow \text{Vect}_k$ to get a fiber functor $\text{for} \circ \nu: \text{Rep}_{\mathbb{Q}}(\Gamma) \rightarrow \text{Vect}_k$. As in the proof of Lemma 2.1 we consider the k -scheme $\mathbf{I}(\omega \otimes k, \text{for} \circ \nu)$. It is a right torsor over Γ viewed as a group scheme over $\text{Spec } k$. By construction, the group scheme G acts on the left on $\mathbf{I}(\omega \otimes k, \text{for} \circ \nu)$ by torsor automorphisms. In other words, we have a homomorphism from G to the group scheme of torsor automorphisms of $\mathbf{I}(\omega \otimes k, \text{for} \circ \nu)$. As explained in Remark 3.48 this group scheme $\mathbf{I}(\omega \otimes k, \text{for} \circ \nu) \times_{\text{Ad}} \Gamma$ is an inner form of Γ as a group scheme over $\text{Spec } k$, and when $\text{for} \circ \nu = F_\varphi$, it is equal to the twist Γ^φ . Thus we get a homomorphism $\psi: G \rightarrow \Gamma^\varphi$ and an identification $\nu = F_{\varphi, \psi}$. This proves the lemma. $\square$

Now observe that the $\text{Mot}^{\mathcal{K}}$ action on $\mathbb{Q}[\text{NE}(X, \mathbb{Z})]/\mathbb{M}^a$ is induced from the $\text{Mot}^{\mathcal{K}}$ -action on $\text{NE}(X, \mathbb{Z})$ which, as we explained above, factors through the homomorphism $\text{Mot}^{\mathcal{K}} \rightarrow \text{Gal}(\bar{\mathcal{K}}/\mathcal{K}) \rightarrow \Gamma_{\mathfrak{X}}$ and the natural action of $\Gamma_{\mathfrak{X}}$ on the lattice $\text{NE}(X, \mathbb{Z}) \cong \mathbb{Z}^{\rho}$. By the previous discussion, the tensor functor

$$\mathfrak{h}_{\mathfrak{X}} : \text{Rep}_{\mathbb{Q}}(\Gamma_{\mathfrak{X}}) \rightarrow \text{Rep}_k(G)$$

corresponds to a pair of homomorphisms $\varphi : \text{Gal}(\bar{k}/k) \rightarrow \Gamma_{\mathfrak{X}}$, and $\psi : G \rightarrow \Gamma_{\mathfrak{X}}^{\varphi}$. In particular we see that as a k -vector space we have

$$\lim_{\substack{\longrightarrow \\ a}} \mathfrak{h}((\mathbb{Q}[\text{NE}(X, \mathbb{Z})]/\mathbb{M}^a) = \bar{k}[\text{NE}(X, \mathbb{Z})]^{\text{Gal}(\bar{k}/k)},$$

where the $\text{Gal}(\bar{k}/k)$ acts tautologically on $\bar{k}$ and acts on $\text{NE}(X, \mathbb{Z})$ via $\varphi : \text{Gal}(\bar{k}/k) \rightarrow \Gamma_{\mathfrak{X}}$. The action of G on $\bar{k}[\text{NE}(X, \mathbb{Z})]^{\text{Gal}(\bar{k}/k)}$ is given by the composition of $\psi : G \rightarrow \Gamma_{\mathfrak{X}}^{\varphi}$ and the canonical action of $\Gamma_{\mathfrak{X}}^{\varphi}$ on $\bar{k}[\text{NE}(X, \mathbb{Z})]^{\text{Gal}(\bar{k}/k)}$.

3.5.2.2.4 Weil cohomology realization of the A-model F-bundle Suppose we are in the above setup. That is we have two fields $\mathcal{K}$ and k , both of characteristic zero, and we have fixed an inclusion $\mathcal{K} \subset \mathbb{C}$ and an algebraic closure $k \subset \bar{k}$. Let G be a given proreductive group over k with a fixed central involution ϵ_G and suppose we are given a (G, ϵ_G) -symmetric Weil cohomology theory on $\mathcal{K}$ -varieties, i.e. a tensor functor

$$\mathfrak{h} : \text{Rep}_{\mathbb{Q}}(\text{Mot}^{\mathcal{K}}) \longrightarrow \text{Rep}_k(G).$$

With these data we can associate a natural $\mathbb{Z}/2$ -graded k -vector space equipped with a G -action, namely $\mathbf{H}^{\bullet}(\mathfrak{X}) := \mathfrak{h}(H_B^{\bullet}(\mathfrak{X}))$. This vector space will serve as a fiber of our F-bundle. As explained in section 2, $\mathbf{H}^{\bullet}(\mathfrak{X})$ comes equipped with a super commutative, associative, G -equivariant quantum product (3.43).

Recall that the Betti cohomology $H_B^{\bullet}(\mathfrak{X})$ is a $\mathbb{Q}$ -representation of $\text{Mot}^{\mathcal{K}}$ which decomposes naturally into a direct sum of rational $\text{Mot}^{\mathcal{K}}$ -subrepresentations

$$H_B^{\bullet}(\mathfrak{X}) = H_B^2(\mathfrak{X})_{\bar{\mathcal{K}}-\text{alg}} \oplus H_B^2(\mathfrak{X})_{\bar{\mathcal{K}}-\text{transc}} \oplus H_B^{*\neq 2}(\mathfrak{X}).$$

so that the action of $\text{Mot}^{\mathcal{K}}$ on $H_B^2(\mathfrak{X})_{\bar{\mathcal{K}}-\text{alg}}$ factors through the finite quotient $\text{Mot}^{\mathcal{K}} \twoheadrightarrow \Gamma_{\mathfrak{X}}$.

As explained in Section 2, the choice of the fiber functor $\mathfrak{h}$ gives rise to a tensor functor

$$\mathfrak{h}_{\mathfrak{X}} : \text{Rep}_{\mathbb{Q}}(\Gamma_{\mathfrak{X}}) \longrightarrow \text{Rep}_k(G),$$

which is equivalent to specifying a pair of homomorphisms $\varphi : \text{Gal}(\bar{k}/k) \rightarrow \Gamma_{\mathfrak{X}}$ and $\psi : G \rightarrow \Gamma_{\mathfrak{X}}^{\varphi}$. Using this we define the following k -representations of G :

- $\mathbf{H}_{\text{alg}}^2(\mathfrak{X}) := \mathfrak{h}_{\mathfrak{X}}(H_B^2(\mathfrak{X})_{\bar{\mathcal{K}}-\text{alg}})$. Explicitly, this k -linear G -representation is given by the k -vector space $(\text{NS}(X, \mathbb{Q}) \otimes \bar{k})^{\text{Gal}(\bar{k}/k)}$, where the $\text{Gal}(\bar{k}/k)$ -action on $\text{NS}(X, \mathbb{Q})$ is given by φ while the $\text{Gal}(\bar{k}/k)$ -action on $\bar{k}$ is tautological. The G -action on this vector space is given by the homomorphism $\psi : G \rightarrow \Gamma_{\mathfrak{X}}^{\varphi}$ and the natural action of $\Gamma_{\mathfrak{X}}^{\varphi}$ on the vector space $(\text{NS}(X, \mathbb{Q}) \otimes \bar{k})^{\text{Gal}(\bar{k}/k)}$.
- $\mathbf{H}_{\text{transc}}^2(\mathfrak{X}) := \mathfrak{h}(H_B^2(\mathfrak{X})_{\text{transc}}) \subset \mathbf{H}^2$

$$\bullet \mathbf{H}^{*\neq 2}(\mathfrak{X}) := \mathfrak{h}\left(H_B^{*\neq 2}(\mathfrak{X})\right).$$

Altogether we get a decomposition of $\mathbf{H}^\bullet(\mathfrak{X})$ as a direct sum of G -representations:

$$(3.51) \quad \mathbf{H}^\bullet(\mathfrak{X}) := \mathbf{H}_{\text{alg}}^2(\mathfrak{X}) \oplus \mathbf{H}_{\text{transc}}^2(\mathfrak{X}) \oplus \mathbf{H}^{*\neq 2}(\mathfrak{X}).$$

Passing to analytifications and choosing suitable admissible domains in the pieces of this decomposition will allow us to describe the base of the A-model F-bundle for $\mathfrak{X}$.

Concretely, let

$$\mathbf{N}_1(X, \mathbb{Z}) = \text{CH}_1^{\text{hom}}(X) = \text{im} [\text{CH}_1 \rightarrow H_2((X \times_{\overline{K}} \mathbb{C})^{\text{an}}, \mathbb{Q})]$$

denote the group of cycle classes of algebraic curves in X . This is again a free abelian group of rank ρ equipped with a $\Gamma_{\mathfrak{X}}$ -action. The cap product between singular homology and cohomology on $(X \times_{\overline{K}} \mathbb{C})^{\text{an}}$ identifies the group $\mathbf{N}_1(X, \mathbb{Z})$ with the dual of the Néron-Severi lattice $\text{NS}(X, \mathbb{Z})_{\text{tf}}$ and this identification is compatible with the natural $\Gamma_{\mathfrak{X}}$ -actions.

Additionally, $\mathbf{N}_1(X, \mathbb{Z})$ contains the $\Gamma_{\mathfrak{X}}$ -stable sub semigroup

$$\text{NE}(X, \mathbb{Z}) \subset \mathbf{N}_1(X, \mathbb{Z})$$

of numerically effective curve classes in X .

The homomorphisms φ and ψ define a non-split affine torus $\mathcal{T}(\mathfrak{X})$ over k on which G acts by automorphisms. Here $\mathcal{T}(\mathfrak{X}) := (\text{NS}(X, \mathbb{Z})_{\text{tf}} \otimes \mathbb{G}_{m, \overline{k}})^{\text{Gal}(\overline{k}/k)}$, where $\mathbb{G}_{m, \overline{k}}$ denotes the pullback of the multiplicative group scheme to $\overline{k}$, $\text{Gal}(\overline{k}/k)$ acts tautologically on $\mathbb{G}_{m, \overline{k}}$, and acts on $\text{NS}(X, \mathbb{Z})_{\text{tf}}$ via the homomorphism $\varphi: \text{Gal}(\overline{k}/k) \rightarrow \Gamma_{\mathfrak{X}}$. As a group scheme over k this torus is equipped with an action of the finite group scheme $\Gamma_{\mathfrak{X}}^\varphi$, and the G action on $\mathcal{T}(\mathfrak{X})$ is given by $\psi: G \rightarrow \Gamma_{\mathfrak{X}}^\varphi$.

Now consider the non-archimedean field $k(\langle \mathbf{y} \rangle)$ of Laurent series in an auxiliary variable $\mathbf{y}$. Let $\mathcal{T}_{k(\langle \mathbf{y} \rangle)}(\mathfrak{X})$ denote the base change of $\mathcal{T}(\mathfrak{X})$ to an affine (still non-split) torus over $k(\langle \mathbf{y} \rangle)$, and let $\mathcal{T}_{k(\langle \mathbf{y} \rangle)}(\mathfrak{X})^{\text{an}}$ be the non-archimedean analytification of the torus $\mathcal{T}_{k(\langle \mathbf{y} \rangle)}(\mathfrak{X})$. Following the same steps as in the construction of the analytic A-model F-bundle for a variety over $\overline{K}$, we define the following rigid analytic spaces over $k(\langle \mathbf{y} \rangle)$:

$B_{\mathfrak{X}, q} \subset \mathcal{T}_{k(\langle \mathbf{y} \rangle)}(\mathfrak{X})$ is the preimage of the ample cone in $\text{NS}(X, \mathbb{R})$ under the valuation map $\mathcal{T}_{k(\langle \mathbf{y} \rangle)}^{\text{an}}(\mathfrak{X}) \rightarrow \text{NS}(X, \mathbb{R})$.

$\mathcal{B}_{\mathfrak{X}, t}^{\text{ev}}$ is the product of the analytic affine line $(\mathbf{H}^0(\mathfrak{X}) \otimes k(\langle \mathbf{y} \rangle))^{\text{an}}$ with an open unit polydisk in the analytification of the $k(\langle \mathbf{y} \rangle)$ -affine space $\mathbf{H}^{2\bullet \geq 2}(\mathfrak{X}) \otimes k(\langle \mathbf{y} \rangle)$

$B_{\mathfrak{X}, t}^{\text{ev}}$ is the product of the analytic affine line $(\mathbf{H}^0(\mathfrak{X}) \otimes k(\langle \mathbf{y} \rangle))^{\text{an}}$ with an open unit polydisk in the analytification of the $k(\langle \mathbf{y} \rangle)$ -affine space $(\mathbf{H}_{\text{transc}}^2(\mathfrak{X}) \oplus \mathbf{H}^{2\bullet > 2}(\mathfrak{X})) \otimes_k k(\langle \mathbf{y} \rangle)$.

$B_{\mathfrak{X}}^{\text{odd}}$ is the super analytic variety corresponding to the purely odd vector space $\mathbf{H}^{\text{odd}}(\mathfrak{X}) \otimes_k k(\langle \mathbf{y} \rangle)$.

Multiplying these together we get two $k(\langle \mathbf{y} \rangle)$ -analytic spaces

$$\mathcal{B}_{\mathfrak{X}} := B_{\mathfrak{X}, q} \times \mathcal{B}_{\mathfrak{X}, t}^{\text{ev}} \times B_{\mathfrak{X}}^{\text{odd}} \quad \text{and} \quad B_{\mathfrak{X}} := B_{\mathfrak{X}, q} \times B_{\mathfrak{X}, t}^{\text{ev}} \times B_{\mathfrak{X}}^{\text{odd}},$$

which are G -stable admissible opens in the $k(\mathbf{y})$ -analytic varieties $\mathcal{T}_{k(\mathbf{y})}(\mathfrak{X})^{\text{an}} \times (\mathbf{H}^\bullet(\mathfrak{X}) \otimes_k k(\mathbf{y}))^{\text{an}}$ and $\mathcal{T}_{k(\mathbf{y})}(\mathfrak{X})^{\text{an}} \times ((\mathbf{H}^2(\mathfrak{X})_{\text{transc}} \oplus \mathbf{H}^{\bullet \neq 2}(\mathfrak{X})) \otimes_k k(\mathbf{y}))^{\text{an}}$ respectively. In particular, we can view $\mathcal{B}_{\mathfrak{X}}$ as a non-archimedean replacement of the formal scheme $\text{Spf Nov}_{\mathfrak{X}}$.

Let D denote the germ at 0 in a $k(\mathbf{y})$ -analytic disk with coordinate u . $\mathcal{H}$ denote the trivial analytic super vector bundle on $\mathcal{B}_{\mathfrak{X}} \times D$ with fiber $\mathbf{H}^\bullet(\mathfrak{X}) \otimes k(\mathbf{y})$. Then, the same argument that we used to prove Lemma 3.29 implies that the formal G -equivariant quantum product (3.43) defines an analytic quantum product

$$\star : \mathcal{H} \otimes \mathcal{H} \longrightarrow \mathcal{H}$$

which is moreover independent of u . Using this product we again define a quantum connection, i.e. a meromorphic flat connection ∇ on $\mathcal{H}$ given by the usual formulas

$$\begin{cases} \nabla_{u\partial_u} = u\partial_u - u^{-1} \text{Eu} \star (-) + \text{Gr}, \\ \nabla_{\partial_{t_i}} = \partial_{t_i} + u^{-1} T_i \star (-), & \text{for } \tau \in \mathbf{H}^\bullet(\mathfrak{X}) \otimes k(\mathbf{y}), \\ \nabla_{\xi q \partial_q} = \xi q \partial_q + u^{-1} \xi \star (-), & \text{for } \xi \in \mathbf{H}^2(\mathfrak{X}) \otimes k(\mathbf{y}), \end{cases}$$

where Eu is the Euler vector field, Gr is the grading operator, and $\xi q \partial_q$ is the derivation of $\mathcal{O}_{\mathcal{B}_{\mathfrak{X}}}$ given by $\xi q \partial_q(q^\beta) = (\beta \cdot \xi) q^\beta$.

Again $(\mathcal{H}, \nabla)/\mathcal{B}_{\mathfrak{X}}$ is an over maximal F-bundle over $\mathcal{B}_{\mathfrak{X}}$ and $B_{\mathfrak{X}} \subset \mathcal{B}_{\mathfrak{X}}$ is a closed analytic subvariety which meets transversally every leaf through a rigid point of the redundancy foliation of $(\mathcal{H}, \nabla)/\mathcal{B}_{\mathfrak{X}}$. Thus the restriction of $(\mathcal{H}, \nabla)|_{B_{\mathfrak{X}} \times D}$ is the associated maximal F-bundle $(\mathcal{H}, \nabla)/B_{\mathfrak{X}}$ over $B_{\mathfrak{X}}$.

Definition 3.52. The F-bundle $(\mathcal{H}, \nabla)/\mathcal{B}_{\mathfrak{X}}$ will be called the *non-archimedean overmaximal A-model F-bundle associated to $\mathfrak{X}$, $\mathcal{K}$, k , and $\mathbf{H}^\bullet$* . The F-bundle $(\mathcal{H}, \nabla)/B_{\mathfrak{X}}$ will be called the *non-archimedean maximal A-model F-bundle associated to $\mathfrak{X}$, $\mathcal{K}$, k , and $\mathbf{H}^\bullet$* .

Remark 3.53. For the applications to atoms we will frequently base change the spaces $\mathcal{B}_{\mathfrak{X}}$ and $B_{\mathfrak{X}}$, and the corresponding F-bundles, to analytic spaces, and analytic F-bundles to an algebraically closed field $\mathbb{k}$ containing $k(\mathbf{y})$, e.g. $\mathbb{k} = \overline{k}(\mathbf{y}^{\mathbb{Q}})$. We do this to ensure that the quantum multiplication by the Euler vector field has well defined eigenvalues. If however the eigenvalues of $\text{Eu} \star (-)$ exist over $k(\mathbf{y})$ one can work with the analytic F-bundles over the smaller field $k(\mathbf{y})$ and get finer obstructions to rationality. Note that the process of base changing to $\mathbb{k}$ definitely loses information, e.g. the base changed torus $\mathcal{T}_{\mathbb{k}}(\mathfrak{X})$ is automatically split.

4 Decomposition theorems

In this section we discuss the basic decomposition theorems for F-bundles. First we recall the spectral decomposition theorem, which works for arbitrary F-bundles. After that we discuss the blowup and projective bundle decomposition theorems that are specific to A-model F-bundles associated to smooth projective varieties.

4.1 Spectral decomposition theorem

Suppose $\mathbb{k}$ is a non-archimedean field of characteristic zero.

Theorem 4.1. *Let B be an admissible open neighborhood of a rigid point b in a $\mathbb{k}$ -analytic affine space, and let $(\mathcal{H}, \nabla)/B$ be an F -bundle over B maximal at b . Assume that we have a decomposition $\mathcal{H}_{b,0} \simeq \bigoplus_{i \in I} H_i$ stable under $\kappa_b = \text{Eu}_b \circ (-)$, and that for any $i \neq j \in I$, the spectra of $\kappa_b|_{H_i}$ and $\kappa_b|_{H_j}$ are disjoint. Then there exists an admissible open neighborhood U of b such that the restricted F -bundle $(\mathcal{H}, \nabla)/U$ decomposes into an external sum of maximal F -bundles $(\mathcal{H}_i, \nabla_i)/U_i$ extending the decomposition of $\mathcal{H}_{b,0}$.*

Proof. See [38, Theorem 1.2]. $\square$

Remark 4.2. The spectral decomposition theorem has obvious versions that hold for formal or $\mathbb{C}$ -analytic F -bundles. Indeed, let $(\mathcal{H}, \nabla)/B$ be a formal maximal F -bundle over an algebraically closed field k of characteristic 0, let $b \in B$ be a closed point, and let $\{\lambda_i(b)\}_{i=1}^s \in k$ be the distinct eigenvalues of κ_b . Then there exists a formal neighborhood U of b in B and maps $\lambda_i: U \rightarrow \text{Spf } k[[t - \lambda_i(b)]]$, sending b to $\lambda_i(b)$, so that λ_i are the distinct eigenvalues of $\kappa|_U$, and $(\mathcal{H}, \nabla)/U$ decomposes canonically into an exterior sum of formal F bundles corresponding to the generalized eigenbundles of $\kappa|_U$. Similarly, if $(\mathcal{H}, \nabla)/B$ is a de Rham part of a nc variation of Hodge structures, i.e. a complex analytic F -bundle, then for any point $b \in B$ there exist disjoint small disk $\mathbf{D}_i \subset \mathbb{C}$, centered at the distinct eigenvalues of κ_b , and $\mathbb{C}$ -analytic functions $\lambda_i: U \rightarrow \mathbf{D}_i \subset \mathbb{C}$ from a small neighborhood U of b in B , so that λ_i are the distinct eigenvalues of κ_U and a decomposition of $(\mathcal{H}, \nabla)/U$ into an external sum of F -bundles $(\mathcal{H}_i, \nabla_i)/U_i$ where U_i is a complex manifold, $(\mathcal{H}_i, \nabla_i)$ is a bundle with connection on $U_i \times \text{Spf } \mathbb{C}[[u]]$, and the decomposition isomorphism is $\mathbb{C}$ -analytic in the U direction and only formal in the u -direction.

4.2 The blowup formula

Let X be a smooth complex projective variety, and $Z \subset X$ a smooth subvariety of pure codimension $r \geq 2$. Let $\hat{X}$ be the blowup of X along Z , and let X' be the disjoint union of X with $(r-1)$ copies of Z . We have A-model maximal F -bundles $(\hat{\mathcal{H}}, \hat{\nabla})/B_{\hat{X}}$, and $(\mathcal{H}', \nabla')/B_{X'}$. The *blowup decomposition* theorem identifies the germs of these F -bundles at appropriately chosen base points. It is extracted from Iritani's blowup formula [44] which gives the same identification for the formal A-model logarithmic F -bundles of $\hat{X}$ and X' .

Before we give a precise formulation and proof of the blowup decomposition, let us look more closely at its shape and structure. The blowup of Z in X gives rise to a standard fibered square

$$(4.3) \quad \begin{array}{ccc} D & \xrightarrow{j} & \hat{X} \\ \pi_D \downarrow & & \downarrow \pi \\ Z & \xrightarrow{\iota} & X \end{array}$$

where $\iota: Z \hookrightarrow X$ is the inclusion, $\pi: \hat{X} \rightarrow X$ is the blowup map, $j: D \hookrightarrow \hat{X}$ is the exceptional divisor, and $\pi_D: D = \mathbb{P}(N_{Z/X}) \rightarrow Z$ is the natural projection. Write $\mathbf{n}: H^\bullet(D, \mathbb{Z}) \rightarrow H^\bullet(D, \mathbb{Z})$ for the operator of (classical) multiplication by $c_1(\mathcal{O}_D(1)) = -c_1(N_{D/\hat{X}})$. Then, there is a natural identification of $\mathbb{Z}/2$ -graded cohomology spaces

$$(4.4) \quad H^\bullet(X', \mathbb{Q}) = H^\bullet(X, \mathbb{Q}) \bigoplus \underbrace{H^\bullet(Z, \mathbb{Q}) \bigoplus \cdots \bigoplus H^\bullet(Z, \mathbb{Q})}_{(r-1)\text{-copies}} \xrightarrow{\sim} H^\bullet(\hat{X}, \mathbb{Q})$$

given by

$$(\alpha, \beta_1, \dots, \beta_{r-1}) \mapsto \pi^*(\alpha) - j_*\pi_D^*(\beta_1) - j_*\mathbf{n}\pi_D^*(\beta_2) - \dots - j_*\mathbf{n}^{r-2}\pi_D^*(\beta_{r-1}).$$

To simplify the discussion suppose $\mathcal{K} = \mathbb{C}$, $k = \mathbb{Q}$, $\mathbb{k} = \overline{\mathbb{Q}}((\mathbf{y}^{\mathbb{Q}}))$, $H^\bullet(Y) = H_B^\bullet(Y(\mathbb{C})^{\text{an}}, \mathbb{Q})$ for any complex projective manifold Y . Let $\text{Amp}(Y) = \text{int}(\text{Nef}(Y)) \subset H^2(Y, \mathbb{R})$ denote the ample cone of Y . Then for any point $\theta \in \text{Amp}(Y) \cap H^2(X, \mathbb{Q})$ we get a rigid point in $B_{Y,q}$ which is given by the homomorphism $\text{NE}(X, \mathbb{Z}) \rightarrow \mathbb{k}^\times$, $\beta \rightarrow \mathbf{y}^{(\beta, \theta)} \in \mathbb{k}^\times$. We will denote this rigid point by $\mathbf{y}^\theta \in B_{Y,q}$.

Let $\omega \in H^2(X, \mathbb{Z})$ be an ample class on X , and let $\hat{\omega} \in H^2(\hat{X}, \mathbb{Q})$ be a class of the form $\hat{\omega} := \epsilon^{-1}\varphi^*\omega - K_X$ for some $\epsilon \in (0, 1) \cap \mathbb{Q}$. For a small enough $\epsilon > 0$ the class $\hat{\omega}$ is ample on $\hat{X}$. Then for every positive $\epsilon \ll 1$ we can consider the point $\hat{Q} = \mathbf{y}^{\hat{\omega}} \in B_{\hat{X},q}$. This is a special point in $B_{\hat{X},q} \subset B_{\hat{X}}$ which is close to the boundary of the tube domain $B_{\hat{X},q}$ and whose valuation is close to ∞ .

In the limit $\epsilon \rightarrow 0$ only stable maps that are vertical for the map $\pi: \hat{X} \rightarrow X$ contribute to the quantum product on $\hat{X}$. Furthermore, in this limit, only points and vertical lines will contribute to the quantum product and the contribution of lines will be $\mathbf{y}^{r-1}$. Consider the operator $\hat{\kappa}$ which is the quantum multiplication in $H^\bullet(\hat{X}, \mathbb{k})$ with the Euler vector field $\text{Eu}_{\hat{X}, \hat{Q}}$, computed at the special point $\hat{Q}$. A direct calculation shows that in the limit $\epsilon \rightarrow 0$ the operator $\hat{k}$ has the following block form with respect to the decomposition (4.4).

$$\hat{\kappa} = \left(\begin{array}{c|c|c|c|c|c|c} \frac{K_X}{-(r-1)\iota^*} & \frac{K_Z - c_1(N)}{-(r-1)} & & & & & \frac{(r-1)\iota_*}{(r-1)(\mathbf{y}^{r-1} + c_{r-1}(N))} \\ \hline & \frac{K_Z - c_1(N)}{-(r-1)} & & & & & \frac{(r-1)c_{r-2}(N)}{(r-1)c_{r-3}(N)} \\ \hline & & \frac{K_Z - c_1(N)}{-(r-1)} & & & & \frac{(r-1)c_{r-3}(N)}{(r-1)c_{r-4}(N)} \\ \hline & & & \frac{K_Z - c_1(N)}{-(r-1)} & & & \frac{(r-1)c_{r-4}(N)}{(r-1)c_{r-5}(N)} \\ \hline & & & & \frac{K_Z - c_1(N)}{-(r-1)} & & \frac{(r-1)c_{r-5}(N)}{(r-1)c_{r-6}(N)} \\ \hline & & & & & \frac{K_Z - c_1(N)}{-(r-1)} & \frac{(r-1)c_{r-6}(N)}{(r-1)c_{r-7}(N)} \end{array} \right)$$

where $N = N_{Z/X}$ is the rank r normal bundle to Z in X , $\iota: Z \hookrightarrow X$ is the natural inclusion, and each coefficient acts by classical multiplication on the cohomology of $\hat{X}$.

This operator preserves the filtration $H^{\geq \bullet}(\hat{X}, \mathbb{k})$ by degree of cohomology and the associated graded with respect to this filtration gives the operator

$$\text{gr } \hat{\kappa} = \left(\begin{array}{c|c|c|c|c|c|c} 0 & & & & & & 0 \\ \hline \frac{0}{-(r-1)\iota^*} & \frac{0}{-(r-1)} & & & & & \frac{(r-1)\hat{Q}}{(r-1)\hat{Q}} \\ \hline & & \frac{0}{-(r-1)} & & & & 0 \\ \hline & & & \frac{0}{-(r-1)} & & & 0 \\ \hline & & & & \frac{0}{-(r-1)} & & 0 \\ \hline & & & & & \frac{0}{-(r-1)} & 0 \end{array} \right)$$

This implies that the eigenvalues of $\hat{\kappa}$ split into r clusters and each cluster is contained in a small analytic disk in $\mathbb{k}$ so that:

- (1) The disks are disjoint, one of them is centered at 0, while the rest are centered at $(r-1)e^{\frac{\pi\sqrt{-1}(2j-1)}{r-1}}$ for $j = 1, \dots, r-1$.
- (2) The sum of the generalized eigenspaces corresponding to the eigenvalues that are close to 0 give a super-vector space isomorphic to $H^\bullet(X, \mathbb{k})$.
- (3) The sum of the generalized eigenspaces corresponding to the eigenvalues close to a rescaled $(m-1)$ -st root of 1 give a super-vector space isomorphic to $H^\bullet(Z, \mathbb{k})$.

By analytic continuation, this picture extends to an appropriate open domain in B_X and yields the desired blowup decomposition. To show rigorously that this reasoning identifies the A-model F-bundles of $\widehat{X}$ and X' we will have to identify the special point ω' in $B_{X'}$ which corresponds to $\widehat{\omega}$. This is quite involved and requires a tricky inductive argument which we will not carry out here. Instead we will extract the blowup decomposition directly by using Iritani's formal blowup decomposition [44] and identifying big open subsets in $B_{\widehat{X}}$ and $B_{X'}$ where Iritani's identification of the F-bundles for $\widehat{X}$ and X' is actually analytic. It is also instructive to note that in the formal setting the above clustering behavior of the eigenvalues is proven rigorously in Iritani's theorem [44, Theorem 5.8(6)].

More precisely we have the following

Theorem 4.5. *There is a canonical isomorphism of maximal F-bundles between $(\widehat{\mathcal{H}}, \widehat{\nabla})$ and $(\mathcal{H}', \nabla')$, over an analytic domain $\widehat{U}$ in $B_{\widehat{X}}$ and an analytic domain U' in $B_{X'}$. The union of different choices of $\widehat{U}$ is connected and nonempty, same for U' .*

Proof. We will deduce the explicit description of the domains and the isomorphism in the statement of Theorem 4.5 from Iritani's blowup formula [44]. First we have the following

Lemma 4.6. *Let k be a field of characteristic zero. Let*

$$\varphi: k[[x_1, \dots, x_l]] \longrightarrow k[p, p^{-1}][[y_1, \dots, y_m]]$$

be a map of k -algebras. Write

$$\varphi(x_j) = \sum_{\lambda \in \mathbb{Z}} \sum_{i_1, \dots, i_m \in \mathbb{N}} a_{i_1, \dots, i_m}^j y_1^{i_1} \cdots y_m^{i_m} p^\lambda.$$

Assume there exist constants $\alpha, \beta, \gamma, \delta \in \mathbb{Z}$ with $\alpha < 0 < \beta$ such that for every term $y_1^{i_1} \cdots y_m^{i_m} p^\lambda$ with nonzero coefficient, we have

$$\alpha d + \gamma < \lambda < \beta d + \delta,$$

where $d = i_1 + \cdots + i_m$ is the total degree. Let $\mathbb{k}$ be an algebraically closed non-archimedean field containing k , such that the absolute value restricted to k is trivial. Then after base change from k to $\mathbb{k}$, for any $\epsilon \in (0, 1)$, the map φ is convergent over the admissible domain given by $|y_i| \in [0, \epsilon)$ for $i = 1, \dots, m$ and $|p| \in (\epsilon^{-\frac{1}{2\alpha}}, \epsilon^{-\frac{1}{2\beta}})$.

Proof. Since the coefficients $a_{i_1, \dots, i_m}^j$ are in k , and the absolute value of $\mathbb{k}$ restricted to k is trivial, it suffices to show that for terms $y_1^{i_1} \cdots y_m^{i_m} p^\lambda$ with nonzero coefficients, $\epsilon^d \cdot |p|^\lambda \rightarrow 0$ as the powers tend to infinity.

As $\lambda \rightarrow +\infty$, we have

$$\epsilon^d \cdot |p|^\lambda < \epsilon^d \cdot \epsilon^{-\frac{\lambda}{2\beta}} = \epsilon^{d - \frac{\lambda}{2\beta}},$$

which converges to 0 since $d - \frac{\lambda}{2\beta} \rightarrow +\infty$.

As $\lambda \rightarrow -\infty$, we have

$$\epsilon^d \cdot |p|^\lambda < \epsilon^d \cdot \epsilon^{-\frac{\lambda}{2\alpha}} = \epsilon^{d - \frac{\lambda}{2\alpha}},$$

which converges to 0 since $d - \frac{\lambda}{2\alpha} \rightarrow +\infty$. This completes the proof. $\square$

To use Iritani's formula from [44] to study the convergence of maps between analytic A-model F-bundles, we will embed the requisite Novikov rings into power series rings.

Let $\iota: Z \rightarrow X$ denote the inclusion, $\varphi: \widehat{X} \rightarrow X$ the projection, $D \subset \widehat{X}$ the exceptional divisor, $\rho_Z = c_1(N_{Z/X})$, and λ the class of a line in the fiber of $E \rightarrow Z$. Recall that r is the codimension of $Z \subset X$. Following [44, (5.11)], we set

$$\mathfrak{s} = \begin{cases} r-1 & \text{if } r \text{ is even,} \\ 2(r-1) & \text{if } r \text{ is odd.} \end{cases}$$

As usual we will write $\mathbf{N}^1(X, \mathbb{Z}) = \mathbf{CH}^{1, \text{num}}(X) = \mathbf{CH}^{1, \text{hom}}(X)$ for the torsion free part of the Néron-Severi group of a smooth projective variety $X/\bar{\mathbb{K}}$.

Choose a basis $\omega_1, \dots, \omega_l$ of the free abelian group $\mathbf{N}^1(X, \mathbb{Z})$, consisting of ample classes, such that the pullbacks $\iota^*\omega_1, \dots, \iota^*\omega_l \in \mathbf{N}^1(Z, \mathbb{Z})$ are contained in a simplicial cone spanned by a basis $\omega'_1, \dots, \omega'_m$ of $\mathbf{N}^1(Z, \mathbb{Z})$ consisting of ample classes. This is possible by choosing $\omega_1, \dots, \omega_l$ sufficient close to a fixed ray in the ample cone of X .

Then we obtain a dual basis $\beta'_1, \dots, \beta'_m \in \mathbf{N}_1(Z, \mathbb{Z})$ such that $\text{NE}(Z, \mathbb{Z})$ is contained in $\text{Cone}(\beta'_1, \dots, \beta'_m)$, as well as a dual basis $\beta_1, \dots, \beta_l \in \mathbf{N}_1(X, \mathbb{Z})$ such that $\text{Cone}(\beta_1, \dots, \beta_l)$ contains $\text{NE}(X, \mathbb{Z})$ and $\iota_*\beta'_i$ for $i = 1, \dots, m$.

Lemma 4.7. *For $i = 1, \dots, l$, there exists a constant $c_i \in \mathbb{Z}_{>0}$ such that for $\tilde{\beta}_i := \beta_i - c_i\lambda$, we have an embedding*

$$\text{NE}(\widehat{X}, \mathbb{Z}) \subset \text{Cone}(\tilde{\beta}_1, \dots, \tilde{\beta}_l, \lambda).$$

This induces embeddings

$$\begin{aligned} k[\text{NE}(X, \mathbb{Z})] &\hookrightarrow k[q_1, \dots, q_l], \\ k[\text{NE}(Z, \mathbb{Z})] &\hookrightarrow k[q'_1, \dots, q'_m], \\ k[\text{NE}(\widehat{X}, \mathbb{Z})] &\hookrightarrow k[\tilde{q}_1, \dots, \tilde{q}_l, \mathfrak{q}]. \end{aligned}$$

Then the inclusion of the Novikov ring of $\widehat{X}$

$$\begin{aligned} k[\text{NE}(\widehat{X}, \mathbb{Z})] &\hookrightarrow k[\mathfrak{q}^{1/\mathfrak{s}}, \mathfrak{q}^{-1/\mathfrak{s}}][\text{NE}(X, \mathbb{Z})] \\ \tilde{Q}^{\tilde{d}} &\mapsto Q^{\varphi_* \tilde{d}} \mathfrak{q}^{-[D] \cdot \tilde{d}} \end{aligned}$$

induces a map

$$(4.8) \quad \begin{aligned} k[\tilde{q}_1, \dots, \tilde{q}_l, \mathfrak{q}] &\longrightarrow k[\mathfrak{q}^{1/\mathfrak{s}}, \mathfrak{q}^{-1/\mathfrak{s}}][q_1, \dots, q_l] \\ \tilde{q}_i &\longmapsto q_i \mathfrak{q}^{-c_i} \\ \mathfrak{q} &\longmapsto \mathfrak{q}. \end{aligned}$$

Similarly, the map from the Novikov ring of Z

$$\begin{aligned} k[\text{NE}(Z, \mathbb{Z})] &\longrightarrow k[\mathfrak{q}^{1/\mathfrak{s}}, \mathfrak{q}^{-1/\mathfrak{s}}][\text{NE}(X, \mathbb{Z})] \\ Q_Z^d &\mapsto Q^{\iota_* d} \mathfrak{q}^{-\rho_Z \cdot d / (r-1)}. \end{aligned}$$

induces a map

$$(4.9) \quad \begin{aligned} k[[q'_1, \dots, q'_l]] &\longrightarrow k[\mathfrak{q}^{1/s}, \mathfrak{q}^{-1/s}][[\tilde{q}_1, \dots, \tilde{q}_l]] \\ q'_i &\longmapsto F_i(\tilde{q}_1, \dots, \tilde{q}_l) \mathfrak{q}^{c'_i}, \end{aligned}$$

where F_i are monomials in $\tilde{q}_1, \dots, \tilde{q}_l$ with non-negative exponents, and $c'_i \in \frac{1}{s}\mathbb{Z}$. Now we can apply Eq. (4.6) to the maps (4.8) and (4.9) to obtain domains of convergence after passing to the non-archimedean field $\mathbb{k}$. The union of such domains of convergence from different choices is connected.

It is shown in [44, Theorem 5.18] that there exists a formal invertible change of variables

$$\begin{aligned} H^\bullet(\widehat{X}) &\longrightarrow H^\bullet(X) \oplus H^\bullet(Z)^{\oplus(r-1)}, \\ \tilde{\tau} &\longmapsto (\tau(\tilde{\tau}), \{\varsigma_j(\tilde{\tau})\}_{0 \leq j \leq r-2}) \end{aligned}$$

defined over $k[\mathfrak{q}^{1/s}, \mathfrak{q}^{-1/s}][\text{NE}(X, \mathbb{Z})]$ and an isomorphism

$$\Psi: \text{QDM}(\widehat{X})^{\text{la}} \longmapsto \tau^* \text{QDM}(X)^{\text{la}} \oplus \bigoplus_{j=0}^{r-2} \varsigma_j^* \text{QDM}(Z)^{\text{la}}$$

that commutes with the quantum connection. Here, for a smooth projective variety X , $\text{QDM}(X)^{\text{la}}$ denotes the formal logarithmic A-model F-bundle of X (called the quantum D -module of X in [44]), after a base change to $k[u, \mathfrak{q}^{1/s}, \mathfrak{q}^{-1/s}][\text{NE}(X, \mathbb{Z})]$. Furthermore, the maps τ , ς_j and Ψ above also satisfy the assumption of Eq. (4.6) by homogeneity, so we obtain domains of convergence after passing to the non-archimedean field $\mathbb{k}$.

Finally, we deduce Eq. (4.5) by locally taking the quotient by the canonical foliations on the base of the over-maximal F-bundles. The canonicity of the isomorphism follows from [38, §5.4]. $\square$

Remark 4.10. The isomorphism in Eq. (4.5) is unique given algebraic initial conditions, see [38, §5.4] and [44, §5.8] for details.

4.3 The Leray-Hirsch formula

Let X be a smooth projective variety over $\overline{\mathbb{K}}$, $E \rightarrow X$ a vector bundle of rank r , and $\mathbb{P}(E) \rightarrow X$ the associated projective bundle. Let X' be the disjoint union of r copies of X . We have A-model maximal F-bundles $(\mathcal{H}, \nabla)/B_{\mathbb{P}(E)}$, and $(\mathcal{H}', \nabla')/B_{X'}$ for $\mathbb{P}(E)$ and X' respectively.

Theorem 4.11. *There is a canonical isomorphism of the maximal F-bundles between $(\mathcal{H}, \nabla)$ and $(\mathcal{H}', \nabla')$, over an analytic domain U in $B_{\mathbb{P}(E)}$ and an analytic domain U' in $B_{X'}$. The union of the different choices of U is connected and nonempty, as is the union of the different choices of U' .*

Proof. As in the proof of Eq. (4.5), Eq. (4.6) reduces the non-archimedean statement to the formal version, which is shown by Iritani-Koto in [45, Theorem 5.1]. $\square$

5 G -atoms

Atoms are elementary F-bundle building blocks which arise from the interplay of Gromov-Witten and motivic information. As it is usually the case, one can build the corresponding theory by

using some universal framework of motivic cohomology, such as the framework of Nori motives [65, 40], or the framework of André motivated cycles [3, 4] (for varieties defined over subfields of $\mathbb{C}$). Alternatively, for a more concrete and constructive description, we can work with some realization of the motives, given by a choice of a suitable Weil cohomology theory.

We will use the conventional notion of a Weil cohomology theory, i.e. a cohomology theory on smooth projective varieties which is functorial for pullbacks, which is equipped with a Tate twist, with trace and cycle class maps, and which satisfies the Künneth formula and Poincaré duality. We will not spell out the details here but the precise definition and axioms can be found in standard sources, e.g [64, Definition 1.2.13] and [6, Chapter 45, tag 0FFG]. To avoid lengthy qualifications we choose to work with André motives and only consider Weil cohomology theories that factor through André motives. In practical terms this means that all Weil cohomology theories we work with, arise as tensor functors from the Karoubi closed symmetric monoidal category $\mathcal{C}^{\mathcal{K}}$ of pure André motives over $\mathcal{K}$ to the monoidal category of graded k -vector spaces or to the monoidal category of graded k -linear G -representations of a reductive group G .

To access the full power and flexibility of the atomic compositions that we will introduce, we would like to be able to treat the A-model F-bundles associated with general Weil cohomology theories in a unified manner. To that end, we will need to keep track of the slightly complex interaction between the symmetries of the field of definition of our varieties and the symmetries of the coefficient field of the particular Weil cohomology theory we are utilizing.

5.1 Mumford-Tate groups in the nc setting

Let again $\mathcal{K} \subset \mathbb{C}$ and let $\mathcal{C}^{\mathcal{K}}$ denote the category of pure André motives of smooth projective varieties over $\mathcal{K}$. Let k be a field of characteristic zero, and let G be a proreductive group over k . We also fix a central element $\epsilon_G \in G$ of order 2.

Definition 5.1. A (G, ϵ_G) -symmetric Weil cohomology theory defined on smooth projective $\mathcal{K}$ -varieties is a tensor functor $H^\bullet(-) : \mathcal{C}^{\mathcal{K}} \rightarrow \text{Rep}_k(G)$ to $\mathbb{Z}$ -graded representations of G , which commutes with duality and satisfies the $\mathbb{Z}/2$ folding conditions

- The cohomology group $H^2(\mathbb{P}_{\mathcal{K}}^1)$ is a trivial rank one G -module.
- For any smooth projective $\mathcal{K}$ -variety $\mathfrak{X}$ and any $i \in \mathbb{Z}$, the element ϵ_G acts on $H^i(\mathfrak{X})$ by $(-1)^i$.

Remark 5.2. (a) As discussed in section 3.5.2.2.1, a (G, ϵ_G) -symmetric Weil cohomology theory can be described equivalently as a tensor functor $H^\bullet(-) : \mathcal{C}^{\mathcal{K}}/\text{Tate} \rightarrow \text{Rep}_k(G)$ from the category of Tate periodized André motives to the category of representations of G .

(b) The notion of (G, ϵ_G) -symmetry of a Weil cohomology theory is designed so that it captures a realization of pure motives over $\mathcal{K}$ in which the cohomological grading is folded to a $\mathbb{Z}/2$ -grading, and the cohomologies are considered modulo Tate twists. In particular (G, ϵ_G) -symmetric Weil cohomology theories are candidates for cohomology theories that could potentially be extended to produce realizations of nc-motives. The following are the standard examples we will be working with.

Example 5.3. Let $\mathcal{K} \subset \mathbb{C}$, and let k be a field with $\text{char } k = 0$. Let $S := \mathbb{Z}/2 = \{0, 1\}$ and $\epsilon_S := 1 \in S$. The category $\text{Rep}_k(S)$ is the category of $\mathbb{Z}/2$ -graded vector spaces. Let $H^\bullet : \mathcal{C}^{\mathcal{K}} \rightarrow \text{Vect}_k$ be any Weil cohomology theory with values in k -vector spaces. We can refine $H^\bullet$ to a functor $H^\bullet : \mathcal{C}^{\mathcal{K}} \rightarrow \text{Rep}_k(S)$ by letting the generator $\epsilon_S \in S$ act by multiplication by $(-1)^i$ on H^i . We will refer to the pair (S, ϵ_S) as the symmetry of $\mathbb{Z}/2$ -graded Weil cohomology theories.

Example 5.4. Let $k = \mathcal{K} \subset \mathbb{C}$, $\mathbf{H} := \mathbb{G}_m$, and $\epsilon_{\mathbf{H}} = -1 \in \mathbf{H} = \mathbb{G}_m$. The category $\text{Rep}_k(\mathbf{H})$ of finite dimensional k -linear representations of $\mathbf{H}$ is the category of $\mathbb{Z}$ -graded k -vector spaces, with the action of $\epsilon_{\mathbf{H}}$ recording the parity of grading. For any smooth projective variety $\mathfrak{X}/\mathcal{K}$, let $H_{\text{Dol}}^i(\mathfrak{X}) = \bigoplus_{p+q=i} H^{p,q}$, where $H^{p,q} := H^q(X, \Omega_X^p)$ and $X = \mathfrak{X} \times_{\mathcal{K}} \mathbb{C}$. Let $\mathbf{H} = \mathbb{G}_m$ act on $H^{p,q}$ with weight $p - q$. We will refer to the pair $(\mathbf{H}, \epsilon_{\mathbf{H}})$ as the symmetry of **Hochschild graded Dolbeault cohomology**.

Example 5.5. Let $\mathcal{K} \subset \mathbb{C}$ and $k = \mathbb{Q}$. Let MT denote **the universal classical Mumford-Tate group** [24, 62], i.e. the group associated with the neutral Tannakian category of polarizable pure $\mathbb{Q}$ -Hodge structures with the forgetful fiber functor. MT is a proreductive algebraic group over $\mathbb{Q}$, whose algebraic quotients are precisely the Mumford-Tate groups of symmetries of polarizable pure $\mathbb{Q}$ -Hodge structures. We have a Lefschetz character $\text{MT} \rightarrow \mathbb{G}_m$ associated to the action of MT on $H_B^2((\mathbb{P}_{\mathbb{C}}^1)^{\text{an}}, \mathbb{Q})$. Let $\text{Hod} = \ker(\text{MT} \rightarrow \mathbb{G}_m)$ be the kernel of the Lefschetz character and let $\epsilon_{\text{Hod}} \in \text{Hod} \subset \text{MT}$ be the central element of order two, which acts as ± 1 on $H^{\text{ev/odd}}$ for each $\mathbb{Z}/2$ -graded Hodge structure. We will refer to the pair $(\text{Hod}, \epsilon_{\text{Hod}})$ as the symmetry of **polarizable pure rational $\mathbb{Z}/2$ -weighted Hodge structures**. The de Rham and Hodge theorems imply that the $\mathbb{Z}/2$ -graded Betti cohomology gives a fiber functor $H_B^{\bullet}(-) : \mathcal{C}^{\mathcal{K}}/\text{Tate} \rightarrow \text{Rep}_{\mathbb{Q}}(\text{Hod})$, i.e. $H_B^{\bullet}(-)$ is a $(\text{Hod}, \epsilon_{\text{Hod}})$ -symmetric Weil cohomology theory.

Example 5.6. Let $\mathcal{K} \subset \mathbb{C}$ and $k = \mathbb{Q}$. Let $\mathcal{C}^{\mathcal{K}}$ be the semisimple neutral Tannakian category of pure motivated motives (see [3, 4]) with fiber functor the forgetful functor to rational vector spaces. Consider the associated group $\mathbf{G}^{\mathcal{K}}$ of tensor automorphisms of the fiber functor is the proreductive group scheme over $\mathbb{Q}$, whose finite dimensional representations Again we have homomorphisms $\iota : \mathbb{G}_m \rightarrow \mathbf{G}^{\mathcal{K}}$ and $\mathbf{G}^{\mathcal{K}} \rightarrow \mathbb{G}_m$ controlling the weights of the motives and the action on $H_B^2(\mathbb{P}_{\mathcal{K}}^1)$. We set $\text{Mot} := \ker(\mathbf{G}^{\mathcal{K}} \rightarrow \mathbb{G}_m)$, and $\epsilon_{\text{Mot}} = \iota(-1)$. Then $(\text{Mot}, \epsilon_{\text{Mot}})$ is the symmetry pair of **André's $\mathbb{Z}/2$ -graded pure motivated motives**. Again the $\mathbb{Z}/2$ -graded Betti cohomology $H^{\bullet}(-)$ is $(\text{Mot}, \epsilon_{\text{Mot}})$ -symmetric.

Example 5.7. Suppose $\mathcal{K} \subset \mathbb{C}$ and $k = \mathbb{Q}$, then by definition we have embeddings of proreductive $\mathbb{Q}$ -groups

$$\mathbf{S} \subset \text{Hod} \subset \text{Mot}$$

so that $\epsilon_{\mathbf{S}} = \epsilon_{\text{Hod}} = \epsilon_{\text{Mot}}$. Furthermore if we consider $\mathcal{K} = \mathbb{C}$ and base-change these groups to $\text{Spec } \mathbb{C}$ we get embeddings of the corresponding complex proreductive groups which can be further augmented to a chain of embeddings

$$\mathbf{S}_{\mathbb{C}} \subset \mathbf{H}_{\mathbb{C}} \subset \text{Hod}_{\mathbb{C}} \subset \text{Mot}_{\mathbb{C}},$$

again with matching central elements of order two. Here the embedding $\mathbf{H}_{\mathbb{C}} \rightarrow \text{Hod}_{\mathbb{C}}$ arises as follows. We have a natural central embedding $\mathbf{H}_{\mathbb{C}} \subset \text{MT}_{\mathbb{C}}$ in which $\lambda \in \mathbf{H}_{\mathbb{C}} = \mathbb{C}^{\times}$ acts as multiplication by λ^{p-q} on $H^{p,q}$ for each $\mathbb{Z}$ -graded $\mathbb{Q}$ pure Hodge structure H . The Lefschetz character is thus trivial on the subgroup $\mathbf{H}_{\mathbb{C}} \subset \text{MT}_{\mathbb{C}}$ and so $\mathbf{H}_{\mathbb{C}} \rightarrow \text{Hod}_{\mathbb{C}}$.

The basic idea for constructing our invariants is the observation that the motivic information captured by a pair (G, ϵ_G) naturally amalgamates the analytic F-bundle data arising from nc-Hodge theory. Since the passage from formal to analytic data will in general break the action of symmetries we will have to exercise some extra care while encoding the motivic information in the analytic setting.

5.2 G -atoms of projective manifolds

In this section we again assume that $\mathcal{K} \subset \mathbb{C}$, k is of characteristic zero, G is a proreductive group over k , $\epsilon_G \in G$ is a central element of order two, and $H^\bullet : \mathcal{C}^{\mathcal{K}} \rightarrow \text{Rep}_k(G)$ is a (G, ϵ_G) -symmetric Weil cohomology theory, as in Definition 5.1. Fix an algebraic closure $\bar{k} \supset k$ and consider the algebraically closed non-archimedean field $\mathbb{k} = \bar{k}((\mathbf{y}^{\mathbb{Q}})) \supset k$.

5.2.1 Germs of analytic Lie groups

We introduce the notion of local Lie groups and germs of Lie groups in the non-archimedean setting, similar to the classical notions in differential geometry. A **$\mathbb{k}$ -analytic local Lie group** consists of data $(U, \iota, \mathbf{e}, V, \mathbf{m})$, where U is a $\mathbb{k}$ -analytic space over $\mathbb{k}$, $\iota : U \rightarrow U$ is an analytic involution, $\mathbf{e} \in U$ is a point fixed by ι , $V \subset U \times U$ is an open subset and $\mathbf{m} : V \rightarrow U$ is an analytic map, so that

- (1) $U \times \{\mathbf{e}\} \cup \{\mathbf{e}\} \times U \subset V$, and $\mathbf{m}(g, \mathbf{e}) = \mathbf{m}(\mathbf{e}, g) = g$ for all $g \in V$.
- (2) $\text{Graph}(\iota) \subset V$, and $\mathbf{m}(\iota g, g) = \mathbf{m}(g, \iota(g)) = \mathbf{e}$.
- (3) If $(g_1, g_2, g_3) \in U^{\times 3}$ are such that the pairs (g_1, g_2) , (g_2, g_3) , $(g_1, \mathbf{m}(g_2, g_3))$, and $(\mathbf{m}(g_1, g_2), g_3)$ are all in V , then $\mathbf{m}(g_1, \mathbf{m}(g_2, g_3)) = \mathbf{m}(\mathbf{m}(g_1, g_2), g_3)$.

Local $\mathbb{k}$ -analytic Lie groups form a category, where a morphism between $(U_1, \iota_1, \mathbf{e}_1, V_1, \mathbf{m}_1)$ and $(U_2, \iota_2, \mathbf{e}_2, V_2, \mathbf{m}_2)$ is given by an analytic map $\varphi : U_1 \rightarrow U_2$ which sends $\mathbf{e}_1$ to $\mathbf{e}_2$, intertwines ι_1 with ι_2 , and is such that $\varphi \times \varphi$ maps V_1 to V_2 and intertwines $\mathbf{m}_1$ and $\mathbf{m}_2$.

A **germ** of a $\mathbb{k}$ -analytic Lie group consists of a $\mathbb{k}$ -analytic local Lie group $(U, \iota, \mathbf{e}, V, \mathbf{m})$ and a (not necessarily analytic) subgroup $H \subset U$. We will denote such a germ by (U, H) when the rest of the data is understood from the context. A strict morphism between two germs (U_1, H_1) and (U_2, H_2) is a morphism $\varphi : U_1 \rightarrow U_2$ of $\mathbb{k}$ -analytic local Lie groups with $\varphi(H_1) \subset H_2$.

The category of germs $\text{LieGerms}_{\mathbb{k}}$ of $\mathbb{k}$ -analytic Lie groups is the localization of the category of germs of $\mathbb{k}$ -analytic Lie groups with strict morphisms along the class of morphisms $\varphi : (U_1, H_1) \rightarrow (U_2, H_2)$, which induce an isomorphism of $\mathbb{k}$ -analytic local Lie groups between U_1 and an open neighborhood of H_2 .

Example 5.8. Suppose $\mathbb{G}$ is a non-archimedean analytic Lie group over $\mathbb{k}$. Let $\mathbb{H} \subset \mathbb{G}$ be a subgroup. Then $(\mathbb{G}, \mathbb{H})$ is an object in $\text{LieGerms}_{\mathbb{k}}$ which is called the analytic germ of $\mathbb{G}$ along $\mathbb{H}$.

Analytic actions of analytic local Lie groups on non-archimedean analytic spaces are defined in the obvious manner. Suppose U is a $\mathbb{k}$ -analytic local Lie group which acts analytically on a non-archimedean analytic space B . Suppose $H \subset U$ is a (not necessarily analytic) subgroup. Suppose S is a closed analytic subvariety in B and suppose $S \subset B^H$, i.e. S is contained in the fixed point locus for the H -action on B . Then for any analytic open neighborhood $S \subset N \subset B$ of S in B we can find an analytic local group neighborhood $H \subset U' \subset U$ of H in U , so that the action map $U \times B \rightarrow B$ restricts to an action map $U' \times N \rightarrow N$. In particular, the analytic Lie group germ of U along H will act on the analytic germ of the space B along S .

5.2.2 Local G -atoms.

Let $\mathfrak{X}$ be a non-empty, smooth projective variety over $\mathcal{K}$ and consider the maximal, non-archimedean A-model F-bundle associated to $\mathfrak{X}$, $\mathcal{K}$, k , and $H^\bullet(-)$, that we described in Definition 3.52. This F-bundle is constructed as an analytic object over $k((\mathbf{y}))$ and we can base change it to an analytic

object over the algebraic closure $\mathbb{k} \supset k(\langle \mathbf{y} \rangle)$. To avoid unnecessary proliferation of notation we will still write $(\mathcal{H}, \nabla)/B_{\mathfrak{X}}$ for the resulting non-archimedean analytic F-bundle over $\mathbb{k}$.

Let G^{an} denote the Berkovich analytification over k with trivial valuation (see [12]). It contains an analytic subgroup $G^{\square}$, which is a compact closed analytic subdomain, consisting of valuations on the function field of G with centers inside G (see [73, Definition 1.3]). Taking base extension from k to $\mathbb{k}$, we obtain an inclusion of $\mathbb{k}$ -analytic groups $G_{\mathbb{k}}^{\square} \subset G_{\mathbb{k}}^{\text{an}}$. Let $\mathcal{G}$ denote the germ of $G_{\mathbb{k}}^{\text{an}}$ along $G_{\mathbb{k}}^{\square}$. By definition $\mathcal{G}$ contains $G(k)$ as a subgroup, and in particular contains ϵ_G .

Since by definition $H^{\bullet}(\mathfrak{X}) \in \text{Rep}_k(G)$, we have $G_{\mathbb{k}}^{\text{an}}$ acting on $(H^{\bullet}(\mathfrak{X}) \otimes_k \mathbb{k})^{\text{an}}$. Since the components $B_{\mathfrak{X},q}$ and $B_{\mathfrak{X},t}^{\text{ev}}$ of the base $B_{\mathfrak{X}}$ are defined by strict inequalities of norms, they are preserved by the germ $\mathcal{G}$. Hence $\mathcal{G}$ acts on the non-archimedean A-model F-bundle $(\mathcal{H}, \nabla)/B_{\mathfrak{X}}$.

Since $H^{\bullet}(\mathfrak{X})$ is a finite dimensional representation of the reductive group G , it follows that under the action of $\mathcal{G}$ on $B_{\mathfrak{X}}$ the fixed points of $G(k)$, $G_{\mathbb{k}}^{\square}$ and $\mathcal{G}$ are all the same, i.e.

$$(5.9) \quad B_{\mathfrak{X}}^{G(k)} = B_{\mathfrak{X}}^{G_{\mathbb{k}}^{\square}} = B_{\mathfrak{X}}^{\mathcal{G}}.$$

To simplify notation we will denote the fixed point locus (5.9) by $B_{\mathfrak{X}}^G$.

Since ϵ_G acts as ± 1 on the even/odd parts of $H^{\bullet}(\mathfrak{X})$, it follows that the fixed point locus $B_{\mathfrak{X}}^G$ is a purely even connected closed non-archimedean smooth $\mathbb{k}$ -analytic subvariety in $B_{\mathfrak{X}}$. We will use the subvariety $B_{\mathfrak{X}}^G \subset B_{\mathfrak{X}}$ to define the G -atoms for $\mathfrak{X}$. Consider

$\tilde{B}_{\mathfrak{X}} \rightarrow B_{\mathfrak{X}}^G$ - the ramified covering given by the spectrum of $\kappa|_{B_{\mathfrak{X}}^G} = \text{Eu}|_{B_{\mathfrak{X}}^G} \star (-)$,

$U_{\mathfrak{X}} \subset B_{\mathfrak{X}}^G$ - the open locus where the number of eigenvalues of $\text{Eu}|_{B_{\mathfrak{X}}^G} \star (-)$ is maximal, i.e. the locus over which the reduced cover $\tilde{B}_{\mathfrak{X},\text{red}} \rightarrow B_{\mathfrak{X}}^G$ is unramified.

$\tilde{U}_{\mathfrak{X}} := \tilde{B}_{\mathfrak{X},\text{red}} \times_{B_{\mathfrak{X}}^G} U_{\mathfrak{X}}$ - the restriction of the cover $\tilde{B}_{\mathfrak{X},\text{red}}$ to $U_{\mathfrak{X}}$.

Definition 5.10. The **set of local G -atoms of $\mathfrak{X}$** is the finite set $\pi_0(\tilde{U}_{\mathfrak{X}})$ of connected components of $\tilde{U}_{\mathfrak{X}}$. The multiplicity $\text{mult}_{\alpha}(\mathfrak{X})$ of an element $\alpha \in \pi_0(\tilde{U}_{\mathfrak{X}})$ is defined as the degree of the cover $\tilde{U}_{\mathfrak{X},\alpha} \rightarrow U_{\mathfrak{X}}$, where we write $\tilde{U}_{\mathfrak{X},\alpha}$ for the connected component of $\tilde{U}_{\mathfrak{X}}$ labeled by α .

The group $\text{Aut}(\mathfrak{X})$ acts on the set $\pi_0(\tilde{U}_{\mathfrak{X}})$, preserving the multiplicity function. The quotient set $\pi_0(\tilde{U}_{\mathfrak{X}})/\text{Aut}(\mathfrak{X})$ depends only on the isomorphism class of $\mathfrak{X}$. Hence we can consider the disjoint union

$$\text{Atoms}_G^{\mathcal{K},\text{loc}} := \bigsqcup_{[\mathfrak{X}]} \pi_0(\tilde{U}_{\mathfrak{X}})/\text{Aut}(\mathfrak{X})$$

where the union runs over all isomorphism classes of non-empty smooth projective varieties defined over $\mathcal{K}$. We will call $\text{Atoms}_G^{\mathcal{K},\text{loc}}$ **the set of local G -atoms of smooth projective $\mathcal{K}$ -varieties**. By definition this set depends only on the base field $\mathcal{K}$.

Next we will define some additional and natural elementary equivalences on local G -atoms.

5.2.3 Elementary equivalences from disjoint unions.

If $\mathfrak{X}_1$ and $\mathfrak{X}_2$ are two non-empty smooth projective varieties defined over $\mathcal{K}$. Then we have

$$\pi_0(\tilde{U}_{\mathfrak{X}_1 \sqcup \mathfrak{X}_2}) = \pi_0(\mathfrak{X}_1) \sqcup \pi_0(\mathfrak{X}_2).$$

In this situation we declare each local atom $\alpha \in \pi_0(\tilde{U}_{\mathfrak{X}_1})/\text{Aut}(\mathfrak{X}_1) \subset \text{Atoms}_G^{\mathcal{K}, \text{loc}}$ to be **disjoint union elementary equivalent** to its image α' under the embedding

$$\pi_0(\tilde{U}_{\mathfrak{X}_1})/\text{Aut}(\mathfrak{X}_1) \hookrightarrow \pi_0(\tilde{U}_{\mathfrak{X}_1 \sqcup \mathfrak{X}_2})/\text{Aut}(\mathfrak{X}_1 \sqcup \mathfrak{X}_2).$$

5.2.4 Elementary equivalences from blowups.

If $\mathfrak{X}$ is a pure-dimensional non-empty smooth projective $\mathcal{K}$ -variety containing a non-empty smooth projective $\mathcal{K}$ -subvariety $\mathfrak{Z} \subset \mathfrak{X}$ of codimension $m \geq 2$, then we have the blowup $\hat{\mathfrak{X}} := \text{Bl}_{\mathfrak{Z}} \mathfrak{X}$ which is also defined over $\mathcal{K}$. Write $\mathfrak{X}'$ for the disjoint union $\mathfrak{X}' := \mathfrak{X} \sqcup \mathfrak{Z} \sqcup \dots \sqcup \mathfrak{Z}$ of $\mathfrak{X}$ and $(m-1)$ -copies of $\mathfrak{Z}$. Consider

- $(\hat{\mathfrak{X}}\mathcal{H}, \hat{\mathfrak{X}}\nabla)/B_{\hat{\mathfrak{X}}}$ - the maximal non-archimedean analytic A-model F-bundle of $\hat{\mathfrak{X}}$;
- $(\mathfrak{X}\mathcal{H}, \mathfrak{X}\nabla)/B_{\mathfrak{X}}, (\mathfrak{Z}\mathcal{H}, \mathfrak{Z}\nabla)/B_{\mathfrak{Z}}$ - the maximal non-archimedean analytic A-model F-bundles for $\mathfrak{X}$ and $\mathfrak{Z}$ respectively;
- $(\mathfrak{X}'\mathcal{H}, \mathfrak{X}'\nabla)/B_{\mathfrak{X}'}$ - the maximal non-archimedean analytic A-model F-bundles for $\mathfrak{X}'$.

Let $\hat{\mathfrak{X}}\kappa : \hat{\mathfrak{X}}\mathcal{H}_o \rightarrow \hat{\mathfrak{X}}\mathcal{H}_o$, $\mathfrak{X}\kappa : \mathfrak{X}\mathcal{H}_o \rightarrow \mathfrak{X}\mathcal{H}_o$, $\mathfrak{Z}\kappa : \mathfrak{Z}\mathcal{H}_o \rightarrow \mathfrak{Z}\mathcal{H}_o$, and $\mathfrak{X}'\kappa : \mathfrak{X}'\mathcal{H}_o \rightarrow \mathfrak{X}'\mathcal{H}_o$ denote the associated operators of quantum multiplication by the respective Euler vector fields.

By definition we have

$$(5.11) \quad (\mathfrak{X}'\mathcal{H}, \mathfrak{X}'\nabla)/B_{\mathfrak{X}'} = (\mathfrak{X}\mathcal{H}, \mathfrak{X}\nabla)/B_{\mathfrak{X}} \boxplus \underbrace{(\mathfrak{Z}\mathcal{H}, \mathfrak{Z}\nabla)/B_{\mathfrak{Z}} \boxplus \dots \boxplus (\mathfrak{Z}\mathcal{H}, \mathfrak{Z}\nabla)/B_{\mathfrak{Z}}}_{(m-1)\text{-times}},$$

and similarly

$$(5.12) \quad (\mathfrak{X}'\mathcal{H}_o, \mathfrak{X}'\kappa) = (\mathfrak{X}\mathcal{H}_o, \mathfrak{X}\kappa) \boxplus \underbrace{(\mathfrak{Z}\mathcal{H}_o, \mathfrak{Z}\kappa) \boxplus \dots \boxplus (\mathfrak{Z}\mathcal{H}_o, \mathfrak{Z}\kappa)}_{(m-1)\text{-times}}.$$

Applying Eq. (4.5), we obtain nonempty connected open subsets $\hat{U} \subset B_{\hat{\mathfrak{X}}}$ and $U' \subset B_{\mathfrak{X}'}$ and a canonical isomorphism

$$(5.13) \quad (\hat{\mathfrak{X}}\mathcal{H}, \hat{\mathfrak{X}}\nabla)/\hat{U} \cong (\mathfrak{X}'\mathcal{H}, \mathfrak{X}'\nabla)/U'$$

of the restricted F-bundles and hence a canonical isomorphism of bundles with κ -operators:

$$(5.14) \quad (\hat{\mathfrak{X}}\mathcal{H}_o, \hat{\mathfrak{X}}\kappa)|_{\hat{U}} \cong (\mathfrak{X}'\mathcal{H}_o, \mathfrak{X}'\kappa)|_{U'} = \left((\mathfrak{X}\mathcal{H}_o, \mathfrak{X}\kappa) \boxplus \underbrace{(\mathfrak{Z}\mathcal{H}_o, \mathfrak{Z}\kappa) \boxplus \dots \boxplus (\mathfrak{Z}\mathcal{H}_o, \mathfrak{Z}\kappa)}_{(m-1)\text{-times}} \right)|_{U'}.$$

The isomorphisms are G -equivariant by the uniqueness of the blowup decomposition (see Eq. (4.10)).

Next, let $\hat{U}^G$ and U'^G be the G -fixed loci. Recall that in the proof of Eq. (4.5), when we apply Eq. (4.6), we get an annulus in the variable $\mathfrak{q}$ and open disks centered at 0 in the other variables.

Note that the G -fixed locus is a linear subspace, and the $\mathfrak{q}$ -direction is fixed because it is algebraic, therefore the loci $\widehat{U}^G$ and U'^G are nonempty. Let

$$\mathbb{U}_{\widehat{\mathfrak{x}}} := U_{\widehat{\mathfrak{x}}} \cap \widehat{U}^G \quad \text{and} \quad \mathbb{U}_{\mathfrak{x}'} := U_{\mathfrak{x}'} \cap U'^G$$

be the maximal open subsets over which the reduced spectral covers of

$$\left(\widehat{\mathfrak{x}} \mathcal{H}_o, \widehat{\mathfrak{x}} \boldsymbol{\kappa} \right) \Big|_{\widehat{U}^G} \quad \text{and} \quad \left(\mathfrak{x}' \mathcal{H}_o, \mathfrak{x}' \boldsymbol{\kappa} \right) \Big|_{U'^G}$$

are unramified. Since $\mathbb{U}_{\widehat{\mathfrak{x}}}$ is the complement in the connected $\mathbb{k}$ -analytic space $\widehat{U}^G$ of the branch subspace for the reduced spectral cover, it is itself connected. By the same token $\mathbb{U}_{\mathfrak{x}'}$ is also a connected open in U'^G .

Following our notational convention, we denote these unramified covers by $\widetilde{\mathbb{U}}_{\widehat{\mathfrak{x}}} \rightarrow \mathbb{U}_{\widehat{\mathfrak{x}}}$ and $\widetilde{\mathbb{U}}_{\mathfrak{x}'} \rightarrow \mathbb{U}_{\mathfrak{x}'}$ respectively. By construction we have

$$\widetilde{\mathbb{U}}_{\widehat{\mathfrak{x}}} = \widetilde{U}_{\widehat{\mathfrak{x}}} \times_{U_{\widehat{\mathfrak{x}}}} \mathbb{U}_{\widehat{\mathfrak{x}}} \quad \text{and} \quad \widetilde{\mathbb{U}}_{\mathfrak{x}'} = \widetilde{U}_{\mathfrak{x}'} \times_{U_{\mathfrak{x}'}} \mathbb{U}_{\mathfrak{x}'},$$

Furthermore, the isomorphism (5.14) induces an isomorphism of covers

$$\begin{array}{ccc} \widetilde{\mathbb{U}}_{\widehat{\mathfrak{x}}} & \xrightarrow{\sim} & \widetilde{\mathbb{U}}_{\mathfrak{x}'} \\ \downarrow & & \downarrow \\ \mathbb{U}_{\widehat{\mathfrak{x}}} & \xrightarrow{\sim} & \mathbb{U}_{\mathfrak{x}'}, \end{array}$$

and hence a bijection of sets of connected components

$$\pi_0(\widetilde{\mathbb{U}}_{\widehat{\mathfrak{x}}}) \cong \pi_0(\widetilde{\mathbb{U}}_{\mathfrak{x}'}).$$

Since $\mathbb{U}_{\widehat{\mathfrak{x}}} \subset U_{\widehat{\mathfrak{x}}}$ and $\mathbb{U}_{\mathfrak{x}'} \subset U_{\mathfrak{x}'}$ are connected open subsets, we also have that the natural maps $\pi_0(\widetilde{\mathbb{U}}_{\widehat{\mathfrak{x}}}) \rightarrow \pi_0(\widetilde{U}_{\widehat{\mathfrak{x}}})$ and $\pi_0(\widetilde{\mathbb{U}}_{\mathfrak{x}'}) \rightarrow \pi_0(\widetilde{U}_{\mathfrak{x}'})$ are surjective. Finally, note that the identification (5.12) gives an identification of unramified covers of $U_{\mathfrak{x}'} = U_{\mathfrak{x}} \times U_3^{\times(m-1)}$:

$$\widetilde{U}_{\mathfrak{x}'} = \widetilde{U}_{\mathfrak{x}} \times U_3^{\times(m-1)} \sqcup U_{\mathfrak{x}} \times \left(\bigsqcup_{a=1}^{m-1} U_3 \times \cdots \times \underset{\text{slot } a}{\widetilde{U}_3} \times \cdots U_3 \right),$$

and thus we get a canonical bijection of sets of connected components

$$\pi_0(\widetilde{U}_{\mathfrak{x}'}) = \pi_0(\widetilde{U}_{\mathfrak{x}}) \sqcup \underbrace{\pi_0(\widetilde{U}_3) \sqcup \cdots \sqcup \pi_0(\widetilde{U}_3)}_{(m-1)\text{-times}}.$$

Altogether we get a correspondence of sets of local G -atoms:

$$(5.15) \quad \begin{array}{ccc} & \pi_0(\widetilde{\mathbb{U}}_{\widehat{\mathfrak{x}}}) \cong \pi_0(\widetilde{\mathbb{U}}_{\mathfrak{x}'}) & \\ \swarrow & & \searrow \\ \pi_0(\widetilde{U}_{\widehat{\mathfrak{x}}}) & & \pi_0(\widetilde{U}_{\mathfrak{x}}) \sqcup \pi_0(\widetilde{U}_3)^{\sqcup(m-1)} \\ \swarrow & & \searrow \\ \pi_0(\widetilde{U}_{\widehat{\mathfrak{x}}}) / \text{Aut}(\widehat{\mathfrak{x}}) & & \pi_0(\widetilde{U}_{\mathfrak{x}'}) / \text{Aut}(\mathfrak{x}'). \end{array}$$

We declare a local atom $\alpha \in \pi_0(\widehat{\mathfrak{X}})/\text{Aut}$ to be **blowup elementary equivalent** to a local atom $\alpha' \in \pi_0(\mathfrak{X}')/\text{Aut}(\mathfrak{X}')$ if α and α' are related by the correspondence (5.15).

5.2.5 Elementary equivalences from projective bundles.

Suppose $\mathfrak{X}/\mathcal{K}$ is a non-empty smooth projective variety and $\mathcal{E}$ is a vector bundle of rank $r \geq 2$ on $\mathfrak{X}$. By the results of Iritani and Koto [45], and Hinault, Yu, Zhang, and Zhang [38, Section 5] reviewed in Section 4.3, we can identify a non-empty connected domain $\mathbb{U}_{\mathbb{P}(\mathcal{E})}$ in $U_{\mathbb{P}(\mathcal{E})}$ with the product of r copies of a domain $\mathbb{U}_{\mathfrak{X}} \subset U_{\mathfrak{X}}$ so that the corresponding F-bundles and operators κ are compatible. Again, this gives a correspondence between the local G -atoms for $\mathbb{P}(\mathcal{E})$ and the local G -atoms for $\mathfrak{X}^{\sqcup r}$. We declare that a local atom $\alpha \in \pi_0(\widetilde{U}_{\mathbb{P}(\mathcal{E})})/\text{Aut}(\mathbb{P}(\mathcal{E}))$ is **Leray-Hirsch elementary equivalent** to a local atom $\alpha' \in \widetilde{U}_{\mathfrak{X}^{\sqcup r}}/\text{Aut}(\mathfrak{X}^{\sqcup r})$ if α and α' are related by the correspondence.

5.2.6 G -atoms of $\mathcal{K}$ -varieties.

With all this in place we can now define G -atoms of projective varieties.

Definition 5.16. The set of G -atoms of smooth projective $\mathcal{K}$ -varieties is the quotient $\text{Atoms}_G^{\mathcal{K}}$ of the set of local atoms $\text{Atoms}_G^{\mathcal{K}, \text{loc}}$ by the equivalence relation generated by the elementary equivalences defined in Sections 5.2.3–5.2.5.

The set $\text{Atoms}_G^{\mathcal{K}}$ of G -atoms is naturally filtered by subsets

$$\text{Atoms}_{G, \dim \leq 0}^{\mathcal{K}} \subset \text{Atoms}_{G, \dim \leq 1}^{\mathcal{K}} \subset \cdots, \quad \text{Atoms}_G^{\mathcal{K}} = \bigcup_{d \geq 0} \text{Atoms}_{G, \dim \leq d}^{\mathcal{K}},$$

where an atom α belongs to $\text{Atoms}_{G, \dim \leq d}^{\mathcal{K}}$ if there exists a smooth projective $\mathcal{K}$ -variety of dimension $\leq d$ whose atomic composition contains α .

Combined with the weak factorization theorem of Abramovich, Karu, Matsuki, and Włodarczyk [81, 1] this makes G -atoms candidates for obstructions to birational equivalence. Concretely we have the following simple

Proposition 5.17. (Non-rationality criterion) *Let $\mathfrak{X}$ be a smooth projective $\mathcal{K}$ -variety of dimension $d \geq 2$. Suppose we have a G -atom $\alpha \in \text{Atoms}_G^{\mathcal{K}}$ which appears in the atomic composition of $\mathfrak{X}$ and is such that $\alpha \notin \text{Atoms}_{G, \dim \leq d-2}^{\mathcal{K}}$. Then $\mathfrak{X}$ can not be birationally equivalent to $\mathbb{P}_{\mathcal{K}}^d$.*

Proof. Suppose $\mathfrak{X}$ is birational to $\mathbb{P}_{\mathcal{K}}^d$. By the weak factorization theorem [81, 1] a birational map between $\mathbb{P}_{\mathcal{K}}^d$ and $\mathfrak{X}$ will be given by a backward/forward sequence of blowups and blow-downs with smooth centers. Since the centers must have codimension at least two, and the atomic composition of $\mathbb{P}_{\mathcal{K}}^d$ consists of $d+1$ copies of the G -atom of a point, the blowup equivalence relation implies that the atomic composition of $\mathfrak{X}$ must be comprised only of atoms in $\text{Atoms}_{G, \dim \leq d-2}^{\mathcal{K}}$. This is a contradiction. $\square$

5.2.6.1 Chemical formulas of $\mathcal{K}$ -varieties. Write $\mathbb{Z}^{\oplus \text{Atoms}_G^{\mathcal{K}}}$ for the free abelian group generated by $\text{Atoms}_G^{\mathcal{K}}$, and $\mathbb{Z}_{\geq 0}^{\oplus \text{Atoms}_G^{\mathcal{K}}}$ for the additive monoid of free linear combinations of atoms with non-negative integer coefficients or. Equivalently, we can think of $\mathbb{Z}^{\oplus \text{Atoms}_G^{\mathcal{K}}}$ (respectively

$\mathbb{Z}_{\geq 0}^{\oplus \text{Atoms}_G^{\mathcal{K}}}$) as the ring (respectively semiring) of finitely supported $\mathbb{Z}$ -valued (respectively $\mathbb{Z}_{\geq 0}$ -valued) functions on $\text{Atoms}_G^{\mathcal{K}}$.

Splitting the product of varieties into its atoms gives rise to a well-defined map

$$\text{Atoms}_G^{\mathcal{K}} \times \text{Atoms}_G^{\mathcal{K}} \longrightarrow \mathbb{Z}_{\geq 0}^{\oplus \text{Atoms}_G^{\mathcal{K}}}.$$

This map endows the additive monoid $\mathbb{Z}_{\geq 0}^{\oplus \text{Atoms}_G^{\mathcal{K}}}$ with the structure of a commutative associative unital semiring, and similarly endows the free abelian group $\mathbb{Z}^{\oplus \text{Atoms}_G^{\mathcal{K}}}$ with the structure of a unital commutative ring.

Note that the assignment which sends an isomorphism class of varieties $[\mathfrak{X}]$ to its atomic composition:

$$\text{CF}_G(\mathfrak{X}) = \sum_{\alpha \in \pi_0(\tilde{U}_{\mathfrak{X}}) / \text{Aut}(\mathfrak{X})} \text{mult}_{\mathfrak{X}}(\alpha) \cdot \delta_{\alpha} : \quad \text{Atoms}_G^{\mathcal{K}} \longrightarrow \mathbb{Z}_{\geq 0} \quad ,$$

is naturally a finite multiset. The multiset $\text{CF}_G(\mathfrak{X})$ can be thought of as the **chemical formula** of $\mathfrak{X}$, i.e. the list of the G -atoms in the atomic composition of $\mathfrak{X}$ together with their multiplicities.

5.2.6.2 Additivity properties of CF. The elementary equivalences we built into the definition of G -atoms imply that the chemical formula multiset satisfies a number of natural additivity properties:

(1) For two smooth projective $\mathcal{K}$ -varieties $\mathfrak{X}_1, \mathfrak{X}_2$, we have

$$\text{CF}_G(\mathfrak{X}_1 \sqcup \mathfrak{X}_2) = \text{CF}_G(\mathfrak{X}_1) + \text{CF}_G(\mathfrak{X}_2).$$

(2) For a smooth projective $\mathcal{K}$ -variety $\mathfrak{X}$, a smooth closed subvariety $\mathfrak{Z} \subset \mathfrak{X}$ of codimension r , and the blowup $\hat{\mathfrak{X}}$ of $\mathfrak{X}$ along $\mathfrak{Z}$, we have

$$\text{CF}_G(\hat{\mathfrak{X}}) = \text{CF}_G(\mathfrak{X}) + (r - 1) \cdot \text{CF}_G(\mathfrak{Z}).$$

(3) For a smooth projective $\mathcal{K}$ -variety $\mathfrak{X}$, a rank r vector bundle V on $\mathfrak{X}$, and the associated projective bundle $\mathbb{P}(V)$, we have

$$\text{CF}_G(\mathbb{P}(V)) = r \cdot \text{CF}_G(\mathfrak{X}).$$

Combined with Bittner's relations [13], these properties imply that the multiset assignment $\mathfrak{X} \mapsto \text{CF}_G(\mathfrak{X})$ induces a well defined ring homomorphism

$$\mathfrak{cf}_G : K_0(\text{Var}_{\mathcal{K}}) \longrightarrow \mathbb{Z}^{\oplus \text{Atoms}_G^{\mathcal{K}}}$$

from the Grothendieck ring of varieties $K_0(\text{Var}_{\mathcal{K}})$ to the ring $\mathbb{Z}^{\oplus \text{Atoms}_G^{\mathcal{K}}}$. The homomorphism $\mathfrak{cf}_G$ can be thought of as the function assigning to each K -class of varieties its chemical formula.

Specific chemical formulas are pretty difficult to compute as they require a detailed knowledge of quantum multiplication. Still we have the following general

Lemma 5.18. *Let X be a connected smooth projective variety over an algebraically closed field $\mathcal{K} = \overline{\mathcal{K}}$ with a numerically effective canonical class. Then the G -atomic composition of X consists of a single atom $\eta(X)$.*

Proof. Let (T_i) be a homogeneous basis of $H^\bullet(X)$, let (t_i) be the associated formal coordinates, and let (T^i) be the dual basis with respect to the Poincaré pairing. Recall that the big quantum product is given by

$$T_i \star T_j = \sum_r \frac{\partial^3 \Phi}{\partial t_i \partial t_j \partial t_r} T^r,$$

where

$$\frac{\partial^3 \Phi}{\partial t_i \partial t_j \partial t_r} = \sum_{n \geq 0, \beta} \frac{q^\beta}{n!} \sum_{i_1, \dots, i_n} \langle T_i T_j T_r T_{i_1} \cdots T_{i_n} \rangle_{0, n+3}^\beta t_{i_1} \cdots t_{i_n}.$$

Since K_X is nef, it follows from the virtual dimension calculation that if T^r has nonzero coefficient in $T_i \star T_j$, then

$$\deg T^r \geq \deg T_i + \deg T_j.$$

We conclude that the spectrum of the Euler vector field action on $H^\bullet(X)$ contains only one element, coming from $H^0(X)$. $\square$

5.2.6.3 Change of group. Let $\iota: \tilde{G} \subset G$ be a reductive subgroup of G defined over $\mathbb{Q}$. Assume also that the central order two element $\epsilon_G \in G$ is contained in $\tilde{G}$. Then the fiber functor $\tilde{H}^\bullet: \mathcal{C}^\mathcal{K} \rightarrow \text{Rep}_k(\tilde{G})$ which is the composition of $H^\bullet: \mathcal{C}^\mathcal{K} \rightarrow \text{Rep}_k(G)$ with the pullback by the inclusion $\iota^*: \text{Rep}_k(G) \rightarrow \text{Rep}_k(\tilde{G})$ is a $(\tilde{G}, \epsilon_G)$ -symmetric Weil cohomology theory.

Suppose $\mathfrak{X}$ is a smooth projective $\mathcal{K}$ -variety. By definition $H^\bullet(\mathfrak{X})$ and $\tilde{H}^\bullet(\mathfrak{X})$ coincide as super k -vector spaces and only differ as representations: $H^\bullet(\mathfrak{X})$ is a representation of G and $\tilde{H}^\bullet(\mathfrak{X})$ is the same vector space viewed as a representation of the subgroup $\tilde{G} \subset G$. Therefore, the constructions of the two maximal A-model $\mathbb{k}$ -analytic F-bundles associated with these symmetric Weil cohomology theories are compatible. In concrete terms, the maximal A-model $\mathbb{k}$ -analytic F-bundle associated with the $(\tilde{G}, \epsilon_G)$ -symmetric Weil cohomology theory $\tilde{H}^\bullet$ equals the maximal A-model $\mathbb{k}$ -analytic F-bundle $(\mathcal{H}, \nabla)/B_{\mathfrak{X}}$ associated with the (G, ϵ_G) -symmetric Weil cohomology theory $H^\bullet$ but instead of considering $(\mathcal{H}, \nabla)/B_{\mathfrak{X}}$ as a G -equivariant F-bundle, one must consider it as a $\tilde{G}$ -equivariant F-bundle. Thus we have an inclusion $B_{\mathfrak{X}}^G \subset B_{\mathfrak{X}}^{\tilde{G}}$ and the spectral cover of $\kappa|_{B_{\mathfrak{X}}^G}$ is the restriction of the spectral cover of $\kappa|_{B_{\mathfrak{X}}^{\tilde{G}}}$ to the submanifold $B_{\mathfrak{X}}^G$.

Tracing the way the components of the unramified part of the reduced spectral cover split under this restriction gives us a natural map of local atoms

$$\iota^{\diamond, \text{loc}}: \text{Atoms}_{\tilde{G}}^{\mathcal{K}, \text{loc}} \longrightarrow \mathbb{Z}_{\geq 0}^{\oplus \text{Atoms}_G^{\mathcal{K}, \text{loc}}},$$

and also a natural map

$$\iota^\diamond: \text{Atoms}_{\tilde{G}}^{\mathcal{K}} \longrightarrow \mathbb{Z}_{\geq 0}^{\oplus \text{Atoms}_G^{\mathcal{K}}}$$

of atoms.

Thus we tautologically get a natural compatibility of the corresponding chemical formula multisets, namely

$$\text{CF}_G(\mathfrak{X}) \circ \iota^\diamond = \text{CF}_{\tilde{G}}(\mathfrak{X}).$$

5.2.7 Reduction to G -invariants.

For computational utility it is very useful to note that the local atoms of a projective variety, are captured completely by the eigenvalues for the action of the Euler vector field on the G -invariant part of the fiber of the F-bundle. To explain this we will need some preliminary statements. Fix a field k of characteristic zero. Suppose G is an algebraic reductive group over k , and $\epsilon_G \in G$ is a central element of order 2.

Suppose $(A, \circ, \mathbf{1}_A)$ is a finite dimensional commutative unital superalgebra over some algebraic closure $\bar{k} \supset k$. Suppose the reductive group G acts linearly on A , preserving the unital algebra structure, and so that ϵ_G acts as the parity operator. Because ϵ_G is central, we will have that the G -fixed part A^G of A is a totally even subalgebra in A . Let $a \in A^G$ be a G -invariant element. Multiplication by a gives a linear operator on the finite dimensional vector space A which automatically will preserve the subalgebras $A^G \subset A^{\text{ev}} \subset A$.

Given a linear operator $T: V \rightarrow V$ on a finite dimensional $\bar{k}$ -vector space V we will write $\text{redspec}(T) \subset \bar{k}$ for the reduced spectrum of T , i.e. for the set of distinct eigenvalues of T .

With this notation we now have the following

Lemma 5.19. *If $a \in A^G$, then multiplication by a has the same eigenvalues on all of A and on the subalgebra of G -invariants A^G . More precisely we have*

$$\text{redspec}(a \circ (-)) = \text{redspec}((a \circ (-))|_{A^{\text{ev}}}) = \text{redspec}((a \circ (-))|_{A^G}).$$

Proof. We have a natural algebra homomorphism $\text{ev}_a: \bar{k}[x] \rightarrow A^G \subset A$, given by $\text{ev}_a(f(x)) = f(a)$. Since A is finite dimensional over $\bar{k}$, the homomorphism ev_a has a non-trivial kernel ideal which will be generated by a monic polynomial $p(x)$, i.e. $\ker(\text{ev}_a) = (p) \triangleleft \bar{k}[x]$.

Since $\bar{k}[x]/(p) \hookrightarrow A^{\text{ev}} \subset A$ is an injective algebra homomorphism, it follows that the map $\text{Spec}(A^{\text{ev}}) \rightarrow \text{Spec}(\bar{k}[x]/(p))$ is dominant. If we decompose p into a product of linear factors $p(x) = \prod_{i=1}^s (x - \lambda_i)^{d_i}$ with $\lambda_i \neq \lambda_j$ if $i \neq j$, then $\bar{k}[x]/(p) = \bigoplus_{i=1}^m \bar{k}[x]/((x - \lambda_i)^{d_i})$ and so $\text{Spec}(A^{\text{ev}})$ dominates the closed zero dimensional subscheme $M = \sqcup_{i=1}^m \text{Spec}(\bar{k}[x]/((x - \lambda_i)^{d_i})) \subset \mathbb{A}_{\bar{k}}^1$. Thus $\text{Spec}(A)$ is a disconnected affine superscheme which is a disjoint union $\text{Spec}(A) = \sqcup_{i=1}^m S_i$, where S_i is the preimage of the i -th connected component $\text{Spec}(\bar{k}[x]/((x - \lambda_i)^{d_i}))$ of M under the map $\text{Spec}(A) \rightarrow \text{Spec}(A^{\text{ev}}) \rightarrow M$.

Let A_i be the algebra of global functions on S_i . Then A decomposes into a direct sum $A = \bigoplus_{i=1}^m A_i$ of unital commutative super algebras. Since a is G -invariant, this decomposition is a decomposition of unital algebras which are G -representations, and so we must have $G \cdot \mathbf{1}_{A_i} = \mathbf{1}_{A_i}$. Similarly, the projection a_i of a into A_i must be G -invariant. But by construction, the minimal polynomial of $(a \circ (-))|_{A_i} = a_i \circ (-): A_i \rightarrow A_i$ is equal to $(x - \lambda_i)^{e_i}$ for some $0 < e_i \leq d_i$, and so for every $i = 1, \dots, m$ we have

$$\text{redspec}((a \circ (-))|_{A_i^G}) = \text{redspec}((a \circ (-))|_{A_i^{\text{ev}}}) = \text{redspec}((a \circ (-))|_{A_i}) = \{\lambda_i\}.$$

This concludes the proof of the lemma. $\square$

Suppose now $\mathfrak{X}/\mathcal{K}$ is a smooth projective variety, and suppose $(\mathcal{H}, \nabla)/B_{\mathfrak{X}}$ is the corresponding A-model non-archimedean $\mathbb{k}$ -analytic F-bundle associated with $\mathfrak{X}$, $k \subset \mathbb{k}$, and a (G, ϵ_G) -symmetric Weil cohomology theory $H^\bullet(-)$ on $\mathcal{K}$ -varieties.

Recall that $(\mathcal{H}, \nabla)/B_{\mathfrak{X}}$ is a $G(k)$ -equivariant maximal F-bundle, such that for $\mathcal{H}_o = \mathcal{H}|_{B_{\mathfrak{X}} \times \{u=0\}}$ the induced quantum product $\star: \mathcal{H}_o^{\otimes 2} \rightarrow \mathcal{H}_o$ and the Euler vector field $\text{Eu} \in H^0(B_{\mathfrak{X}}, T_{B_{\mathfrak{X}}}) =$

$H^0(B_{\mathfrak{X}}, \mathcal{H}_o)$ are preserved by $G(k)$. The group $G(k)$ acts on the restriction $\mathcal{H}_o|_{B_{\mathfrak{X}}^G}$ by bundle automorphisms and so the quantum multiplication by $\text{Eu}|_{B_{\mathfrak{X}}^G}$ preserves the subbundle $(\mathcal{H}_o|_{B_{\mathfrak{X}}^G})^{G(k)} \subset \mathcal{H}_o|_{B_{\mathfrak{X}}^G}$ of $G(k)$ -invariants, which is also equal to the bundle of $\mathcal{G}$ -invariants⁴. Let $\mathfrak{eu}$ denote the restriction of the operator $\text{Eu}|_{B_{\mathfrak{X}}^G} \star (-)$ to the subbundle $(\mathcal{H}_o|_{B_{\mathfrak{X}}^G})^{G(k)}$. Let $\mathfrak{S}_{\mathfrak{X}} \rightarrow B_{\mathfrak{X}}^G$ be the spectral cover parametrizing the eigenvalues of the operator

$$\mathfrak{eu} : (\mathcal{H}_o|_{B_{\mathfrak{X}}^G})^{G(k)} \rightarrow (\mathcal{H}_o|_{B_{\mathfrak{X}}^G})^{G(k)}.$$

By construction $\mathfrak{S}_{\mathfrak{X}} \subset \tilde{B}_{\mathfrak{X}}$ is a closed analytic subspace in the spectral cover $\tilde{B}_{\mathfrak{X}}$ parametrizing the eigenvalues of $\text{Eu}|_{B_{\mathfrak{X}}^G} \star (-)$ acting on all of $\mathcal{H}_o|_{B_{\mathfrak{X}}^G}$. However, since $\mathcal{H}_o$ is a bundle of unital commutative super-algebras and since Eu is even, Lemma 5.19 implies that the inclusion $\mathfrak{S}_{\mathfrak{X}} \subset \tilde{B}_{\mathfrak{X}}$ induces an identification $\mathfrak{S}_{\mathfrak{X}, \text{red}} = \tilde{B}_{\mathfrak{X}, \text{red}}$ of the associated reduced $\mathbb{k}$ -analytic spaces. Therefore we can also describe $\tilde{U}_{\mathfrak{X}} \rightarrow U_{\mathfrak{X}}$ as the unramified part of the reduced spectral cover $\mathfrak{S}_{\mathfrak{X}, \text{red}} \rightarrow B_{\mathfrak{X}}^G$ for $\text{Eu} \star (-)$ acting on the bundle of $G(k)$ -invariants, or equivalently the bundle of $\mathcal{G}$ -invariants.

5.3 G -atomic F-bundles

The abstract definition of atoms in Section 5.2.6 is universal but hard to use in practice. In this section, we introduce the notion of G -atomic F-bundle, which will serve as a simpler version of G -atom for the purpose of this paper.

5.3.1 Definition of an atomic F-bundle

Next we will consider actions of germs of analytic Lie groups on F-bundles. Let k be a field of characteristic zero, and let $\mathbb{k}$ be an algebraically closed non-archimedean field containing k , extending the trivial valuation on k . Fix a pair (G, ϵ_G) as in Section 5.1. More precisely, we replace G by any finite dimensional quotient.

Let $G, G^{\square} \subset G^{\text{an}}, G_{\mathbb{k}}^{\square} \subset G_{\mathbb{k}}^{\text{an}}$ and $\mathcal{G}$ be as in Section 5.2.2. Let $(\mathcal{H}, \nabla)/B$ be a $\mathcal{G}$ -equivariant $\mathbb{k}$ -analytic F-bundle where the $\mathcal{G}$ -action on $\mathbb{D}$ is trivial. Let $b \in B$ be a rigid point fixed by $\mathcal{G}$ and (B, b) the germ of B at b . Restricting $(\mathcal{H}, \nabla)/B$ to (B, b) , we obtain a **$\mathcal{G}$ -equivariant $\mathbb{k}$ -analytic F-bundle germ**.

Definition 5.20. A (G, ϵ_G) -**atomic F-bundle** (or a **G -atomic F-bundle** for short) is a $\mathcal{G}$ -equivariant $\mathbb{k}$ -analytic F-bundle germ $(\mathcal{A}, \nabla)/(B, b)$ such that

- (1) $(\mathcal{A}, \nabla)/(B, b)$ is maximal,
- (2) the action of ϵ_G on the super vector bundle $\mathcal{A}$ is the $\mathbb{Z}/2$ -grading action,
- (3) the Eu -action on $\mathcal{A}|_{(B^G, b), u=0}$ has a single eigenvalue.

⁴Note that since $G(k)$ acts trivially on the base $B_{\mathfrak{X}}^G$ and by linear analytic automorphisms on $\mathcal{H}_o$, it follows that the linear action of $G(k)$ along the fibers of $\mathcal{H}_o|_{B_{\mathfrak{X}}^G}$ extends to the analytic action of $\mathcal{G}$ corresponding to the linear action of $G_{\mathbb{k}}$ on $H^{\bullet}(\mathfrak{X}) \otimes_{\mathbb{k}} \mathbb{k}$. Again since $H^{\bullet}(\mathfrak{X})$ is a finite dimensional representation of the proreductive group $G(k)$, the fixed points of the $\mathcal{G}$ -action on $(H^{\bullet}(\mathfrak{X}) \otimes_{\mathbb{k}} \mathbb{k})^{\text{an}}$ will be equal to $(H^{\bullet}(\mathfrak{X})^{G(k)} \otimes_{\mathbb{k}} \mathbb{k})^{\text{an}}$, i.e. to the fixed points of the $G(k)$ -action on $(H^{\bullet}(\mathfrak{X}) \otimes_{\mathbb{k}} \mathbb{k})^{\text{an}}$.

In the rest of this section we will discuss the G -atomic F-bundles coming from Gromov-Witten theory. Ultimately the invariants we need will be universal $\mathbf{A}$ -model G -atomic F-bundles, i.e. $\mathbf{A}$ -model G -atomic F-bundles associated with some Tannakian theory of motives or some particular realization of motives. Such G -atomic F-bundles are dubbed G -atoms and we discuss the most useful examples below.

5.3.2 Geometric G -atomic F-bundles

We use the notations from Section 5.2.2. By the spectral decomposition theorem (Theorem 4.1), for every rigid point $b \in U_{\mathfrak{X}}$, the restriction of the F-bundle $(\mathcal{H}, \nabla)$ to the germ $(U_{\mathfrak{X}}, b)$ splits into a collection of G -atomic F-bundles $\mathcal{A}_{\mathfrak{X}, b, \tilde{b}}$ indexed by the points $\tilde{b}$ of the fiber $\tilde{U}_{\mathfrak{X}, b}$.

Definition 5.21. A **geometric G -atomic F-bundles** of $\mathcal{K}$ -varieties is a G -atomic F-bundle $\mathcal{A}_{\mathfrak{X}, b, \tilde{b}}$ arising from the above construction. We consider the equivalence relation on geometric G -atomic F-bundles generated by the following:

- (1) Two geometric G -atomic F-bundles are equivalent if the G -atomic F-bundles are isomorphic.
- (2) Two geometric G -atomic F-bundles $\mathcal{A}_{\mathfrak{X}, b_1, \tilde{b}_1}$ and $\mathcal{A}_{\mathfrak{X}, b_2, \tilde{b}_2}$ arising from the same variety $\mathfrak{X}$ are equivalent if $\tilde{b}_1$ and $\tilde{b}_2$ belong to the same connected component of $\tilde{U}_{\mathfrak{X}}$.

We denote by $\text{Atoms}_G^{\mathcal{K}, \text{F}}$ the set of geometric G -atomic F-bundles modulo the above equivalence relation. Given a smooth projective $\mathcal{K}$ -variety $\mathfrak{X}$, let $\text{CF}_G^{\text{F}}(\mathfrak{X})$ denote the multiset of geometric G -atomic F-bundles associated to $\mathfrak{X}$. Formally, $\text{CF}_G^{\text{F}}(\mathfrak{X})$ is a multiplicity function $\text{CF}_G^{\text{F}}(\mathfrak{X}): \text{CF}_G^{\mathcal{K}, \text{F}} \rightarrow \mathbb{Z}_{\geq 0}$ with finite support.

Proposition 5.22. *We have a map from the set $\text{Atoms}_G^{\mathcal{K}}$ of G -atoms of varieties over $\mathcal{K}$ to the set $\text{Atoms}_G^{\mathcal{K}, \text{F}}$ of geometric G -atomic F-bundles sending each connected component $\tilde{U}_{\mathfrak{X}, \alpha}$ to a G -atomic F-bundle $\mathcal{A}_{\mathfrak{X}, b, \tilde{b}}$ for some rigid point $b \in U_{\mathfrak{X}}$ and $\tilde{b} \in \tilde{U}_{\mathfrak{X}, b}$.*

Proof. The equivalences (1) and (2) in Definition 5.21, imply that there is a well defined map from local G -atoms to geometric G -atomic F-bundles. Indeed, suppose $\alpha \in \pi_0(\tilde{U}_{\mathfrak{X}})$, let $b \in U_{\mathfrak{X}}$ be a rigid point, and let $x \in \tilde{U}_{\mathfrak{X}}$ be a rigid point sitting over b . Then in the spectral decomposition (see Theorem 4.1) of the F-bundle germ $(\mathcal{H}, \nabla)/(B_{\mathfrak{X}}, b)$ there is a factor $(\mathcal{A}_{\mathfrak{X}, b, x}, \nabla)/(^x B_{\mathfrak{X}}^x b)$ corresponding to the generalized κ -eigenspace with eigenvalue x . This factor is by construction a geometric G -atomic F-bundle whose equivalence class depends only on α but not the points b and x .

This gives a map from the set $\text{Atoms}_G^{\mathcal{K}, \text{loc}}$ of local G -atoms of varieties over $\mathcal{K}$ to the set $\text{Atoms}_G^{\mathcal{K}, \text{F}}$. It remains to check that this map factors through the equivalence relations defined in Sections 5.2.3–5.2.5. Concretely, we need to check the additivity properties:

- (1) For two smooth projective $\mathcal{K}$ -varieties $\mathfrak{X}_1, \mathfrak{X}_2$, we have

$$\text{CF}_G^{\text{F}}(\mathfrak{X}_1 \sqcup \mathfrak{X}_2) = \text{CF}_G^{\text{F}}(\mathfrak{X}_1) + \text{CF}_G^{\text{F}}(\mathfrak{X}_2).$$

- (2) For a smooth projective $\mathcal{K}$ -variety $\mathfrak{X}$, a smooth closed subvariety $\mathfrak{Z} \subset \mathfrak{X}$ of codimension r , and the blowup $\hat{\mathfrak{X}}$ of $\mathfrak{X}$ along $\mathfrak{Z}$, we have

$$\text{CF}_G^{\text{F}}(\hat{\mathfrak{X}}) = \text{CF}_G^{\text{F}}(\mathfrak{X}) + (r-1) \cdot \text{CF}_G^{\text{F}}(\mathfrak{Z}).$$

(3) For a smooth projective $\mathcal{K}$ -variety $\mathfrak{X}$, a rank r vector bundle V on $\mathfrak{X}$, and the associated projective bundle $\mathbb{P}(V)$, we have

$$\mathrm{CF}_G^{\mathbf{F}}(\mathbb{P}(V)) = r \cdot \mathrm{CF}_G^{\mathbf{F}}(\mathfrak{X}).$$

These hold almost tautologically. The $\mathbf{A}$ -model $\mathbf{F}$ -bundle associated to a disjoint union $\mathfrak{X}_1 \sqcup \mathfrak{X}_2$ is isomorphic to the external direct sum of the respective $\mathbf{A}$ -model $\mathbf{F}$ -bundles. This implies (1). The remaining statements (2) and (3) follow again from Theorems 4.5 and 4.11. $\square$

Proposition 5.23. *For a G -atom of $\mathcal{K}$ -varieties represented by a G -atomic $\mathbf{F}$ -bundle $(\mathcal{A}, \nabla)$ over a germ (B, b) , the isomorphism class of the $G_{\mathbb{K}}$ -representation $\mathcal{A}|_{b, u=0}$ is well-defined and independent of the choice of the representative.*

Proof. It suffices to check the independence for the equivalence in Definition 5.21(1). Using the notations therein, if $\tilde{b}_1$ and $\tilde{b}_2$ belong to a same connected component $\mathfrak{U}$ of $\tilde{U}_{\mathfrak{X}}$, we obtain a subbundle $\mathcal{A} \subset \mathcal{H}|_{U_{\mathfrak{X}} \times \{u=0\}}$ over $U_{\mathfrak{X}}$ given by generalized eigenspaces of the Eu -action with eigenvalues in V . Since G is a proreductive group, its finite dimensional representations are rigid (see e.g. [61, Proposition 15.15]).

Therefore, every fiber of $\mathcal{A}$ gives the same isomorphism class of G -representations. $\square$

Again, the set $\mathrm{Atoms}_G^{\mathcal{K}}$ of G -atoms is naturally filtered by subsets

$$\mathrm{Atoms}_{G, \dim \leq 0}^{\mathcal{K}, \mathbf{F}} \subset \mathrm{Atoms}_{G, \dim \leq 1}^{\mathcal{K}, \mathbf{F}} \subset \cdots, \quad \mathrm{Atoms}_G^{\mathcal{K}, \mathbf{F}} = \bigcup_{d \geq 0} \mathrm{Atoms}_{G, \dim \leq d}^{\mathcal{K}, \mathbf{F}},$$

where an atom α belongs to $\mathrm{Atoms}_{G, \dim \leq d}^{\mathcal{K}, \mathbf{F}}$ if there exists a smooth projective $\mathcal{K}$ -variety of dimension $\leq d$ whose atomic composition contains α .

Splitting the product of varieties into its atoms gives rise to a well-defined map

$$\mathrm{Atoms}_G^{\mathcal{K}, \mathbf{F}} \times \mathrm{Atoms}_G^{\mathcal{K}, \mathbf{F}} \longrightarrow \mathbb{Z}_{\geq 0}^{\oplus \mathrm{Atoms}_G^{\mathcal{K}, \mathbf{F}}}.$$

This map endows the additive monoid $\mathbb{Z}_{\geq 0}^{\oplus \mathrm{Atoms}_G^{\mathcal{K}, \mathbf{F}}}$ with the structure of a commutative associative unital semiring, and similarly endows the abelian group $\mathbb{Z}^{\oplus \mathrm{Atoms}_G^{\mathcal{K}, \mathbf{F}}}$ with the structure of a commutative ring.

Note that Bittner's relations [13] ensure that the assignment which sends an isomorphism class of varieties to its chemical formula

$$[\mathfrak{X}] \longrightarrow \mathrm{CF}_G^{\mathbf{F}}(\mathfrak{X}),$$

gives rise to a well defined ring homomorphism

$$\mathrm{cf}_G^{\mathbf{F}}: K_0(\mathrm{Var}_{\mathcal{K}}) \longrightarrow \mathbb{Z}^{\oplus \mathrm{Atoms}_G^{\mathcal{K}, \mathbf{F}}}$$

from the Grothendieck ring of varieties $K_0(\mathrm{Var}_{\mathcal{K}})$ to the ring $\mathbb{Z}^{\oplus \mathrm{Atoms}_G^{\mathcal{K}, \mathbf{F}}}$. Again the homomorphism $\mathrm{cf}_G^{\mathbf{F}}$ can be thought of as the function assigning to each K -class of varieties its **atomic $\mathbf{F}$ -bundle valued chemical formula**, i.e. the list of its G -atomic $\mathbf{F}$ -bundles and their multiplicities.

5.4 Hodge atoms

In this section we discuss a particularly versatile example of (G, ϵ_G) -atoms which we call Hodge atoms. These are the atoms corresponding to the symmetry pair from Example 5.5. In other words, we take $\mathcal{K} = \mathbb{C}$, $k = \mathbb{Q}$ and we consider the Mumford-Tate pair $(\text{Hod}, \epsilon_{\text{Hod}})$ controlling the polarizable pure rational $\mathbb{Z}/2$ -graded Hodge structures. We will also fix the rational Betti cohomology $H_B^\bullet(-, \mathbb{Q})$ as our $(\text{Hod}, \epsilon_{\text{Hod}})$ -symmetric Weil cohomology theory. Let $\mathbb{k} = \overline{\mathbb{Q}}((\mathbf{y}^\mathbb{Q}))$ be the algebraically-closed non-archimedean field of Puiseux series in an auxiliary variable $\mathbf{y}$. For a smooth complex projective variety X we take $(\mathcal{H}, \nabla)/B_X$ to be the maximal non-archimedean A-model F-bundle associated with X , $\mathcal{K} = \mathbb{C}$, $\mathbb{k} = \overline{\mathbb{Q}}((\mathbf{y}^\mathbb{Q})) \supset k = \mathbb{Q}$, and $H^\bullet(-, \mathbb{Q})$, that we described in Definition 3.32. By construction $(\mathcal{H}, \nabla)/B_X$ is $(\text{Hod}(\mathbb{Q}), \epsilon_{\text{Hod}})$ -equivariant, and so by specializing Definition 5.21 to this context we arrive at the notion of **Hod-atoms** which we will call **Hodge atoms**. Since Hodge atoms are our main invariant let us explicate.

Let $B_X^{\text{Hod}} \subset B_X$ be the locus of $\text{Hod}(\mathbb{Q})$ -fixed points in the non-archimedean super $\mathbb{k}$ -analytic variety B_X . Let $\tilde{B} \rightarrow B^{\text{Hod}}$ be the ramified covering given by the spectrum of the Eu -action, $U_X \subset B^{\text{Hod}}$ the locus where the number of eigenvalues of the Eu -action is maximal, and $\tilde{U}_X := \tilde{B}_{\text{red}} \times_B U_X$. The set of **local Hodge atoms** associated to X is again $\pi_0(\tilde{U}_X)$ and the multiplicity of an $\alpha \in \pi_0(\tilde{U}_X)$ is defined to be the degree of the corresponding connected component $\tilde{U}_{X,\alpha}$ of $\tilde{U}_X$ as a covering of U_X .

The set **HAtoms** of all Hodge atoms of smooth complex projective varieties is the quotient

$$\text{HAtoms} := \left(\bigsqcup_{[X]} \pi_0(\tilde{U}_X) / \text{Aut}(X) \right) / \sim,$$

where the union is taken over isomorphism classes of complex smooth projective varieties, and the equivalence relation $\sim$ is generated by the elementary equivalences corresponding to disjoint unions, blowups with smooth centers, and projective bundles.

For future reference let us record a few elementary but useful properties of Hodge atoms.

Lemma 5.24. *Let X be a connected complex smooth projective variety with a numerically effective canonical class. Then $\text{HAtoms}(X)$ consists of a single atom $\eta(X)$.*

Proof. This is a special case of Lemma 5.18. □

Lemma 5.25. *Let X be a smooth complex projective variety. For any rigid point $b \in U_X$, the $\text{Hod}_{\mathbb{k}}(\mathbb{k})$ -representation $(\mathcal{H}_o)_b = \mathcal{H}_{(b,0)}$ is induced from a finite dimensional $\overline{\mathbb{Q}}$ -linear representation of $\text{Hod}(\overline{\mathbb{Q}})$ whose isomorphism class is independent of the choice of b .*

Proof. This is a special case of Proposition 5.23. By definition

$$(\mathcal{H}_o)_b \cong H_B^\bullet(X, \mathbb{k}) = H_B^\bullet(X, \mathbb{Q}) \otimes_{\mathbb{Q}} \mathbb{k}$$

is equipped with the action of $\text{Hod}_{\mathbb{k}}(\mathbb{k})$ induced from the action of $\text{Hod}(\mathbb{Q})$ on $H_B^\bullet(X, \mathbb{Q})$ by extension of scalars.

Since $\mathbb{Q} \subset \overline{\mathbb{Q}} \subset \overline{\mathbb{Q}}((\mathbf{y}^\mathbb{Q})) = \mathbb{k}$, we can also view the $\mathbb{k}$ -linear action of $\text{Hod}_{\mathbb{k}}(\mathbb{k})$ on $H_B^\bullet(X, \mathbb{k}) = H_B^\bullet(X, \overline{\mathbb{Q}}) \otimes_{\overline{\mathbb{Q}}} \mathbb{k}$ as being induced by extension of scalars from the natural action of $\text{Hod}_{\overline{\mathbb{Q}}}(\overline{\mathbb{Q}})$ on $H_B^\bullet(X, \overline{\mathbb{Q}})$. But $\text{Hod}_{\overline{\mathbb{Q}}}$ is a proreductive group over the algebraically closed field $\overline{\mathbb{Q}}$, and the vector

space $H_B^\bullet(X, \overline{\mathbb{Q}})$ is finite dimensional, so the action ψ_b of $\text{Hod}_{\overline{\mathbb{Q}}}(\overline{\mathbb{Q}})$ on $H_B^\bullet(X, \overline{\mathbb{Q}})$ factors through a completely reducible action of a finite dimensional reductive group over $\overline{\mathbb{Q}}$. The irreducible representations of this finite dimensional group are classified by their highest weights and so can not vary continuously with b . Therefore, the isomorphism class of the representation ψ_b is independent of the choice of b . $\square$

Using this we can define some basic numerical invariants of Hodge atoms. Let $\alpha \in \text{HAtoms}$ be a Hodge atom. Then there is a smooth projective variety X so that the equivalence class α is represented by a local atom $\alpha \in \text{HAtoms}^{\text{loc}}(X)$, i.e. by a choice of a connected component $\mathfrak{U}_\alpha$ of the unramified part $\tilde{U}_X$ of the reduced spectral cover of the operator $\kappa = \text{Eu} \star (-) : \mathcal{H}_o|_{B_X^{\text{Hod}}} \rightarrow \mathcal{H}_o|_{B_X^{\text{Hod}}}$ for the $\mathbb{k}$ -analytic A-model maximal F-bundle $(\mathcal{H}, \nabla)/B_X$.

The subbundle $\mathcal{E}^\alpha \subset \mathcal{H}_o|_{U_X}$ of generalized eigenspaces of $\text{Eu} \star (-)$ corresponding to the eigenvalues parametrized by $\tilde{U}_{X, \alpha}$ is a $\text{Hod}_{\mathbb{k}}(\mathbb{k})$ -invariant. By Lemma 5.25, for any rigid point $b \in U_X$ the representation of $\text{Hod}_{\mathbb{k}}(\mathbb{k})$ on $\mathcal{E}_b^\alpha$ is induced by extension of scalars from some finite dimensional $\overline{\mathbb{Q}}$ -linear representation E^α of $\text{Hod}_{\overline{\mathbb{Q}}}$ whose isomorphism class is independent of b . As we have indicated by the notation, the isomorphism class of the $\text{Hod}_{\overline{\mathbb{Q}}}$ -module E^α depends only on the atom α and not on the particular local atom α representing the equivalence class α . Indeed, this follows immediately from the fact that the elementary equivalences used defining Hodge atoms all respect the action of Hod .

Thus to every Hodge atom α we have assigned a finite dimensional $\overline{\mathbb{Q}}$ -linear representation E^α of $\text{Hod}_{\overline{\mathbb{Q}}}$ which is well defined up to isomorphism. With this notation we now have the following

Definition 5.26. For every Hodge atom $\alpha \in \text{HAtoms}$ we define

- (1) the **dimension** ρ_α of the space of Hodge classes in α as $\dim_{\mathbb{C}}(E^\alpha \otimes \mathbb{C})^{\text{Hod}_{\mathbb{C}}}$, and
- (2) the **Hodge polynomial** $P_\alpha(t)$ of α as the polynomial $P_\alpha(t) \in \mathbb{Z}[t, t^{-1}]$ whose coefficient at t^k is equal to $\dim_{\mathbb{C}}(E^\alpha \otimes \mathbb{C})^k$, where $(E^\alpha \otimes \mathbb{C})^k$ is the subspace in $E^\alpha \otimes \mathbb{C}$ on which the subgroup $\mathbb{C}^\times \cong \text{H}_{\mathbb{C}}(\mathbb{C}) \subset \text{Hod}_{\mathbb{C}}(\mathbb{C})$ acts with weight k .

Remark 5.27. Suppose the $\overline{\mathbb{Q}}$ -linear representation E^α of $\text{Hod}_{\overline{\mathbb{Q}}}$ is obtained by extension of scalars from a $\mathbb{Q}$ -linear representation of Hod . But a $\mathbb{Q}$ -linear representation of Hod is a pure $\mathbb{Z}/2$ -weighted $\mathbb{Q}$ -Hodge structure, i.e. an ordinary pure $\mathbb{Q}$ -Hodge structures whose bigrading has been folded to the single Hochschild grading. Therefore if $E^\alpha = V \otimes_{\mathbb{Q}} \overline{\mathbb{Q}}$ for some pure $\mathbb{Q}$ -Hodge structure, then $(E^\alpha \otimes \mathbb{C})^k = \bigoplus_{p,q: p-q=k} V^{p,q}$. In other words, $P_\alpha(t)$ is obtained from the usual two variable Hodge polynomial of $\sum_{p,q} \dim_{\mathbb{C}} V^{p,q} u^p v^q$ of V after the substitution $u = t, v = t^{-1}$. The symmetry of Hodge numbers of V then implies that in this case the Hodge polynomial $P_\alpha(t)$ will be reciprocal (= invariant under $t \mapsto t^{-1}$).

Of course, for a general $\text{Hod}_{\overline{\mathbb{Q}}}$ -representation the associated Hodge polynomial will not have any symmetry, so for a general Hodge atom the Hodge polynomial $P_\alpha(t)$ will not be reciprocal.

Proposition 5.28. For each Hodge atom α the dimension ρ_α of the space of Hodge classes in α satisfies $\rho_\alpha \geq 1$.

Proof. This follows immediately from the fact that fiber of the Hodge atom at a rigid point of U_X is of the form $A_\alpha \otimes_{\overline{\mathbb{Q}}} \mathbb{k}$ where A_α is a non-zero unital commutative associative superalgebra over $\overline{\mathbb{Q}}$ which is endowed with a $\text{Hod}_{\overline{\mathbb{Q}}}(\overline{\mathbb{Q}})$ -action. In particular there is a non-zero even $\text{Hod}_{\overline{\mathbb{Q}}}(\overline{\mathbb{Q}})$ -invariant vector, namely the unit 1_{A_α} for the multiplication in A_α . $\square$

Remark 5.29. Let $\alpha \in \text{HAtoms}$ be a Hodge atom and let $\alpha \in \pi_0(\tilde{U}_X)$ be a local atom representative of α . Let $x \in \tilde{U}_{X,\alpha}$ be a rigid point, let $b_x \in U_X$ be its image, and let $({}^x\mathcal{H}, {}^x\nabla)$ the associated Hod-atomic F-bundle over the germ $({}^xB, b_x)$. Then $\dim({}^xB)^{\text{Hod}(\overline{\mathbb{Q}})} = \rho_\alpha$ and the fiber ${}^x\mathcal{H}_{(b_x,0)}$ is isomorphic to $E^\alpha \otimes_{\overline{\mathbb{Q}}} \mathbb{k}$ for a finite dimensional $\overline{\mathbb{Q}}$ -linear representation E^α of $\text{Hod}_{\overline{\mathbb{Q}}}(\overline{\mathbb{Q}})$ which has Hodge polynomial $P_\alpha(t)$ and has a space $(E^\alpha)^{\text{Hod}(\overline{\mathbb{Q}})}$ of Hodge classes of dimension $\dim(E^\alpha)^{\text{Hod}(\overline{\mathbb{Q}})} = \rho_\alpha$.

Finally, note that as in the case of the general G -atoms, the set of Hodge atoms is filtered by the minimal dimension of a smooth projective variety in whose atomic composition a given atom can appear. Again, this filtration together with the weak factorization theorem [81, 1] imply that Hodge atoms can obstruct birational equivalence. Concretely we have the following

Proposition 5.30. (Non-rationality criterion) *Let X be a smooth complex projective variety of dimension $d \geq 2$. Suppose we have a Hodge atom α which appears in the atomic decomposition of X and is such that $\alpha \notin \text{HAtoms}_{\dim \leq d-2}$. Then X can not be a rational variety.*

Proof. This is a special case of Proposition 5.17. $\square$

5.5 Atoms from motivated motives

Suppose that we are in the setting of Example 5.6, i.e. $\mathcal{K} \subset \mathbb{C}$, $k = \mathbb{Q}$, and $H^\bullet(-)$ is Betti cohomology. Take $\mathbb{k} := \mathbb{Q}((\mathbf{y}^\mathbb{Q}))$. As discussed in section 3.5.2.2 the corresponding motivic Galois group $\text{Mot}^\mathcal{K}$ of the category $\mathcal{C}^\mathcal{K}/\text{Tate}$ of André's pure $\mathbb{Z}/2$ -graded motivated motives is a proreductive algebraic group over $\mathbb{Q}$.

Let $\mathfrak{X}$ be a smooth projective variety over $\mathcal{K}$. View the Betti cohomology as a $(\text{Mot}^\mathcal{K}, \epsilon_{\text{Mot}^\mathcal{K}})$ -symmetric fiber functor $H^\bullet(-) : \mathcal{C}^\mathcal{K} \rightarrow \text{Rep}_{\mathbb{Q}}(\text{Mot}^\mathcal{K})$. Then we have a $\text{Mot}^\mathcal{K}$ -equivariant maximal non-archimedean $\mathbb{k}$ -analytic A-model F-bundle $(\mathcal{H}, \nabla)/B_{\mathfrak{X}}$ corresponding to $\mathfrak{X}$, $\mathcal{K}$, k , and $H^\bullet(\mathfrak{X})$, obtained by base changing the maximal $k((\mathbf{y}))$ -analytic F-bundle introduced in Definition 3.52 from $k((\mathbf{y}))$ to $\mathbb{k}$. The atomic F-bundles arising from such a $(\mathcal{H}, \nabla)/B_{\mathfrak{X}}$ are by definition $\text{Mot}^\mathcal{K}$ -equivariant F-bundles defined over the locus $B_{\mathfrak{X}}^{\text{Mot}^\mathcal{K}}$ corresponding to algebraic cycles on $\mathfrak{X}$ that are defined over $\mathcal{K}$. The corresponding multiset of atoms $\text{Atoms}_{\text{Mot}^\mathcal{K}}(\mathfrak{X})$ is the most sensitive obstruction to $\mathcal{K}$ -rationality our atom formalism provides. Note also that if $\mathfrak{X}$ is such that the eigenvalues of $\text{Eu} \star (-)$ happen to be already defined over $\mathbb{Q}((\mathbf{y}))$, then by section 3.5.2.2.4, the base of the corresponding atomic F-bundle contains a non-split torus which is twisted by a homomorphism $\text{Gal}(\overline{\mathcal{K}}/\mathcal{K}) \rightarrow \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ from the Galois group of the field of definition, to the Galois group of the coefficient field of our cohomology theory.

More invariantly, without making a choice of a fiber functor, we can define the **intrinsic motivated atoms** of smooth projective varieties over $\mathcal{K}$. The category of nc-André motives $\mathcal{C}^\mathcal{K}/\text{Tate}$ is a $\mathbb{Z}/2$ -graded $\mathbb{Q}$ -linear semisimple tensor category. Let $\mathbb{1} \in \text{ob}(\mathcal{C}^\mathcal{K}/\text{Tate})$ be the zero dimensional Tate motive. As before, for a smooth projective variety $\mathfrak{X}$ we denote the André motive of $\mathfrak{X}$ by $h(\mathfrak{X})$. Then the $\mathbb{Z}/2$ -graded rational vector space $M(\mathfrak{X}) := \text{Hom}_{\mathcal{C}^\mathcal{K}/\text{Tate}}(\mathbb{1}, h(\mathfrak{X}))$ is intrinsically associated with $\mathfrak{X}$, and the standard construction of Gromov-Witten classes gives again a quantum product

$$\star : M(\mathfrak{X}) \otimes M(\mathfrak{X}) \longrightarrow M(\mathfrak{X}) \widehat{\otimes} \mathbb{Q}[\text{NE}(\mathfrak{X}, \mathbb{Z})] \widehat{\otimes} \widehat{\text{Sym}} M(\mathfrak{X})^\vee.$$

Since $\text{NE}(\mathfrak{X}, \mathbb{Z})$ parametrizes numerically effective classes of curves in $\mathfrak{X}$ that are defined over $\mathcal{K}$ and $M(\mathfrak{X})$ parametrizes motivated cycles in $\mathfrak{X}$ defined over $\mathcal{K}$, it follows that the actions of $\text{Gal}(\overline{\mathcal{K}}/\mathcal{K})$

on $\mathrm{NE}(\mathfrak{X}, \mathbb{Z})$ and $M(\mathfrak{X})$ are both trivial. The construction from section 3.5.2.2 now produces an overmaximal $\mathbb{k}$ -analytic A-model F-bundle $(\mathcal{M}, \nabla)/b_{\mathfrak{X}}$ and a maximal $\mathbb{k}$ -analytic A-model F-bundle $(\mathcal{M}, \nabla)/b_{\mathfrak{X}}$. Here $b_{\mathfrak{X}}$ is a product of an $\mathbb{k}$ -analytic affine line (for the motivated cycles of cohomological degree zero), a polydisk (for the motivated cycles of cohomological degree > 0), and a tube domain in the analytic torus $(\mathrm{NS}(\mathfrak{X}, \mathbb{Z})_{\mathrm{tf}} \otimes \mathbb{G}_{m, \mathbb{k}})^{\mathrm{an}}$. Similarly $b_{\mathfrak{X}}$ is a product an $\mathbb{k}$ -analytic affine line (for the motivated cycles of cohomological degree zero), a polydisk (for the motivated cycles of cohomological degree > 2), and a tube domain in the analytic torus $(\mathrm{NS}(\mathfrak{X}, \mathbb{Z})_{\mathrm{tf}} \otimes \mathbb{G}_{m, \mathbb{k}})^{\mathrm{an}}$.

Note that if we use the Betti cohomology fiber functor, then these F-bundles are realized by the $\mathrm{Mot}^{\mathcal{K}}$ -invariant part of the F-bundles we described in section 3.5.2.2, i.e. passing to Betti cohomology identifies $(\mathcal{M}, \nabla)/b_{\mathfrak{X}}$ with $(\mathcal{H}^{\mathrm{Mot}^{\mathcal{K}}}, \nabla)/\mathcal{B}_{\mathfrak{X}}^{\mathrm{Mot}^{\mathcal{K}}}$ and similarly identifies $(\mathcal{M}, \nabla)/b_{\mathfrak{X}}$ with $(\mathcal{H}^{\mathrm{Mot}^{\mathcal{K}}}, \nabla)/B_{\mathfrak{X}}^{\mathrm{Mot}^{\mathcal{K}}}$. In particular, by section 5.2.7 the unramified part $\tilde{u}_{\mathfrak{X}}$ of the reduced spectral cover for the operator κ associated with $(\mathcal{H}, \nabla)/B_{\mathfrak{X}}^{\mathrm{Mot}^{\mathcal{K}}}$ is isomorphic to the unramified part $\tilde{u}_{\mathfrak{X}}$ of the reduced spectral cover of the operator κ associated with $(\mathcal{H}^{\mathrm{Mot}^{\mathcal{K}}}, \nabla)/B_{\mathfrak{X}}^{\mathrm{Mot}^{\mathcal{K}}} \cong (\mathcal{M}, \nabla)/b_{\mathfrak{X}}$.

Using $(\mathcal{M}, \nabla)/b_{\mathfrak{X}}$ as a basic invariant of $\mathfrak{X}$ we can define the set of intrinsic motivated local atoms of $\mathfrak{X}$ as the set $\pi_0(\tilde{u}_{\mathfrak{X}})$ of connected components of $\tilde{u}_{\mathfrak{X}}$ and the set of intrinsic atoms of $\mathcal{K}$ -varieties as the quotient

$$\mathrm{Atoms}_{\mathrm{intr}}^{\mathcal{K}} := \left(\bigsqcup_{[X]} \pi_0(\tilde{u}_{\mathfrak{X}}) / \mathrm{Aut}(\mathfrak{X}) \right) / \sim,$$

where the union is taken over isomorphism classes of complex smooth projective varieties, and the equivalence relation $\sim$ is generated by the elementary equivalences corresponding to disjoint unions, blowups with smooth centers, and projective bundles.

As explained in the previous paragraph, the choice of Betti cohomology as a fiber functor on $\mathcal{C}^{\mathcal{K}}$, gives a bijection between the set of intrinsic motivated atoms $\mathrm{Atoms}_{\mathrm{intr}}^{\mathcal{K}}$ and the set $\mathrm{Atoms}_{\mathrm{Mot}^{\mathcal{K}}}^{\mathcal{K}}$ of $\mathrm{Mot}^{\mathcal{K}}$ -atoms. The intrinsic viewpoint however, has the advantage that the set of atoms $\mathrm{Atoms}_{\mathrm{intr}}^{\mathcal{K}}$ does not depend on any choices and by construction is equipped with an action of the Galois group $\mathrm{Gal}(\mathbb{k}/\mathbb{Q})$ which is an extension of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ by the group $\mathbb{Z}$ of profinite integers.

6 New obstructions to rationality

In this section we will use Hodge atoms and more general G -atoms to prove new non-rationality results. We will use the setup of Sections 5.2, 5.3, and 5.4. Specifically, we consider the pair $(\mathrm{Hod}, \epsilon_{\mathrm{Hod}})$ defined in Example 5.5, where Hod is the Galois group of the Tannakian category of $\mathbb{Z}/2$ -weighted, polarizable, pure $\mathbb{Q}$ -Hodge structures, and $\epsilon_{\mathrm{Hod}} \in \mathrm{Hod}$ is the $\mathbb{Z}/2$ -grading element. The atoms for this symmetry group are the Hodge atoms and we will use them as obstructions to rationality.

Before we can discuss the atomic composition of specific smooth projective varieties, we need to revisit Givental's calculation of the quantum cohomology and quantum differential equation of complete intersections and recast it in the F-bundle language.

6.1 F-bundles and Givental's theorem

Let X be a smooth complex complete intersection of $k \geq 1$ hypersurfaces of degrees $d_1, \dots, d_k$ in $\mathbb{P}^{N-1}$. We will assume that $d_i > 1$ for all $i = 1, \dots, k$, and that $d_{\mathrm{tot}} = d_1 + \dots + d_k \leq N$, i.e.

that X is either a Fano or a Calabi-Yau. Under these assumptions we have $1 \leq k < d_{\text{tot}} \leq N$, $\dim X = N - 1 - k \geq 1$, and $-K_X = \mathcal{O}_X(N - d_{\text{tot}})$.

The image $\mathbf{R} \subset H^\bullet(X, \mathbb{Q})$ of the rational cohomology of $\mathbb{P}^{N-1}$ in the rational cohomology of X is the $(N - k)$ -dimensional subalgebra spanned over $\mathbb{Q}$ by the classes $\{\mathbf{P}^i\}_{i=0}^{N-k-1}$, where $\mathbf{P} = c_1(\mathcal{O}(1)) \in H^2(X, \mathbb{Q})$ is the hyperplane class.

Let $(\mathcal{H}, \nabla)/B_X$ be the maximal non-archimedean, analytic A-model F-bundle associated with X . Consider the subgroup $\mathbb{Z}\mathbf{P} \subset \text{NS}(X, \mathbb{Z})_{\text{tf}}$. It gives rise to a subtorus

$$\mathfrak{b}_X := (\mathbb{Z}\mathbf{P} \otimes \mathbb{G}_{m, \mathbb{k}})^{\text{an}} \subset (\text{NS}(X, \mathbb{Z})_{\text{tf}} \otimes \mathbb{G}_{m, \mathbb{k}})^{\text{an}} = B_{X, q}.$$

The torus $\mathfrak{b}_X$ is one dimensional with natural coordinate q which is dual to $\mathbf{P}$.

The restriction of $(\mathcal{H}, \nabla)|_{\mathfrak{b}_X \times \mathbb{D}}$ is an analytic F-bundle. The comparison [28] of the Gromov-Witten invariants of X and the equivariant Gromov-Witten invariants of $\mathbb{P}^{N-1}$ implies that the restricted connection $\nabla|_{\mathfrak{b}_X \times \mathbb{D}}$ preserves the subbundle

$$\mathfrak{h} := \mathbf{R} \otimes \mathcal{O}_{\mathfrak{b}_X \times \mathbb{D}} \subset H^\bullet(X, \mathbb{Q}) \otimes \mathcal{O}_{\mathfrak{b}_X \times \mathbb{D}} = \mathcal{H}|_{\mathfrak{b}_X \times \mathbb{D}}.$$

This gives a new smaller analytic F-bundle $(\mathfrak{h}, \nabla)/\mathfrak{b}_X$ of rank $N - k = \dim X + 1$ over the one dimensional base $\mathfrak{b}_X$. If we write q for the coordinate on the analytic affine line $(\mathbb{k} \cdot \mathbf{P})^{\text{an}}$ that is dual to basis vector $\mathbf{P}$, then the meromorphic connection ∇ on $\mathfrak{h}$ becomes

$$(6.1) \quad \begin{cases} \nabla_{\partial_u} = \partial_u - u^{-2}\mathbf{K} + u^{-1}\mathbf{G}, \\ \nabla_{\partial_q} = \partial_q + u^{-1}q^{-1}\mathbf{A}, \end{cases}$$

where $\mathbf{G}$ is the grading operator on $\mathfrak{h}$, $\mathbf{A}$ is the operator of quantum multiplication by $\mathbf{P}$, and $\mathbf{K} = (N - d_{\text{tot}})\mathbf{A}$ is the operator of quantum multiplication by $c_1(T_X)$. These operators are coefficients of the restricted quantum $\nabla|_{\mathfrak{b}_X \times \mathbb{D}}$ acting on the subbundle $\mathfrak{h}$ and so we have that $\mathbf{K}$ is equal to the action of $\kappa|_{\mathfrak{b}_X}$ on the subbundle $\mathfrak{h}|_{u=0}$, $\mathbf{G}$ is the restriction of Gr to the subbundle $\mathfrak{h}|_{u=0}$, and $\mathbf{A}$ is the restriction quantum multiplication by $\mathbf{P}$ to the subbundle $\mathfrak{h}|_{u=0} \subset \mathcal{H}|_{\mathfrak{b}_X \times \{0\}}$.

In [28, Section 9] Givental described the sheaf of horizontal sections of $(\mathfrak{h}, \nabla)/\mathfrak{b}_X$ in terms of the fundamental solution of an explicit auxiliary linear ODE of order $N - k$, which was later dubbed the **quantum differential equation**. Givental used his description to give a recursive procedure for computing the Gromov-Witten invariants of X . For us however, the main takeaway from Givental's theorem is that along the one dimensional base $\mathfrak{b}_X$ the action of the operator $(\text{Eu} \circ (-))|_{\mathfrak{b}_X} : \mathfrak{h} \rightarrow \mathfrak{h}$ can be computed from the quantum differential equation.

To unpack this, let us recall Givental's calculation from [28, Section 9]. Consider the following setup:

(1) Let $\mathbf{d}$ be a non-negative integer and let $\overline{\mathcal{M}}_{0,2}(\mathbb{P}^{N-1}, \mathbf{d})$ denote the moduli stack of genus zero degree $\mathbf{d}$ stable maps $\varphi : (C, p_1, p_2) \rightarrow \mathbb{P}^{N-1}$ with two marked points.

(2) Let $e_{\mathbf{d}}$ be the top Chern class of the vector bundle on $\overline{\mathcal{M}}_{0,2}(\mathbb{P}^{N-1}, \mathbf{d})$ whose fiber at (C, p_1, p_2, φ) is $H^0(C, \varphi^*(\mathcal{O}(d_1) \oplus \cdots \mathcal{O}(d_k)))$.

(3) Let $c \in H^2(\overline{\mathcal{M}}_{0,2}(\mathbb{P}^{N-1}, \mathbf{d}), \mathbb{Q})$ denote the first Chern class of the universal tangent line at the first marked point.

(4) Write $\text{ev}_1 : \overline{\mathcal{M}}(\mathbb{P}^{N-1}, \mathbf{d}) \rightarrow \mathbb{P}^{N-1}$ for the evaluation at the first marked point.

In [28, Section 9] Givental argued that the equivariant version of [28, Corollary 6.4] implies that the series

$$S(t, \hbar) := e^{P^t/\hbar} \sum_{d=0}^{\infty} (X \hookrightarrow \mathbb{P}^{N-1})^* \text{ev}_{1*} \left(\frac{1}{\hbar + c} e_d \right) \in \mathbf{R}((\hbar)) \llbracket e^t \rrbracket [t],$$

is a horizontal section of the formal quantum connection for X . Givental's variables $(t, \hbar)$ are related to our variables (q, u) by the change of variables $q = e^t$, $u = -\hbar$. Therefore, we have that S as a function of (q, u) belongs to $\mathbf{R}((u)) \llbracket q \rrbracket [\log q]$, and viewed as a formal section of $\mathfrak{h}$ is annihilated by the connection ∇ given by (6.1).

Furthermore, in [28, Theorem 9.1 and Corollary 9.2] Givental proved:

(i) $S(t, \hbar)$ is given by the explicit formula

$$S(t, \hbar) = e^{tP/\hbar} \sum_{d=0}^{\infty} e^{dt} \frac{\prod_{i=1}^k \left(\prod_{m_i=1}^{d_i} (d_i P + m_i \hbar) \right)}{\prod_{m=1}^d (P + m \hbar)^N} \in \mathbf{R}((\hbar)) \llbracket e^t \rrbracket [t],$$

(ii) $S(t, \hbar)$ satisfies the so called **quantum differential equation**

$$(6.2) \quad (\hbar \partial_t)^{N-k} S = e^t \prod_{i=1}^k \left(d_i \prod_{m_i=1}^{d_i-1} \hbar (d_i \partial_t + m_i) \right) S,$$

and the components $S_i(t, \hbar) \in \mathbb{Q}((\hbar)) \llbracket e^t \rrbracket [t]$ of $S = \sum_{i=0}^{\dim X} S_i \cdot P^i$ form a basis of the space of solutions of the linear ODE (6.2).

Remark 6.3. (a) Givental's description implies that in fact $S(t, \hbar)$ belongs to the subring $\mathbf{R}[\hbar, \hbar^{-1}] \llbracket e^t \rrbracket [t]$ and so its specialization $S(t, 1)$ as $\hbar \rightsquigarrow 1$ makes sense. Thus we have the specialized series

$$S(t, 1) = e^{Pt} \sum_{d=0}^{\infty} e^{dt} \frac{\prod_{i=1}^k \left(\prod_{m_i=1}^{d_i} (d_i P + m_i) \right)}{\prod_{m=1}^d (P + m)^N} \in \mathbf{R} \llbracket e^t \rrbracket [t]$$

or, in terms of the variables (q, u) , the $u \rightsquigarrow -1$ specialized series

$$S(q, -1) = e^{P \log(q)} \sum_{d=0}^{\infty} q^d \frac{\prod_{i=1}^k \left(\prod_{m_i=1}^{d_i} (d_i P + m_i) \right)}{\prod_{m=1}^d (P + m)^N} \in \mathbf{R} \llbracket q \rrbracket [\log q].$$

(b) Rewriting the equation (6.2) in terms of the variables (q, u) we get that S as a series in $\mathbf{R}[u, u^{-1}] \llbracket q \rrbracket [\log q]$ satisfies

$$(6.4) \quad (u q \partial_q)^{N-k} S = (-1)^{N-d_{\text{tot}}} q \prod_{i=1}^k \left(d_i \prod_{m_i=1}^{d_i-1} u (d_i q \partial_q + m_i) \right) S.$$

Note that the component $S_0(q, u)$ solving (6.4) is particularly simple. It is given by the hypergeometric series

$$S_0 = e^{P \log(q)} \sum_{d=0}^{\infty} q^d \frac{\prod_{i=1}^k \left(\prod_{m_i=1}^{d_i} (d_i P + m_i) \right)}{\prod_{m=1}^d (P + m)^N} \in \mathbf{R} \llbracket q \rrbracket [\log(q)].$$

Consider now the operator $\mathbf{A}: \mathbf{R}((\hbar))[[e^t]][t] \rightarrow \mathbf{R}((\hbar))[[e^t]][t]$ of quantum multiplication by $\mathbf{P}$. Its matrix in the standard basis $\mathbf{1}, \mathbf{P}, \dots, \mathbf{P}^{\dim X}$ has a **diagonal-jump-shape** with the step of the jump equal to the index $(N - d_{\text{tot}})$ of the Fano variety X (in the example $\dim X = 8$, $N - d_{\text{tot}} = 3$)

$$\mathbf{A} = \begin{pmatrix} 0 & 0 & *e^t & 0 & 0 & *e^{2t} & 0 & 0 & *e^{3t} \\ \color{red}{1} & 0 & 0 & *e^t & 0 & 0 & *e^{2t} & 0 & 0 \\ 0 & \color{red}{1} & 0 & 0 & *e^t & 0 & 0 & *e^{2t} & 0 \\ 0 & 0 & \color{red}{1} & 0 & 0 & *e^t & 0 & 0 & *e^{2t} \\ 0 & 0 & 0 & \color{red}{1} & 0 & 0 & *e^t & 0 & 0 \\ 0 & 0 & 0 & 0 & \color{red}{1} & 0 & 0 & *e^t & 0 \\ 0 & 0 & 0 & 0 & 0 & \color{red}{1} & 0 & 0 & *e^t \\ 0 & 0 & 0 & 0 & 0 & 0 & \color{red}{1} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \color{red}{1} & 0 \end{pmatrix}.$$

Here the unknown coefficients $*$ are integers, and $\mathbf{A}$ is symmetric with respect to a flip along the anti-diagonal. Equivalently, in the variables (q, u) , the matrix of $\mathbf{A}: \mathbf{R}((u))[[q]][\log q] \rightarrow \mathbf{R}((u))[[q]][\log q]$ is given by the same matrix but with the exponentials $e^t, e^{2t}, e^{3t}, \dots$ replaced by $q, q^2, q^3, \dots$, respectively.

Consider now the second horizontality condition $\nabla_{\partial_q} \Psi = 0$ for the connection ∇ . Explicitly it reads $uq\partial_q \Psi = -\mathbf{A}\Psi$, or

$$(6.5) \quad uq\partial_q \begin{bmatrix} \psi_0 \\ \psi_1 \\ \vdots \\ \vdots \\ \psi_{\dim X} \end{bmatrix} = - \begin{pmatrix} 0 & \dots & 0 & *q & 0 & \dots & 0 & *q^2 & 0 & \dots \\ \color{red}{1} & 0 & \dots & 0 & *q & 0 & \dots & 0 & *q^2 & \dots \\ 0 & \color{red}{1} & 0 & \dots & 0 & *q & 0 & \dots & 0 & \dots \\ 0 & 0 & \color{red}{1} & 0 & \dots & 0 & *q & 0 & \dots & \dots \\ 0 & 0 & 0 & \color{red}{1} & 0 & \dots & 0 & *q & 0 & \dots \\ 0 & 0 & 0 & 0 & \color{red}{1} & 0 & \dots & 0 & *q & \dots \\ 0 & 0 & 0 & 0 & 0 & \color{red}{1} & 0 & \dots & 0 & \dots \\ 0 & 0 & 0 & 0 & 0 & 0 & \color{red}{1} & 0 & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \end{pmatrix} \cdot \begin{bmatrix} \psi_0 \\ \psi_1 \\ \vdots \\ \vdots \\ \psi_{\dim X} \end{bmatrix},$$

where the coordinates ψ_i of the vector function Ψ are functions of the variable of differentiation q , depending on the parameter u .

The connection between (6.5) and Givental's quantum differential equation (6.4) arises as follows. The last coordinate function $\psi_{\dim X}$ obeys the quantum differential equation, i.e. is a $\mathbb{C}$ -linear combination of the functions $S_0, S_1, \dots, S_{\dim X-1}$. Now, starting with a vector valued solution $\Psi = (\psi_0, \dots, \psi_{\dim X})^T$ of (6.5), we can use the diagonal-jump-shape of $\mathbf{A}$ to solve recursively for the coordinate functions $\psi_{\dim X-1}, \psi_{\dim X-2}, \dots, \psi_1, \psi_0$ in terms of the last coordinate function $\psi_{\dim X}$. Thus the system (6.5) becomes equivalent to $\psi_{\dim X}$ obeying an ODE of order $N - k$ which has to be Givental's quantum differential equation. Comparing the coefficients of the ODE of order $N - k$ that we obtain from the recursive procedure with the coefficients of (6.4) we obtain a system of algebraic equations on the unknown entries $*$ in the matrix $\mathbf{A}$. This gives an algorithm for computing $\mathbf{A}$ and $\mathbf{K} = (N - d_{\text{tot}})\mathbf{A}$ from the quantum differential equation.

Example 6.6. (i) To illustrate the process, let us compute $\mathbf{A}$ in the case when X is a 3-dimensional quadric, i.e. $N = 5$, $k = 1$, $d = 2$. In this case the system (6.5) reads

$$uq\partial_q \begin{bmatrix} \psi_0 \\ \psi_1 \\ \psi_2 \\ \psi_3 \end{bmatrix} = - \begin{pmatrix} 0 & 0 & a_1q & 0 \\ 1 & 0 & 0 & a_2q \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \cdot \begin{bmatrix} \psi_0 \\ \psi_1 \\ \psi_2 \\ \psi_3 \end{bmatrix}$$

where a_1, a_2 are some unknown integers. If we simplify notation by setting $\phi = \psi_3$ we get

$$\begin{aligned} \psi_2 &= -uq\partial_q\phi, \\ \psi_1 &= -uq\partial_q\psi_2, \\ \psi_0 &= -uq\partial_q\psi_1 - a_2q\phi, \\ uq\partial_q\psi_0 &= -a_1q\psi_2. \end{aligned}$$

Solving recursively for all ψ_i in terms of ϕ we arrive at the 4-th order linear ODE

$$(uq\partial_q)^4\phi = -qu((a_1 + a_2)q\partial_q + a_2)\phi.$$

Comparing this with the quantum differential equation

$$(uq\partial_q)^4\phi = -qu(2(2q\partial_q + 1))\phi,$$

we get that $a_1 + a_2 = 4$, $a_2 = 2$. Hence for this case we get

$$\mathbf{A} = \begin{pmatrix} 0 & 0 & 2q & 0 \\ 1 & 0 & 0 & 2q \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

Following this strategy, we can compute all coefficients in the connection (6.1) of the F-bundle $(\mathfrak{h}, \nabla)/\mathfrak{b}_X$ for any complete intersection. In particular we have the following

(ii) Suppose X is a smooth 3-dimensional cubic, i.e. $N = 5$, $k = 1$, $d = 3$. Then the coefficients of the connection (6.1) defining the F-bundle $(\mathfrak{h}, \nabla)/\mathfrak{b}_X$ are

$$\mathbf{A} = \begin{pmatrix} 0 & 6q & 0 & 0 \\ 1 & 0 & 15q & 0 \\ 0 & 1 & 0 & 6q \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad \mathbf{K} = 2 \begin{pmatrix} 0 & 6q & 0 & 0 \\ 1 & 0 & 15q & 0 \\ 0 & 1 & 0 & 6q \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad \mathbf{G} = \begin{pmatrix} -3/2 & 0 & 0 & 0 \\ 0 & -1/2 & 0 & 0 \\ 0 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & 3/2 \end{pmatrix}.$$

(iii) Suppose X is a smooth 4-dimensional cubic, i.e. $N = 6$, $k = 1$, $d = 3$. The coefficients of the connection (6.1) defining the F-bundle $(\mathfrak{h}, \nabla)/\mathfrak{b}_X$ are

$$\mathbf{A} = \begin{pmatrix} 0 & 0 & 6q & 0 & 0 \\ 1 & 0 & 0 & 15q & 0 \\ 0 & 1 & 0 & 0 & 6q \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}, \quad \mathbf{K} = 3 \begin{pmatrix} 0 & 0 & 6q & 0 & 0 \\ 1 & 0 & 0 & 15q & 0 \\ 0 & 1 & 0 & 0 & 6q \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}, \quad \mathbf{G} = \begin{pmatrix} -2 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{pmatrix}.$$

Remark 6.7. In this section we used Givental's theorem to compute the action of the Euler vector field for the A-model F-bundle of a projective manifold, which can in turn be used to analyze the atomic decomposition of Fano complete intersections. It is important to note that even though Givental's original results for toric complete intersections do not work directly for other types of Fano varieties, the current state of the art comprises many other methods that can be used to compute the quantum spectrum virtually in any specific example. Three of these techniques deserve special attention. The first is the quantum Lefschetz/Serre package, which includes the quantum Lefschetz theorems of Lee, Kim, and Coates-Givental [58, 50, 18], and the quantum Serre theorems of Coates-Givental and Iritani-Mann-Mignon [18, 46]. The second is the toric degeneration technique which, as demonstrated in the work of Ciocan-Fontanine, Kim, and Sabbah [17], can be used to derive the exact form of the quantum differential equation for complete intersections in homogeneous spaces. Last but not least, one can use Homological Mirror Symmetry and the explicit forms of mirrors as provided by the Hori-Vafa construction [39] to compute the quantum spectrum as the critical values of the mirror Landau-Ginzburg potential.

6.2 Four-dimensional cubics

In this section we will apply the theory to some classical rationality problems.

Theorem 6.8. *A very general 4-dimensional cubic hypersurface in $\mathbb{CP}^5$ is not rational.*

Proof. Let X be a smooth cubic hypersurface in $\mathbb{CP}^5$. Assume that X is Noether-Lefschetz general in the sense that the only rational Hodge classes on X are the powers of the hyperplane class $\mathbf{P} = c_1(\mathcal{O}_{\mathbb{P}^5}(1))|_X$. Such cubic fourfolds exist e.g. by Voisin's proof [75] of the Torelli theorem for cubic fourfolds.

The odd degree cohomology of X is zero and the total dimension of the even cohomology is 27. Following the construction explained in Section 3.5.2 we choose a homogeneous basis $\{T_i\}_{i=1}^{27}$ of $H_B^\bullet(X, \mathbb{Q})$, so that $T_1 = \mathbf{1} \in H^0(X, \mathbb{Q})$, $T_2 = \mathbf{P}$, and $\deg T_i > 2$ for $i > 2$. We write q for the coordinate on $\text{NS}(X, \mathbb{Q}) = \mathbb{Q} \cdot \mathbf{P}$ to $\mathbf{P}$ and we write t_i for the coordinates dual to T_i for $i \neq 2$. Let $(\mathcal{H}, \nabla)/B_X$ be the maximal analytic A-model F-bundle for X defined in Section 3.5.2 with connection ∇ given by

$$(6.9) \quad \begin{cases} \nabla_{\partial_u} = \partial_u - u^{-2}(\text{Eu} \star (-)) + u^{-1} \frac{\text{Deg} - 4 \cdot \text{id}}{2}, \\ \nabla_{\partial_q} = \partial_q + u^{-1} q^{-1}(\mathbf{P} \star (-)), \\ \nabla_{\partial_{t_i}} = \partial_{t_i} + u^{-1}(T_i \star (-)), \text{ for } i \neq 2. \end{cases}$$

Let $b \in B_X$ be the rigid point with coordinates $q = 1$, and $t_i = 0$, for all $i \neq 2$. Then $b \in \mathfrak{b}_X \subset B_X^{\text{Hod}} \subset B_X$ and by Lemma 5.19 we have that the reduced spectrum of the operator $(\text{Eu} \star (-))_b : \mathcal{H}_{(b,0)} \rightarrow \mathcal{H}_{(b,0)}$ is the same as the reduced spectrum of the action of $(\text{Eu} \star (-))_b$ on the subspace of Hodge classes $\mathcal{H}_{(b,0)}^{\text{Hod}}$. By our genericity assumption

$$\mathcal{H}_{(b,0)}^{\text{Hod}} = \bigoplus_{a=0}^4 \mathbb{K} \cdot \mathbf{P}^a = \mathfrak{h}_{(b,0)}.$$

Thus

$$(\text{Eu} \star (-))_b|_{\mathfrak{h}_{(b,0)}} = \mathbf{K}|_{q=1},$$

where $\mathbf{K}$ is the operator introduced in the previous section.

As we saw in Section 6.1, the operator $\mathbf{K}: \mathfrak{h}|_{\mathfrak{b}_X \times \{0\}} \rightarrow \mathfrak{h}|_{\mathfrak{b}_X \times \{0\}}$ can be computed from Givental's theorem and from Example 6.6 (iii) we have

$$\mathbf{K} = 3 \begin{pmatrix} 0 & 0 & 6q & 0 & 0 \\ 1 & 0 & 0 & 15q & 0 \\ 0 & 1 & 0 & 0 & 6q \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix} \quad \text{and so} \quad \mathbf{K}|_{q=1} = 3 \begin{pmatrix} 0 & 0 & 6 & 0 & 0 \\ 1 & 0 & 0 & 15 & 0 \\ 0 & 1 & 0 & 0 & 6 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

The characteristic polynomial of this matrix is $\lambda^5 - 3^6 \lambda^2$ and so $\mathbf{K}|_{q=1}$ has eigenvalues 0 (appearing with multiplicity 2), and $9, 9\zeta, 9\zeta^2$ where ζ is a primitive third root of unity. Therefore the reduced spectrum of $\kappa_b := \text{Eu}_b \circ (-)$ acting on all of $\mathcal{H}_{(b,0)}$ is also equal to $\{0, 9, 9\zeta, 9\zeta^2\}$.

By the spectral decomposition theorem Theorem 4.1 we then have that over some admissible open neighborhood $b \in U \subset B_X^{\text{Hod}}$ the A-model F-bundle $(\mathcal{H}, \nabla)/B_X^{\text{Hod}}$ decomposes into an external direct sum of maximal F-bundles:

$$(\mathcal{H}, \nabla)/U = \bigoplus_{\lambda \in \{0, 9, 9\zeta, 9\zeta^2\}} (\mathcal{H}^\lambda, \nabla^\lambda)/U^\lambda$$

compatible with the decomposition of $\mathcal{H}_{(b,0)}$ into a direct sum of generalized eigenspaces for $\kappa_b = \text{Eu}_b \circ (-)$.

Let now $\alpha \in \text{HAtoms}$ be a Hodge atom appearing in the atomic composition of X . By definition α is represented by some $\alpha \in \pi_0(\tilde{U}_X)$. Let $U_{X,\alpha}$ be the corresponding connected component of $\tilde{U}_X$. Since U_X is open and dense in B_X^{Hod} , the intersection $U \cap U_X$ will be non-trivial and so the equivalence class of the atomic F-bundle associated with α must be a generalized eigenbundle for the action of $\text{Eu} \star (-)$ on just one of the four pieces $(U \times \mathbb{D} \rightarrow U^\lambda \times \mathbb{D})^* (\mathcal{H}^\lambda, \nabla^\lambda)$.

This implies that if E^α is the finite dimensional $\overline{\mathbb{Q}}$ -linear $\text{Hod}_{\overline{\mathbb{Q}}}$ -representation corresponding to α , then we have

$$\dim_{\overline{\mathbb{Q}}}(E^\alpha)^{\text{Hod}(\overline{\mathbb{Q}})} \leq \min_{\lambda \in \{0, 9, 9\zeta, 9\zeta^2\}} \text{rank}(\mathcal{H}_{u=0}^\lambda)^{\text{Hod}} \leq \min_{\lambda \in \{0, 9, 9\zeta, 9\zeta^2\}} \dim_{\mathbb{K}} \mathfrak{h}_{(b,0)}^\lambda,$$

where $\mathfrak{h}_{(b,0)}^\lambda$ denotes the generalized eigenspace of the operator $\mathbf{K}|_{q=1}$ acting on $\mathfrak{h}_{(b,0)} = \bigoplus_{i=0}^4 \mathbb{K} \cdot \mathbf{P}^i$.

Therefore we get that for every Hodge atom α appearing in the atomic composition of X , the dimension ρ_α of Hodge classes in α satisfies

$$\rho_\alpha = \dim_{\overline{\mathbb{Q}}}(E^\alpha)^{\text{Hod}(\overline{\mathbb{Q}})} \leq 2.$$

On the other hand, by our assumption of Noether-Lefschetz genericity of X the Hodge pieces in the degree 4 primitive cohomology of X corresponding to the row

$$(h_{\text{prim}}^{4,0}(X), h_{\text{prim}}^{3,1}(X), h_{\text{prim}}^{2,2}(X), h_{\text{prim}}^{1,3}(X), h_{\text{prim}}^{0,4}(X)) = (0, 1, 20, 1, 0)$$

in the primitive Hodge diamond of X all consist of transcendental cycles. Since by definition the atomic composition of the cohomology of X is compatible with the Hochschild grading, this implies that in the atomic composition of X we must have an atom α with $\text{Coeff}_{t^2}(P_\alpha(t)) = 1$.

Thus, according to Proposition 5.30, it only remains to show that a Hodge atom α with $\text{Coeff}_{t^2}(P_\alpha(t)) = 1$ and $\rho_\alpha \leq 2$ can not appear in the atomic composition of a smooth projective

variety of dimension ≤ 2 , i.e. can not come from points, curves, or surfaces. Moreover, for each birational class of surfaces it is sufficient to consider the atomic composition of one representative in the class.

But if α is an atom of a point or a curve, then $\text{Coeff}_{t^2}(P_\alpha(t)) = 0$. If, on the other hand, α is an atom appearing in the atomic composition of a smooth projective surface S , then $\text{Coeff}_{t^2}(P_\alpha(t)) \leq H^{2,0}(S) = p_g(S)$ and by the classification of surfaces S must be either an abelian surface, a K3 surface, an elliptic surface with $\kappa = 1$ and $p_g = 1$, or a surface of general type. But every such surface has a nef K_S , and so by Lemma 5.24 the atomic composition of S consists of a single atom $\eta(S)$, and the corresponding representation $E^{\eta(S)}$ of $\text{Hod}_{\overline{\mathbb{Q}}}$ is just equal to $H_B^\bullet(S, \overline{\mathbb{Q}})$. Since S is projective it has at least one algebraic cycle of dimension 2 and two more algebraic cycles of dimension 0 and 4 respectively. Thus $\rho_{\eta(S)} \geq 3$. This shows that α can not be an atom of a surface and so X can not be rational. $\square$

Remark 6.10. In the proof of the previous theorem we used the fact that Hodge atoms of algebraic surfaces for which the $p - q = 2$ part of Hodge structure is nontrivial must at least a 3 dimensional space Hodge cycles spanned by the 0-th, 1-st, and 2-nd powers of the hyperplane class. In fact the Enriques-Kodaira classification of surfaces gives a complete list of Hodge atoms for varieties of dimension ≤ 2 . We have summarized this list in the following table, where again we have simplified the notation by labeling the Hodge atoms by the corresponding subspaces in the cohomology $H^\bullet$ of the respective variety.

dim	0	1	1	2	2	2	2	2
X	pt	$\mathbb{P}^1$	curve C , $g(C) \geq 1$	$\mathbb{P}^2$	$\mathbb{P}^1 \times C$ $g(C) \geq 1$	abelian surface S or K3 surface	Enriques or elliptic or bielliptic	general type with ADE singularity $\rightarrow$ blow up $\tilde{S}$
Hodge atoms	$H^\bullet(pt)$ $= \mathbb{C}$	$H^\bullet(pt \cup pt)$	$H^\bullet(C)$	$H^\bullet(pt \cup pt \cup pt)$	$H^\bullet(C) \oplus H^\bullet(C)$	$H^\bullet(S)$	$ p - q \leq 1$ in nc-Hodge structure	$H^\bullet(\tilde{S})$

Table 1: Hodge atoms for varieties of dimension ≤ 2

Let X be a smooth cubic fourfold which is Noether-Lefschetz general as in the proof of Theorem 6.8. The coarse analysis of the atomic composition of X that we used in the previous theorem, can be made more precise. First we have the following

Lemma 6.11. *For any $\phi \in H_{\text{prim}}^4(X)$, we have $\kappa_b(\phi) = 0$.*

Proof. It suffices to show that for any $\psi \in H_B^\bullet(X)$, and any curve class $\beta \in H_2(X)$, the Gromov-Witten invariant $\langle \mathbf{P}, \phi, \psi \rangle_{0,n,\beta}^X = 0$. By the virtual dimension calculation, if $\langle \mathbf{P}, \phi, \psi \rangle_{0,n,\beta}^X \neq 0$, then $\text{codim } \psi \geq 4$. The only possibility is $\psi \in H^8(X)$. Now both $\mathbf{P}$ and ψ come from the ambient projective space. Since $\phi \in H_{\text{prim}}^4(X)$, we deduce that $\mathbf{P} \smile \phi \smile \psi = 0$, hence $\langle \mathbf{P}, \phi, \psi \rangle_{0,n,\beta}^X = 0$, completing the proof. $\square$

Corollary 6.12. *Let X be a very general cubic fourfold over $\mathbb{C}$, then X has 4 Hodge atoms in its atomic composition. Three of those are one dimensional and correspond to the eigenvectors with eigenvalues 9 , 9ζ , and $9\zeta^2$ for the action of κ_b on $\mathcal{H}_{(b,0)}$, and one is 24-dimensional and corresponds to the generalized eigenspace of κ_b for the eigenvalue 0 .*

Proof. By Lemma 6.11 the action of $\kappa_b = \mathbf{K}|_{q=1}$ on $H_B^\bullet(X, \mathbb{k})$ will have a single eigenvalue 0 on the subspace $H^4(X)_{\text{prim}} \otimes \mathbb{k}$. Therefore, the piece $(\mathcal{H}^0, \nabla^0)/U^0$ in the spectral decomposition of $(\mathcal{H}, \nabla)/U$ is a maximal F-bundle of rank 24.

The Torelli theorem [75, 59, 15] for cubic 4-folds also implies that for a very general X the transcendental part of the cohomology of X is an irreducible representation of $\text{Hod}_{\overline{\mathbb{Q}}}$. Furthermore, the operator $\mathbf{K}|_{q=1}$ has a size 2 Jordan block corresponding to the eigenvalue 0, and since by Proposition 5.28 a Hodge atom must have dimension of Hodge classes ≥ 1 , it follows that a Hodge atom α in the atomic composition of X which comes from a generalized eigenspace factor in $(\mathcal{H}^0, \nabla^0)/U^0$ must be the whole thing, i.e. have $\dim_{\overline{\mathbb{Q}}} E^\alpha = 24$. This proves the corollary. $\square$

Remark 6.13. We expect that the 24 dimensional representation of Hod associated with the atom is the cohomology of the Kuznetsov component [54] of X .

6.3 Birational maps of Calabi-Yau manifolds

Hodge atoms can also be used effectively to study the motivic properties of birational equivalences between smooth projective varieties. In particular, Hodge atoms can be used to give a new proof of Batyrev's theorem [7, Corollary 6.29] asserting that birationally equivalent Calabi-Yau manifolds have equal Hodge numbers. The traditional proofs of this result [7, 80, 47] use either motivic integration, or p -adic Hodge theory and Čebotarev density. The proof via atoms has a completely different flavor as it uses Gromov-Witten theory and nc-Hodge theory.

To show the equality of Hodge numbers we begin with the following observation

Lemma 6.14. *Let X_1 and X_2 be two complex birational Calabi-Yau manifolds of dimension d . Then the Betti cohomology spaces $H_B^\bullet(X_1, \mathbb{Q})$ and $H_B^\bullet(X_2, \mathbb{Q})$ are isomorphic as representations of $\text{Hod}_{\overline{\mathbb{Q}}}(\mathbb{Q})$. In particular the $\mathbb{Z}/2$ -folded Hodge polynomials of X_1 and X_2 are equal, or equivalently*

$$\sum_{p-q=k} h^{p,q}(X_1) = \sum_{p-q=k} h^{p,q}(X_2), \quad \text{for all } k = -d, \dots, d.$$

Proof. Since K_{X_i} is numerically trivial for $i = 1, 2$, Lemma 5.24 implies that the atomic compositions of X_1 and X_2 each consist of a single atom. Let us denote these respective atoms by $\eta(X_1)$ and $\eta(X_2)$. Since X_1 and X_2 are birational, the weak factorization theorem [1] implies that X_1 and X_2 are related by a sequence of forward/backward blowups with smooth centers. But the blowup decomposition Theorem 4.5 implies that each such blowup adds to the atomic composition only atoms of smooth projective varieties of dimension $\leq d - 2$. In particular for every such atom α we must have $\text{Coeff}_{t^d}(P_\alpha(t)) = 0$. On the other hand $h^{d,0}(X_i) = 1$ for $i = 1, 2$ and so $\text{Coeff}_{t^d}(\eta(X_1)) = \text{Coeff}_{t^d}(\eta(X_2)) = 1$. Thus we must have $\eta(X_1) = \eta(X_2)$ which proves the lemma. $\square$

Next we will use some special features of the A-model F-bundle in the Calabi-Yau case. Let X be a d -dimensional complex projective manifold. Let $(\mathcal{H}, \nabla)/B_X$ be the $\mathbb{k}$ -analytic maximal A-model F-bundle of X . Fix any rigid point $b \in B_X^{\text{ev}} \subset B_X$, so that the coordinate of b corresponding to $1 \in H_B^0(X, \mathbb{k})$ vanishes. Consider the restriction of $(\mathcal{H}, \nabla)|_{\{b\} \times \text{Spf } \mathbb{k}[[u]]}$. Then we have

$$\begin{aligned} \mathcal{H}|_{\{b\} \times \text{Spf } \mathbb{k}[[u]]} &= H^\bullet(X)[[u]] \\ \nabla|_{\{b\} \times \text{Spf } \mathbb{k}[[u]]} &\text{ is given by } \nabla_{\partial_u} = \partial_u - u^{-2} \text{Eu} \star (-) + u^{-1} \text{Gr}, \end{aligned}$$

where

$$\text{Gr} = \frac{\text{Deg} - d \cdot \text{id}}{2}$$

is the grading operator, and we write $\text{Eu} \star (-)$ as a shortcut for $\text{Eu}_b \star_b (-)$ and $H^\bullet(X)$ as a shortcut for $H_B^\bullet(X, \mathbb{k})$. To keep track of the half integer eigenvalues of Gr , let us also introduce the operator $\mathbf{T}: H^\bullet(X) \rightarrow H^\bullet(X)$ which is the identity on each $H^a(X)$ with $a - d$ odd and is zero on each $H^a(X)$ with $a - d$ even. Note that by definition the operator $\mathbf{g} = \text{Gr} + \frac{1}{2}\mathbf{T}$ is semisimple with integral eigenvalues. Therefore, we have a meromorphic gauge transformation $u^{\mathbf{g}}$.

With this notation we now have the following

Claim 6.15. *Suppose K_X is nef. Then the connection*

$$\nabla_{\partial_u} = \partial_u - u^{-2} \text{Eu} \star (-) + u^{-1} \text{Gr} : H^\bullet(X)[[u]] \longrightarrow H^\bullet(X)((u))$$

has a regular singularity at $u = 0$. More precisely, the gauge transformed connection

$$u^{\mathbf{g}} \circ \left(\nabla + \frac{1}{2} u^{-1} \mathbf{T} du \right) \circ u^{-\mathbf{g}}$$

has a first order pole and a nilpotent residue at $u = 0$.

Proof. Let $\tilde{\nabla} = \nabla + \frac{1}{2} u^{-1} \mathbf{T} du$, then $\tilde{\nabla}_{u\partial_u} = \nabla_{u\partial_u} + \frac{1}{2} \mathbf{T}$, and since $\mathbf{g}$ commutes with Gr we get

$$\begin{aligned} u^{\mathbf{g}} \circ \tilde{\nabla}_{u\partial_u} \circ u^{-\mathbf{g}} &= u\partial_u - \mathbf{g} + \frac{1}{2} \mathbf{T} + u^{-1} u^{\mathbf{g}} \circ (\text{Eu} \star (-)) \circ u^{-\mathbf{g}} + u^{\mathbf{g}} \circ \text{Gr} \circ u^{-\mathbf{g}} \\ &= u\partial_u - u^{-1} u^{\mathbf{g}} \circ (\text{Eu} \star (-)) \circ u^{-\mathbf{g}}. \end{aligned}$$

Since $K_X \geq 0$, and $b \in B_X^{\text{ev}}$ has a vanishing $H^0(X)$ -coordinate, the operator $\text{Eu} \star (-)$ will move the eigenspace of Gr corresponding to an eigenvalue $m \in \frac{1}{2}\mathbb{Z}$ to the sum of eigenspaces for eigenvalues $\geq m + 1$. Thus if we let $\kappa_{ij}: H^i(X, \mathbb{k}) \rightarrow H^{i+2j}(X, \mathbb{k})$ denote the respective block matrix component of $\text{Eu} \star (-)$, then $\kappa_{ij} = 0$ for $j \leq 0$. The corresponding block matrix component of the gauged transformed operator $u^{-1} u^{\mathbf{g}} \circ (\text{Eu} \star (-)) \circ u^{-\mathbf{g}}$ is given by the composition

$$H^i(X) \xrightarrow{u^{\text{const} - \lfloor \frac{j}{2} \rfloor}} H^i(X) \xrightarrow{u^{-1} \kappa_{ij}} H^{i+2j}(X) \xrightarrow{u^{\lfloor \frac{j}{2} \rfloor + j - \text{const}}} H^{i+2j}(X),$$

and $\text{const} \in \mathbb{Z}$. In other words the block matrix component of the transformed operator is $u^{-1+j} \kappa_{ij}$ and so the gauge transformed connection operator $u^{\mathbf{g}} \circ \tilde{\nabla}_{u\partial_u} \circ u^{-\mathbf{g}}$ has no pole at $u = 0$ and the $u = 0$ residue of $u^{\mathbf{g}} \circ \tilde{\nabla}_{u\partial_u} \circ u^{-\mathbf{g}}$ is strictly lower triangular as a block matrix with entries labeled by the degree of cohomology. $\square$

Theorem 6.16. *Let X be a smooth projective complex Calabi-Yau. Then the Hodge numbers of X can be reconstructed from the atomic F-bundle associated with the Hodge atom $\eta(X)$ of X . In particular, birationally equivalent Calabi-Yau manifolds have equal Hodge numbers.*

Proof. Let $(\mathcal{H}, \nabla)/B_X$ be the maximal analytic A-model F-bundle of X . Let $b \in B_X^{\text{Hod}}$ be a general point with vanishing component in $H^0(X)$. Then, since $\eta(X)$ is the unique Hodge atom in the atomic composition of X , it follows that the germ $(\mathcal{H}, \nabla)/B_X(b)$ represents the atomic F-bundle associated with $\eta(X)$. The restriction of this atomic F-bundle to $\{b\} \times \text{Spf } \mathbb{k}[[t]]$ gives rise to a $k((u))$ -vector space

$H^\bullet(X)((u))$ equipped with a meromorphic connection ∇ which by Claim 6.15 has a regular singularity at $u = 0$. Thus the $k((u))$ -vector space $H^\bullet(X)((u))$ comes equipped with two natural lattices, i.e. two $\mathbb{k}[[u]]$ -submodules $H^{\text{int}}, H^{\text{can}}$, such that $H^{\text{int}} \otimes_{\mathbb{k}[[u]]} \mathbb{k}((u)) = H^{\text{can}} \otimes_{\mathbb{k}[[u]]} \mathbb{k}((u)) = H^\bullet(X)((u))$. Here H^{int} is the intrinsic lattice, i.e. the module of global sections of the original bundle $\mathcal{H}|_{\{b\} \times \text{Spf } \mathbb{k}[[u]]}$, and H^{can} is the module given by the Deligne's canonical extension [22, 23] of the regular singular connection $(H^\bullet(X)((u)), \nabla)$, i.e. the extension uniquely characterized by the property that in this extension the connection has a pole of order ≤ 1 with a nilpotent residue.

This characterization of H^{can} implies that what we showed in Claim 6.15 is that the automorphism $u^g: H^\bullet(X)((u)) \rightarrow H^\bullet(X)((u))$ moves the lattice H^{int} to H^{can} . Thus an automorphism of $H^\bullet(X)((u))$ which moves H^{int} isomorphically to H^{can} will induce an additional filtration labeled by cohomological degree, i.e. by $p+q$. Combined with the grading by $p-1$ this gives the (p, q) bigrading on $H^\bullet(X, \mathbb{k})$. $\square$

6.4 Enhanced atoms and further examples

Hodge atoms can be enhanced in several ways to produce stronger obstructions to rationality. In particular, we can consider F-bundles equipped with additional algebraic structures, such as pairings, duality automorphisms, and integral structures. For the analytic A-model F-bundle associated with a smooth projective variety X , these structures arise naturally from natural structures on the cohomology of X - the Mukai pairing (which is non-symmetric in general), the Serre functor automorphism, and the Γ -integral structure. This leads to enhanced versions of the Hodge atoms - Hodge atoms with pairings, Hodge atoms with duality automorphisms, and Hodge atoms with integral structures.

In more detail, let $\nu: B \times \mathbb{D} \rightarrow B \times \mathbb{D}$ be the involution given by $\nu(b, u) = (b, -u)$. A **non-degenerate pairing** on an analytic F-bundle $(\mathcal{H}, \nabla)/B$ is an even $\mathcal{O}_{B \times \mathbb{D}}$ -linear map

$$\psi: \mathcal{H} \otimes \nu^* \mathcal{H} \longrightarrow \mathcal{O}_{B \times \mathbb{D}},$$

which is symmetric (in the sense that $\psi(\xi_1, \xi_2) = (\nu^* \psi)(\xi_2, \xi_1)$ for all local sections $\xi_1 \in \mathcal{H}$, $\xi_2 \in \nu^* \mathcal{H}$), non-degenerate (in the sense that it induces an isomorphism $\mathcal{H} \cong \nu^* \mathcal{H}^\vee$), and preserved by ∇ (in the sense that $d\psi(\xi_1, \xi_2) = \psi(\nabla \xi_1, \xi_2) + \psi(\xi_1, \nu^* \nabla \xi_2)$). Alternatively, a non-degenerate pairing can be specified by an isomorphism $\Psi: \mathcal{H} \xrightarrow{\sim} \nu^* \mathcal{H}^\vee$ of super vector bundles on $B \times \mathbb{D}$, which intertwines the connections on the source and target, and is self-dual in the sense that $\nu^* \Psi^\vee = \Psi$. Note that the by definition non-degenerate pairings are compatible with the spectral decomposition of F-bundles. This immediately gives induced pairings on G -atoms.

A duality automorphism of an F-bundle $(\mathcal{H}, \nabla)/B$ equipped with a non-degenerate pairing ψ is an automorphism $S: \mathcal{H}|_{B \times \{u=1\}} \rightarrow \mathcal{H}|_{B \times \{u=1\}}$ of the restricted bundle $\mathcal{H}|_{B \times \{u=1\}}$ which satisfies appropriate compatibility conditions with ∇ and ψ . It takes some work to formulate these compatibilities since they are inherently transcendental and require the Riemann-Hilbert correspondence which is not readily available in the non-archimedean analytic setting. We will discuss this in more detail in [49] but for now let us give a brief topological discussion of this interpretation.

Suppose that $S^1 = \{u : |u| = 1\} \subset \mathbb{C}$ is the unit circle, and let $\nu: S^1 \rightarrow S^1$ be the negation map $\nu(u) = -u$. Let $\mathcal{H} \rightarrow S^1$ be a local system of finite dimensional complex super vector spaces. As before we define a symmetric non-degenerate pairing on $\mathcal{H}$ as an isomorphism $\Psi: \mathcal{H} \xrightarrow{\sim} \nu^* \mathcal{H}^\vee$, such that $\nu^* \Psi^\vee = \Psi$. Similarly, if V is a finite dimensional complex super vector space, then we define a (not necessarily symmetric) non-degenerate pairing on V as a linear isomorphism $G: V \xrightarrow{\sim} V^\vee$.

$$\mathcal{H} : \begin{array}{ccc} & V & \\ G \swarrow & & \searrow G^\vee \\ & V^\vee & \end{array} .$$

Note that with this definition, if we identify $\mathcal{H}_1$ with V , then the monodromy operator $\text{mon}_1: \mathcal{H}_1 \rightarrow \mathcal{H}_1$ of $\mathcal{H}$ at the point $1 \in S^1$ (for going once around S^1 in the counterclockwise direction), is given by $\text{mon}_1 = (G^\vee)^{-1} \circ G: V \rightarrow V$.

This description immediately yields analogous explicit presentations of $\nu^*\mathcal{H}$ and $\nu^*\mathcal{H}^\vee$:

$$\nu^* \mathcal{H} : \quad (G^\vee)^{-1} \begin{array}{c} V \\ \downarrow \quad \downarrow \\ V^\vee \end{array} G^{-1} \quad \text{and} \quad \nu^* \mathcal{H}^\vee : \quad G \begin{array}{c} V \\ \downarrow \quad \downarrow \\ V^\vee \end{array} G^\vee$$

and we define the map $\Psi : \mathcal{H} \rightarrow \nu^* \mathcal{H}^\vee$ to be

$$\begin{array}{ccc} V & \xrightarrow{\text{id}} & V \\ G \downarrow & & \downarrow G \\ V^V & \xrightarrow{\text{id}} & V^V \end{array}$$

The two assignments $(\mathcal{H}, \Psi) \mapsto (V, G)$ and $(V, G) \mapsto (\mathcal{H}, \Psi)$ are inverse to each other and establish the equivalence of pairing data.

Let now X be a complex smooth projective variety. Assume for simplicity that the quantum product is convergent and so we have a complex analytic version $(\mathcal{H}, \nabla)/B_X$ of the A-model maximal F-bundle for X . In this case we have $\mathcal{H}_u := \mathcal{H}|_{B \times \{u\}} = H_B^\bullet(X, \mathbb{C}) \otimes \mathcal{O}_B$, and we can define $\psi_u: \mathcal{H}_u \otimes_{\mathcal{O}_B} \mathcal{H}_{-u} \rightarrow \mathcal{O}_B$ to be the pairing between $\mathcal{H}|_{B \times \{u\}} = H_B^\bullet(X, \mathbb{Q}) \otimes \mathcal{O}_B$ to be the $\mathcal{O}_B$ -linear extension of the Poincaré pairing on $H_B^\bullet(X, \mathbb{Q})$. The pairing ψ defined in this way is compatible with geometric data:

(a) The pairing ψ is preserved by the connection ∇ . Indeed, in the u -direction we have $\nabla_{\partial_u} = \partial_u - u^{-2} \text{Eu} \star (-) + u^{-1} \text{Gr}$, the operators $\text{Eu} \star (-)$ and Gr are independent of u , and by definition $\text{Eu} \star (-)$ is selfadjoint with respect to the Poincaré pairing while Gr is anti-selfadjoint with respect to the Poincaré pairing. Similarly, along B , the connection is given by covariant derivatives $\nabla_{\partial_\tau} = \partial_\tau + u^{-1} \tau \star (-)$ or $\nabla_{\xi q \partial_q} = \xi q \partial_q + u^{-1} \xi \star (-)$, and since the operators $\tau \star (-)$ or $\xi \star (-)$ are all self adjoint, it follows that ψ is horizontal along B as well.

(b) The pairing ψ takes integral values on the Γ -lattice in $\mathcal{H}$. Indeed, recall from [48, Section 3.1] that if $(\mathcal{H}, \nabla)/B_X$ is of exponential type, one can construct a special $\mathbb{Z}$ -local subsystem $\mathcal{E} \subset (\mathcal{H}|_{B_X \times \mathbf{D}^\times})^\nabla$ in the local system of ∇ -horizontal sections in the restriction of $\mathcal{H}$ to the complement of the divisor $\{u = 0\}$. The local system $\mathcal{E}$ has the property, that $\mathcal{E} \otimes_{\mathbb{Z}} \mathcal{O} \cong \mathcal{H}$ over $B_X \times \mathbf{D}^\times$, and conjecturally, for every $b \in B_X$ its restriction to $\{b\} \times \mathbf{D}^\times$ is compatible with the Deligne-Malgrange-Stokes filtration for the wildly ramified connection $(\mathcal{H}, \nabla)|_{\{b\} \times \mathbf{D}}$. For any point $(b, u) \in B_X \times \mathbf{D}^\times$, the fiber $\mathcal{E}_{(b, u)}$ of the local system $\mathcal{E}$ is defined as a particular embedding $\iota_{(\tilde{b}, \log u)}: K_{\text{top}}(X) \rightarrow \mathcal{H}_{(b, u)}$, which depends on a lift $(\tilde{b}, \log u)$ of (b, u) in the universal cover of $B_X \times \mathbf{D}^\times$, and is given by a multiplication of the Chern character map by the Gamma class of X . The compatibility of ψ and $\mathcal{E}$ is then expressed in the property that for every (b, u) , any two elements $s_1, s_2 \in \mathcal{E}_{(b, u)}$, and any choice of a branch of the logarithm which is well defined on the upper semicircle arc^+ , we have that

$$\psi_u(\iota_{(\tilde{b}, \log u)}(s_1), \iota_{(\tilde{b}, \log(-u))}(\tau_{\text{arc}^+}(s_2))) \in \mathbb{Z}.$$

Furthermore, the value of this pairing does not depend on the lifts and if s_1 and s_2 are the Chern characters of two objects E_1 and E_2 in the derived category $D^b(X)$, then we have

$$\psi_u(\iota_{(\tilde{b}, \log u)}(s_1), \iota_{(\tilde{b}, \log(-u))}(\tau_{\text{arc}^+}(s_2))) = \chi(\text{RHom}(E_1, E_2)).$$

(c) Equivalently, the compatibility described in (b) can be reformulated as follows. Consider the Euler pairing defined by

$$\chi: H_B^\bullet(X, \mathbb{C}) \otimes H_B^\bullet(X, \mathbb{C}) \rightarrow \mathbb{C}, \quad \chi(a, b) := \left\langle \sqrt{td(X)} \smile a, \sqrt{td(X)} \smile b \right\rangle,$$

where $\langle -, - \rangle$ is the **Mukai pairing**

$$\langle v, v' \rangle = \int_X (v^\vee \smile v') \smile \exp(c_1(X)/2),$$

and the **dual** $v^\vee$ of a cohomology class $v = \sum_k v_k \in \oplus_k H_B^k(X, \mathbb{C})$ is defined by $v^\vee = \sum_k (\sqrt{-1})^k v_k$. As usual, here the Euler pairing is normalized so that if E and F are objects in $D^b(X)$, then $\chi(ch(E), ch(F)) = \sum (-1)^k \dim_{\mathbb{C}} \text{Ext}^k(E, F)$.

Let $V = H_B^\bullet(X, \mathbb{C})$, and let $G: V \rightarrow V^\vee$ be the map $a \mapsto \chi(a, -)$. Then composing G with the clockwise half circle parallel transport in $(\mathcal{H}, \nabla)$ we get a non-degenerate symmetric pairing $\mathcal{H} \otimes \nu^* \mathcal{H}^\vee \rightarrow \mathcal{O}$ which we will call the **F-bundle Euler pairing**. When $(\mathcal{H}, \nabla)/B_X$ is of exponential type, the F-bundle Euler pairing is identified with the Poincaré duality non-degenerate pairing defined above. This follows from the observation that for every cohomology class $s \in H_B^{2\text{odd}}(X)$, we have that the operator on $H_B^\bullet(X)$ of classical multiplication by $\exp(s)$ preserves the Euler pairing $\chi(-, -)$. Since the multiplication by $\sqrt{td(X)}$ and the multiplication by the Gamma class differ by multiplications by such $\exp(s)$ we conclude that the non-symmetric pairing attached to ψ and the map given by $a \mapsto \chi(a, -)$ will coincide.

Extra work is required to carry out this comparison for the non-archimedean A-model F-bundles, and this will be discussed in more detail in [49].

(d) Finally, we have the duality automorphism $S: H_B^\bullet(X, \mathbb{C}) \rightarrow H_B^\bullet(X, \mathbb{C})$ given by the cup product with $(-1)^d \exp(-c_1(x))$, which by Serre duality satisfies $\chi(a, b) = \chi(b, S(a))$ for all a, b . In

particular $S = (G^\vee)^{-1} \circ G$, i.e. S equals the monodromy operator on $\mathcal{H}_1$ for the loop in the u plane going once in the counterclockwise direction around the origin. This gives an analytic automorphism of $\mathcal{H}_1$ which will again denote by S . By construction S will be preserved by the connection along B . Therefore, if $(\mathcal{H}, \nabla)/B$ is a complex analytic F-bundle of exponential type with a non-degenerate pairing ψ we can define the **F-bundle duality automorphism** $S : \mathcal{H}|_{B \times \{u=1\}} \rightarrow \mathcal{H}|_{B \times \{u=1\}}$ to be the monodromy in the u -direction. Its compatibility with the pairing is expressed by the property $\chi(a, b) = \chi(b, S(a))$, where χ is the non-symmetric pairing on $\mathcal{H}|_{B \times \{u=1\}}$ associated with ψ via the above procedure. Again, the definition in the case of non-archimedean analytic F-bundle requires extra work, and will be discussed in [49].

The theory of Hodge atoms enhanced with Euler pairings and Serre automorphisms is completely straightforward and is just a repeat of the theory of undecorated Hodge atoms. The theory of Hodge atoms enhanced with an integral structure is much more difficult and requires substantial work and new ideas. Already the definition of an integral structure on a $\mathbb{k}$ -analytic F-bundle is non-trivial due to the limited understanding of the Riemann-Hilbert correspondence we have in the equal characteristic non-archimedean analytic setting. Additionally one needs to prove the existence of the Γ -corrected integral structure for analytic A-model F-bundles of projective manifolds, and has to show that the blowup decomposition from Theorem 4.5 is compatible with Γ -corrected integral structures. We deal with all these issues in forthcoming work [49].

Here we just note that with the enhancements in place, we get a stronger version of the non-rationality criterion Proposition 5.30. In this version we conclude that if a smooth projective variety X has an enhanced atomic composition containing an enhanced atom that does not come from a smooth projective variety of dimension $\leq \dim X - 2$, then X can not be rational.

In the remainder of this section we will discuss some interesting applications of this theory.

Example 6.17. Let X be a very general cubic fourfold containing a plane $P \subset X$. Then the linear projection centered at P gives a quadric surface fibration on $Y \rightarrow \mathbb{P}^2$ on the fourfold $Y = \text{Bl}_P X$. The fibration $Y \rightarrow \mathbb{P}^2$ gives rise to a double cover $S \rightarrow \mathbb{P}^2$ - the $K3$ surface parametrizing the rulings of the quadrics and a $\mathbb{P}^1$ -bundle $\mathcal{R} \rightarrow S$ parametrizing the lines in the quadrics.

If $\alpha \in H^2(S, \mathcal{O}^\times)$ is the characteristic class of the Brauer-Severi variety $\mathcal{R}$, then α defines a 2-torsion gerbe ${}_\alpha S$ on S , whose weight one derived category is the derived category $D^b(S, \alpha)$ of α -twisted sheaves on S .

In [54] Kuznetsov proved that for a general cubic with a plane the category $D^b(S, \alpha)$ can not be equivalent to the derived category of any smooth projective surface. In fact, he showed that the numerical K -group $K_{\text{num}}(S, \alpha)$ of $D^b(S, \alpha)$ viewed as a lattice equipped with the α -twisted Mukai pairing, can not be isomorphic to the numerical K -lattice of any smooth projective surface. On the other hand the spectrum calculation in the proof of Theorem 6.8 and formalism of Hodge atoms enhanced with pairings and integral structures immediately gives that the α -twisted Hodge structure on $H^\bullet(S)$ together with its integral structure and Mukai pairing gives an atom in the enhanced atomic composition of X . Since enhanced atoms give obstructions to rationality this implies that X is not rational.

Example 6.18. Let $X = \cap_{t \in \mathbf{A}} Q_t$ be the base locus of a general net $\mathbf{A} \cong \mathbb{P}^2 \subset |\mathcal{O}_{\mathbb{P}^7}(2)|$ of quadrics in $\mathbb{P}^7$. Then X contains a line $\ell \subset X$, and

- (1) $Y = \{(t, \Pi) \mid \ell \subset \Pi \subset Q_t\} \subset \text{Gr}(3, 8)$ is a smooth fourfold Y , which is birational to X .
- (2) $Y \rightarrow \mathbf{A}$ is a quadric surface fibration which gives rise to

- a double cover $S \rightarrow \mathbb{P}^2$ - a surface of general type parametrizing the rulings of the quadrics.
- a $\mathbb{P}^1$ -bundle $\mathcal{R} \rightarrow S$ parametrizing the lines in the fibers of $Y \rightarrow \mathbb{P}^2$.

If $\alpha \in H^2(S, \mathcal{O}^\times)$ is the characteristic class of the Brauer-Severi variety $\mathcal{R}$, then again we get a 2-torsion gerbe ${}_\alpha S$ on S , and a twisted derived category $D^b(S, \alpha)$.

Again the α -twisted Hodge structure of S together with its Mukai pairing and integral structure gives rise of an enhanced Hodge atom in the atomic composition of X . Similarly to the case of a cubic with a plane, it can be shown [25] that the numerical K -lattice of $D^b(X, \alpha)$ is not isomorphic to the numerical K -lattice of any smooth projective surface, and so $D^b(X, \alpha)$ can not be equivalent to the derived category of any smooth projective surface. Thus the very general X can not be rational. This argument gives a new proof of a non-rationality theorem of Hassett-Pirutka-Tschinkel [33].

Example 6.19. Let $X \subset \mathbb{P}^5$ be a very general smooth quartic hypersurface. Its non-rationality is known by the works of Kollár [51], Schreieder [72], and Totaro [74]. We outline an atom theory argument also proving that very general X is not rational.

Suppose X is Noether-Lefschetz general so that it has Picard rank one. By repeating the argument in the proof of Theorem 6.8 we see that the Hodge atom corresponding to the eigenvalue zero for the quantum multiplication by the Euler vector field looks like the Hodge atom of a surface of general type.

At the point b in B_X corresponding to the hyperplane class, the atomic composition of $H^\bullet(X)$ looks like

$$(6.20) \quad H^\bullet(X) = \alpha(X) \oplus \alpha_+(X) \oplus \alpha_-(X).$$

Here $\alpha(X)$ is the atom corresponding to eigenvalue zero of $\text{Eu} \star (-)$ and is shaped like an atom of a surface of a general type, while $\alpha_+(X)$ and $\alpha_-(X)$ are one dimensional atoms corresponding to eigenvalues which are scalings of ± 1 . The Witt algebra argument from Remark 3.14 shows that there is no further splitting if we move in a small analytic neighborhood of b .

As explained in Claim 6.15 the quantum connection for a surface of general type has regular singularities and so the conjugacy class of its monodromy is well defined and is unipotent on the even cohomology of the surface.

However, from the point of view of the decomposition (6.20), in the cohomological grading the Serre automorphism S associated with the $\alpha(X)$ has a graded minimal polynomial $S^3 = [4]$. This contradicts the unipotency of the monodromy on the even part of an atom of a surface of general type. Thus $\alpha(X)$ cannot be an enhanced atom associated with surface of general type which proves the non rationality of very general X .

Example 6.21. Consider a smooth three dimensional cubic $X \subset \mathbb{P}^4$. Again for the point $b \in B_X$ corresponding to the hyperplane class, the atomic composition of X consists of three atoms - two one dimensional atoms corresponding to non-zero eigenvalues of $\text{Eu} \star (-)$, and one atom $\alpha(X)$ corresponding to the eigenvalue 0. The Witt algebra argument from Remark 3.14 shows that there is no further splitting in a neighborhood of b .

By Claim 6.15 the quantum connection for a higher genus curve will have regular singularities, and a well defined conjugacy class of the monodromy which will be unipotent on the even cohomology and quasi-unipotent with eigenvalue -1 on the odd cohomology.

The atom $\alpha(X)$ looks like the atom of a smooth projective curve of genus 5 but when viewed as a Hodge atom enhanced with a Serre automorphism, the corresponding Serre automorphism S in

the cohomological $\mathbb{Z}$ -grading satisfies the graded minimal polynomial $S^5 = [3]$. Thus S can not have an eigenvalue -1 and so S cannot be a Serre automorphism for a smooth genus 5 curve. So we get that all smooth three dimensional cubics are not rational. Similar approach applies to other three dimensional Fanos.

References

- [1] Dan Abramovich, Kalle Karu, Kenji Matsuki, and Jarosław Włodarczyk. Torification and factorization of birational maps. *J. Amer. Math. Soc.*, 15(3):531–572, 2002.
- [2] Nicolas Addington and Richard Thomas. Hodge theory and derived categories of cubic fourfolds. *Duke Math. J.*, 163(10):1885–1927, 2014.
- [3] Yves André. Pour une théorie inconditionnelle des motifs. *Inst. Hautes Études Sci. Publ. Math.*, 83:5–49, 1996.
- [4] Yves André. *Une introduction aux motifs (motifs purs, motifs mixtes, périodes)*, volume 17 of *Panoramas et Synthèses [Panoramas and Syntheses]*. Société Mathématique de France, Paris, 2004.
- [5] Asher Auel, Christian Böhning, and Alena Pirutka. Stable rationality of quadric and cubic surface bundle fourfolds. *Eur. J. Math.*, 4(3):732–760, 2018.
- [6] The Stacks Project Authors. Stacks project, 2025.
- [7] Victor V. Batyrev. Stringy Hodge numbers of varieties with Gorenstein canonical singularities. In *Integrable systems and algebraic geometry (Kobe/Kyoto, 1997)*, pages 1–32. World Sci. Publ., River Edge, NJ, 1998.
- [8] Arnaud Beauville. Variétés de Prym et jacobiniennes intermédiaires. *Ann. Sci. École Norm. Sup. (4)*, 10(3):309–391, 1977.
- [9] Arnaud Beauville and Ron Donagi. La variété des droites d’une hypersurface cubique de dimension 4. *C. R. Acad. Sci. Paris Sér. I Math.*, 301(14):703–706, 1985.
- [10] Kai Behrend. Gromov-Witten invariants in algebraic geometry. *Invent. Math.*, 127(3):601–617, 1997.
- [11] Kai Behrend and Barbara Fantechi. The intrinsic normal cone. *Invent. Math.*, 128(1):45–88, 1997.
- [12] Vladimir G. Berkovich. *Spectral theory and analytic geometry over non-Archimedean fields*, volume 33 of *Mathematical Surveys and Monographs*. American Mathematical Society, Providence, RI, 1990.
- [13] Franziska Bittner. The universal Euler characteristic for varieties of characteristic zero. *Compos. Math.*, 140(4):1011–1032, 2004.
- [14] Leonardo Cavenaghi, Lino Grama, Ludmil Katzarkov, and Maxim Kontsevich. Atoms meet symbols, 2025. in preparation.
- [15] François Charles. A remark on the Torelli theorem for cubic fourfolds, 2012.
- [16] Zihong Chen. On the exponential type conjecture, 2024.
- [17] Ionuț Ciocan-Fontanine, Bumsig Kim, and Claude Sabbah. The abelian/nonabelian correspondence and Frobenius manifolds. *Invent. Math.*, 171(2):301–343, 2008.
- [18] Tom Coates and Alexander Givental. Quantum Riemann-Roch, Lefschetz and Serre. *Ann. of Math. (2)*, 165(1):15–53, 2007.
- [19] Olivier Debarre and Alexander Kuznetsov. Gushel-Mukai varieties: classification and birationalities. *Algebr. Geom.*, 5(1):15–76, 2018.
- [20] Olivier Debarre and Alexander Kuznetsov. Gushel-Mukai varieties: intermediate Jacobians. *Épjournal Géom. Algébrique*, 4:Art. 19, 45, 2020.
- [21] Olivier Debarre and Giovanni Mongardi. Gushel-Mukai varieties with many symmetries and an explicit irrational Gushel-Mukai threefold. *Boll. Unione Mat. Ital.*, 15(1-2):133–161, 2022.
- [22] Pierre Deligne. *Équations différentielles à points singuliers réguliers*, volume Vol. 163 of *Lecture Notes in Mathematics*. Springer-Verlag, Berlin-New York, 1970.
- [23] Pierre Deligne. Local behavior of Hodge structures at infinity. In *Mirror symmetry, II*, volume 1 of *AMS/IP Stud. Adv. Math.*, pages 683–699. Amer. Math. Soc., Providence, RI, 1997.

- [24] Pierre Deligne, James S. Milne, Arthur Ogus, and Kuang-yen Shih. *Hodge cycles, motives, and Shimura varieties*, volume 900 of *Lecture Notes in Mathematics*. Springer-Verlag, Berlin-New York, 1982.
- [25] Ron Donagi and Tony Pantev. The intersection of three quadrics in $\mathbb{P}^7$ revisited, 2025. preprint.
- [26] Boris Dubrovin. Geometry of 2D topological field theories. In *Integrable systems and quantum groups (Montecatini Terme, 1993)*, volume 1620 of *Lecture Notes in Math.*, pages 120–348. Springer, Berlin, 1996.
- [27] Alexander B. Givental. Homological geometry and mirror symmetry. In *Proceedings of the International Congress of Mathematicians, Vol. 1, 2 (Zürich, 1994)*, pages 472–480. Birkhäuser, Basel, 1995.
- [28] Alexander B. Givental. Equivariant Gromov-Witten invariants. *Internat. Math. Res. Notices*, 13:613–663, 1996.
- [29] Hans Grauert and Reinhold Remmert. *Theory of Stein spaces*. Classics in Mathematics. Springer-Verlag, Berlin, 2004. Translated from the German by Alan Huckleberry, Reprint of the 1979 translation.
- [30] Brendan Hassett. Special cubic fourfolds. *Compositio Math.*, 120(1):1–23, 2000.
- [31] Brendan Hassett. Cubic fourfolds, K3 surfaces, and rationality questions. In *Rationality problems in algebraic geometry*, volume 2172 of *Lecture Notes in Math.*, pages 29–66. Springer, Cham, 2016.
- [32] Brendan Hassett, Alena Pirutka, and Yuri Tschinkel. Intersections of three quadrics in $\mathbb{P}^7$. In *Surveys in differential geometry 2017. Celebrating the 50th anniversary of the Journal of Differential Geometry*, volume 22 of *Surv. Differ. Geom.*, pages 259–274. Int. Press, Somerville, MA, 2018.
- [33] Brendan Hassett, Alena Pirutka, and Yuri Tschinkel. Intersections of three quadrics in $\mathbb{P}^7$. In *Surveys in differential geometry 2017. Celebrating the 50th anniversary of the Journal of Differential Geometry*, volume 22 of *Surv. Differ. Geom.*, pages 259–274. Int. Press, Somerville, MA, 2018.
- [34] Brendan Hassett, Alena Pirutka, and Yuri Tschinkel. Stable rationality of quadric surface bundles over surfaces. *Acta Math.*, 220(2):341–365, 2018.
- [35] Brendan Hassett, Alena Pirutka, and Yuri Tschinkel. A very general quartic double fourfold is not stably rational. *Algebr. Geom.*, 6(1):64–75, 2019.
- [36] C. Hertling. *Frobenius manifolds and moduli spaces for singularities*, volume 151 of *Cambridge Tracts in Mathematics*. Cambridge University Press, Cambridge, 2002.
- [37] Claus Hertling. tt^* geometry, Frobenius manifolds, their connections, and the construction for singularities. *J. Reine Angew. Math.*, 555:77–161, 2003.
- [38] Thorgal Hinault, Tony Yue Yu, Chi Zhang, and Shaowu Zhang. Decomposition and framing of F-bundles and applications to quantum cohomology. *arXiv preprint arXiv:2411.02266*, 2024.
- [39] Kentaro Hori and Cumrun Vafa. Mirror symmetry. *arXiv preprint hep-th/0002222*, 2000.
- [40] Annette Huber and Stefan Müller-Stach. *Periods and Nori motives*, volume 65 of *Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge. A Series of Modern Surveys in Mathematics [Results in Mathematics and Related Areas. 3rd Series. A Series of Modern Surveys in Mathematics]*. Springer, Cham, 2017. With contributions by Benjamin Friedrich and Jonas von Wangenheim.
- [41] Daniel Huybrechts. The K3 category of a cubic fourfold. *Compos. Math.*, 153(3):586–620, 2017.
- [42] Daniel Huybrechts. *The geometry of cubic hypersurfaces*, volume 206 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, Cambridge, 2023.
- [43] Hiroshi Iritani. Quantum D -modules and generalized mirror transformations. *Topology*, 47(4):225–276, 2008.
- [44] Hiroshi Iritani. Quantum cohomology of blowups, 2023.
- [45] Hiroshi Iritani and Yuki Koto. Quantum cohomology of projective bundles, 2024.
- [46] Hiroshi Iritani, Etienne Mann, and Thierry Mignon. Quantum Serre theorem as a duality between quantum D -modules. *Int. Math. Res. Not. IMRN*, 9:2828–2888, 2016.
- [47] Tetsushi Ito. Stringy Hodge numbers and p -adic Hodge theory. *Compos. Math.*, 140(6):1499–1517, 2004.
- [48] Ludmil Katzarkov, Maxim Kontsevich, and Tony Pantev. Hodge theoretic aspects of mirror symmetry. In *From Hodge theory to integrability and TQFT tt^* -geometry*, volume 78 of *Proc. Sympos. Pure Math.*, pages 87–174. Amer. Math. Soc., Providence, RI, 2008.

- [49] Ludmil Katzarkov, Maxim Kontsevich, Tony Pantev, and Tony Yue YU. Enhanced atoms and integrality, 2025. in preparation.
- [50] Bumsig Kim. Quantum hyperplane section theorem for homogeneous spaces. *Acta Math.*, 183(1):71–99, 1999.
- [51] János Kollár. Nonrational hypersurfaces. *J. Amer. Math. Soc.*, 8:241–249, 1995.
- [52] Maxim Kontsevich and Yuri Manin. Gromov-Witten classes, quantum cohomology, and enumerative geometry. *Comm. Math. Phys.*, 164(3):525–562, 1994.
- [53] Viktor S. Kulikov. A remark on the nonrationality of a generic cubic fourfold. *Mat. Zametki*, 83(1):61–68, 2008.
- [54] Alexander Kuznetsov. Derived categories of cubic fourfolds. In *Cohomological and geometric approaches to rationality problems*, volume 282 of *Progr. Math.*, pages 219–243. Birkhäuser Boston, Boston, MA, 2010.
- [55] Alexander Kuznetsov. Derived categories view on rationality problems. In *Rationality problems in algebraic geometry*, volume 2172 of *Lecture Notes in Math.*, pages 67–104. Springer, Cham, 2016.
- [56] Alexander Kuznetsov and Yuri Prokhorov. Rationality of Mukai varieties over non-closed fields. In *Rationality of varieties*, volume 342 of *Progr. Math.*, pages 249–290. Birkhäuser/Springer, Cham, [2021] ©2021.
- [57] Alexander Kuznetsov and Yuri Prokhorov. Rationality of Fano threefolds over non-closed fields. *Amer. J. Math.*, 145(2):335–411, 2023.
- [58] Yuan-Pin Lee. Quantum Lefschetz hyperplane theorem. *Invent. Math.*, 145(1):121–149, 2001.
- [59] Eduard Looijenga. The period map for cubic fourfolds. *Invent. Math.*, 177(1):213–233, 2009.
- [60] Yuri I. Manin. *Frobenius manifolds, quantum cohomology, and moduli spaces*, volume 47 of *American Mathematical Society Colloquium Publications*. American Mathematical Society, Providence, RI, 1999.
- [61] James S. Milne. *Algebraic groups*, volume 170 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, Cambridge, 2017. The theory of group schemes of finite type over a field.
- [62] James S. Milne. Classification of the Mumford–Tate groups of rational polarizable Hodge structures, 2023.
- [63] James S. Milne. Tannakian categories, 2025.
- [64] Jacob P. Murre, Jan Nagel, and Chris A. M. Peters. *Lectures on the theory of pure motives*, volume 61 of *University Lecture Series*. American Mathematical Society, Providence, RI, 2013.
- [65] Madhav Nori. TIFR notes on motives, 1998.
- [66] Alena Pirutka. Varieties that are not stably rational, zero-cycles and unramified cohomology. In *Algebraic geometry: Salt Lake City 2015*, volume 97.2 of *Proc. Sympos. Pure Math.*, pages 459–483. Amer. Math. Soc., Providence, RI, 2018.
- [67] Daniel Pomerleano and Paul Seidel. The quantum connection, Fourier-Laplace transform, and families of A-infinity-categories, 2024.
- [68] Claude Sabbah. *Isomonodromic deformations and Frobenius manifolds*. Universitext. Springer-Verlag London, Ltd., London; EDP Sciences, Les Ulis, french edition, 2007. An introduction.
- [69] Kyoji Saito. The higher residue pairings $K_F^{(k)}$ for a family of hypersurface singular points. In *Singularities, Part 2 (Arcata, Calif., 1981)*, volume 40 of *Proc. Sympos. Pure Math.*, pages 441–463. Amer. Math. Soc., Providence, RI, 1983.
- [70] Kyoji Saito. Period mapping associated to a primitive form. *Publ. Res. Inst. Math. Sci.*, 19(3):1231–1264, 1983.
- [71] Morihiko Saito. On the structure of Brieskorn lattice. *Ann. Inst. Fourier (Grenoble)*, 39(1):27–72, 1989.
- [72] Stefan Schreieder. Stably irrational hypersurfaces of small slopes. *J. Amer. Math. Soc.*, 32:1171–1199, 2019.
- [73] Amaury Thuillier. Géométrie toroïdale et géométrie analytique non archimédienne. Application au type d’homotopie de certains schémas formels. *Manuscripta Math.*, 123(4):381–451, 2007.
- [74] Burt Totaro. Hypersurfaces that are not stably rational. *J. Amer. Math. Soc.*, 29:883–891, 2016.
- [75] Claire Voisin. Théorème de Torelli pour les cubiques de $\mathbb{P}^5$. *Invent. Math.*, 86(3):577–601, 1986.

- [76] Claire Voisin. Integral Hodge classes, decompositions of the diagonal, and rationality questions. In *Trends in contemporary mathematics*, volume 8 of *Springer INdAM Ser.*, pages 137–149. Springer, Cham, 2014.
- [77] Claire Voisin. Unirational threefolds with no universal codimension 2 cycle. *Invent. Math.*, 201(1):207–237, 2015.
- [78] Claire Voisin. Stable birational invariants and the Lüroth problem. In *Surveys in differential geometry 2016. Advances in geometry and mathematical physics*, volume 21 of *Surv. Differ. Geom.*, pages 313–342. Int. Press, Somerville, MA, 2016.
- [79] Claire Voisin. Birational invariants and decomposition of the diagonal. In *Birational geometry of hypersurfaces*, volume 26 of *Lect. Notes Unione Mat. Ital.*, pages 3–71. Springer, Cham, [2019] ©2019.
- [80] Chin-Lung Wang. K -equivalence in birational geometry. In *Second International Congress of Chinese Mathematicians*, volume 4 of *New Stud. Adv. Math.*, pages 199–216. Int. Press, Somerville, MA, 2004.
- [81] Jarosław Włodarczyk. Birational cobordisms and factorization of birational maps. *J. Algebraic Geom.*, 9(3):425–449, 2000.

Ludmil Katzarkov, IMSA, University of Miami, USA; ICMS, IMI, Bulgaria; HSE University, Russia

Email: `l.katzarkov@math.miami.edu`

Maxim Kontsevich, Institut des Hautes Études Scientifiques, France

Email: `maxim@ihes.fr`

Tony Pantev, University of Pennsylvania, USA

Email: `tpantev@math.upenn.edu`

Tony Yue Yu, California Institute of Technology, USA

Email: `yuyuetony@gmail.com`